Predicting Formal Protest Petitions Against Housing Development: Evidence from Austin, Texas

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Abstract

This thesis investigates whether formal protest petitions against proposed housing developments in Austin, Texas can be anticipated before public hearings, using only the administrative and neighborhood records available at the time of application filing. The central question is whether parcel characteristics, neighborhood demographics, and application features observable by planning staff at filing can meaningfully identify which cases are likely to attract organized opposition.

Using discretionary zoning cases from Austin's pre-HB 24 petition regime (2007–2024), predictive models are trained and evaluated on a temporal holdout that mimics forward administrative use. The results are qualified: filing-date features can rank petition risk above chance—the best model achieves moderate out-of-distribution discrimination with a wide uncertainty interval (PR-AUC = 0.718, 95% CI: [0.48, 0.91])—but the rarity of petitioned cases and the model's limited external validity mean predictions are best used as relative triage rankings, not as case-level probabilities for high-stakes decisions.

The core finding is that administrative data at filing carries real predictive signal for petition risk, yet that signal is too uncertain and context-dependent to support uncritical institutional deployment. The broader contribution is a methodology—time-stamped features, temporal holdout evaluation, and explainability audits—that is portable to other discretionary review systems, even if the specific parameter estimates are Austin-specific.

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1 Introduction

The United States faces a persistent housing affordability crisis, with local barriers to new housing supply drawing increasing scrutiny from policymakers and researchers. While zoning laws are often cited as primary constraints, much of the friction occurs through discretionary political processes, including public hearings, protest petitions, and city council negotiations, that can delay or block individual projects. Understanding how these mechanisms operate in practice is essential for diagnosing the roots of housing scarcity and for designing effective policy interventions. Austin, Texas, provides a compelling case study: rapid population and employment growth between 2010 and 2024 intensified supply constraints, fueling conflict between neighborhood preservation and regional development goals. Austin's contemporary land-use politics also sit atop a longer history of exclusionary planning, racial segregation, and uneven redevelopment stretching from the 1928 city plan into the present [35, 64, 65, 51].

One of the most consequential mechanisms for neighborhood opposition in Austin has been the formal protest petition, enabled under Texas Local Government Code Section 211.006 [58]. Prior to 2025, if property owners representing at least 20% of the land area within 200 feet of a proposed zoning change signed a petition, and the municipal clerk certified its validity, the proposal required a three-fourths supermajority vote in city council to proceed. This statutory process created a powerful tool for delaying or blocking new housing, reinforcing the influence of organized local opposition [21]. In 2025, House Bill 24 replaced this system with a differentiated, multi-tier framework codified under Texas Local Government Code Section 211.0061 [59, 31, 57], but this study focuses on the pre-HB 24 regime, when the protest petition was a central feature of Austin's land use politics.

This thesis asks whether it is possible to predict, at the time of filing, which housing development proposals in Austin will trigger a measured threshold-crossing protest petition, specifically, whether administrative, spatial, and neighborhood features available before public hearings can identify cases likely to face this form of organized opposition. The focus is on forecasting the statutory petition event, not all forms of opposition or the final legal certification by the clerk.

The outcome of interest is a "measured threshold-crossing petition," defined as an event where the reconstructed petition-share calculation exceeds the statutory 20% threshold based on public records available at filing. This measure is distinct from the clerk-certified petition, which is not directly observed. The thesis does not attempt to predict all forms of opposition or the final legal outcome, but rather assesses whether administrative, spatial, and economic features at the time of application can meaningfully signal the risk of this specific institutional event. The main contribution is to clarify the predictive value and limitations of these features for anticipating organized opposition in a high-stakes land use context.

To supplement the quantitative analysis, the study incorporates semi-structured interviews with planning professionals. These interviews provide institutional context, highlighting informal negotiations, strategic petition timing, and neighborhood mobilization, that administrative data alone cannot capture. While not used to validate model performance, these perspectives help situate the findings within the realities of planning practice and underscore the ethical and governance considerations surrounding predictive tools in land use decision-making.

2 Literature Review

This literature review synthesizes three core themes that motivate the thesis's methodological approach.

First, land-use regulation is widely recognized as a binding constraint on housing supply, driving a wedge between construction costs and market prices [25, 24, 28, 49]. Measures of regulatory stringency such as the Wharton Residential Land Use Regulatory Index document substantial cross-metropolitan variation in local land-use controls [29, 27]. Fischel's "homevoter hypothesis" [20] posits that homeowners mobilize politically to protect property values, often leveraging zoning to preserve neighborhood exclusivity; McCabe [41] extends this logic by linking local politics to household wealth accumulation through homeownership. Empirical work by Einstein, Palmer, and Glick [17] documents "participatory distortion," where older, wealthier, long-tenured homeowners dominate public processes and systematically oppose new housing. Hankinson [32] finds that renters, too, may resist nearby development due to displacement fears, reinforcing the spatially localized nature of opposition. Recent syntheses of NIMBY and YIMBY politics show that these conflicts are embedded in broader coalitions over land use, growth, and neighborhood change rather than reducible to a single homeowner motive [7]. At the same time, supply-side reform arguments coexist with cautions that new supply alone will not fully resolve affordability and displacement pressures in high-cost markets [30, 4].

Second, the power of participatory distortion is magnified in discretionary review systems, where individual projects require site-specific approval rather than proceeding "by right" [39, 33]. Research on housing opposition shows that neighborhood resistance is heterogeneous, often mixing status-quo protection, procedural claims, environmental review, and distributive conflict rather than a single undifferentiated NIMBY logic [48, 53]. In Texas, the protest petition statute empowered adjacent property owners to trigger supermajority council votes, making it a potent institutional veto point [58, 21]. Supermajority requirements systematically favor the status quo [10], amplifying the influence of organized opposition. In Austin, measured threshold-crossing petitions have been especially common for rezonings that increase residential density, highlighting the asymmetric impact of these institutional tools on housing supply.

Third, while much of the urban planning literature focuses on causal inference, estimating the effects of zoning changes or protest petitions on outcomes like land value or project delay, addressing discretionary gridlock also requires anticipating where opposition will arise. Shmueli [55] and Mullainathan and Spiess [44] distinguish between explanatory and predictive modeling, emphasizing that predictive models can inform policy and administrative triage even when causal mechanisms are complex or uncertain. Kleinberg et al. [34] formalize the "predictive policy question," where optimal decisions depend on forecasting rather than explanation. In the context of Austin's protest petition regime, the relevant question is not just why opposition occurs, but whether it can be predicted in advance using only the information available at the time of filing. This predictive framing aligns with emerging best practices in public policy research [1]. It also raises public-sector governance questions about opacity, proxy discrimination, and sociotechnical context that cannot be resolved by predictive performance alone [18, 47, 2, 54].

3 Outcome Definition and As-of Information Setup

The empirical foundation is a time-stamped spatial panel that reconstructs what information was available at each historical prediction point. Every record anchors to an explicit as-of date, ensuring that backtests use only Census releases, appraisal rolls, and case documents published before the prediction date.

3.1 Analytical Panel Structure

The core analytics rely on the filing-date analytic sample, which merges spatial and demographic variables into single case-level observations (with temporally matched covariates):

1. **Zoning Application Records:** Case ID, milestone dates (filing, notice, petition deadline, planning commission hearing, council reading, withdrawal), requested zoning changes, unit limits, target FAR, and disposition status [15].

2. **Spatial Geometries:** Parcel IDs, geometries, acreage, frontage, and 200ft buffers used to aggregate nearby properties (council districts and overlapping geometries) [15].

3. **Parcel Appraisals:** Annually indexed TCAD appraisal variables for parcels used to compute variables within the 200-foot buffer: land value, structure age, homestead exemption rates, and owner-occupancy shares [61].

4. **Neighborhood Demographics:** Tract and block group ACS variables constrained to their historical 5-year release dates: rent burdens, tenure splits, demographic composition, and displacement vulnerability indicators [63].

5. **Institutional Calendar:** Council-term and policy-timing indicators, including the 2022 council turnover, Austin's 2023 HOME Initiative adoption [14], and the HB 24 effective date (September 1, 2025) [57].

These components are assembled from City of Austin open-data records, Travis Central Appraisal District appraisal files, U.S. Census Bureau ACS 5-year estimates, Travis County certified election results, and statutory materials governing the protest-petition regime [15, 61, 63, 62, 58, 57].

Because municipal databases frequently overwrite preliminary timestamps with final disposition dates, milestone dates are reconstructed through two pathways. First, explicit dates are parsed from scraped historical council agendas and PDF case files. Second, where explicit text is unavailable, dates are imputed by back-calculating from end-dates using minimum procedural constraints mandated by Texas Local Government Code and Austin zoning ordinances (for example, statutory mail-notice minima). This parsed-versus-imputed distinction is carried into interpreting the results, as timing uncertainty introduces measurement error.

Supporting tables log meeting events (staff and commission recommendations), speech comments (speaker stance and segmented text), petition records (validity and signature counts), and vote records (case-by-councilmember dynamics).

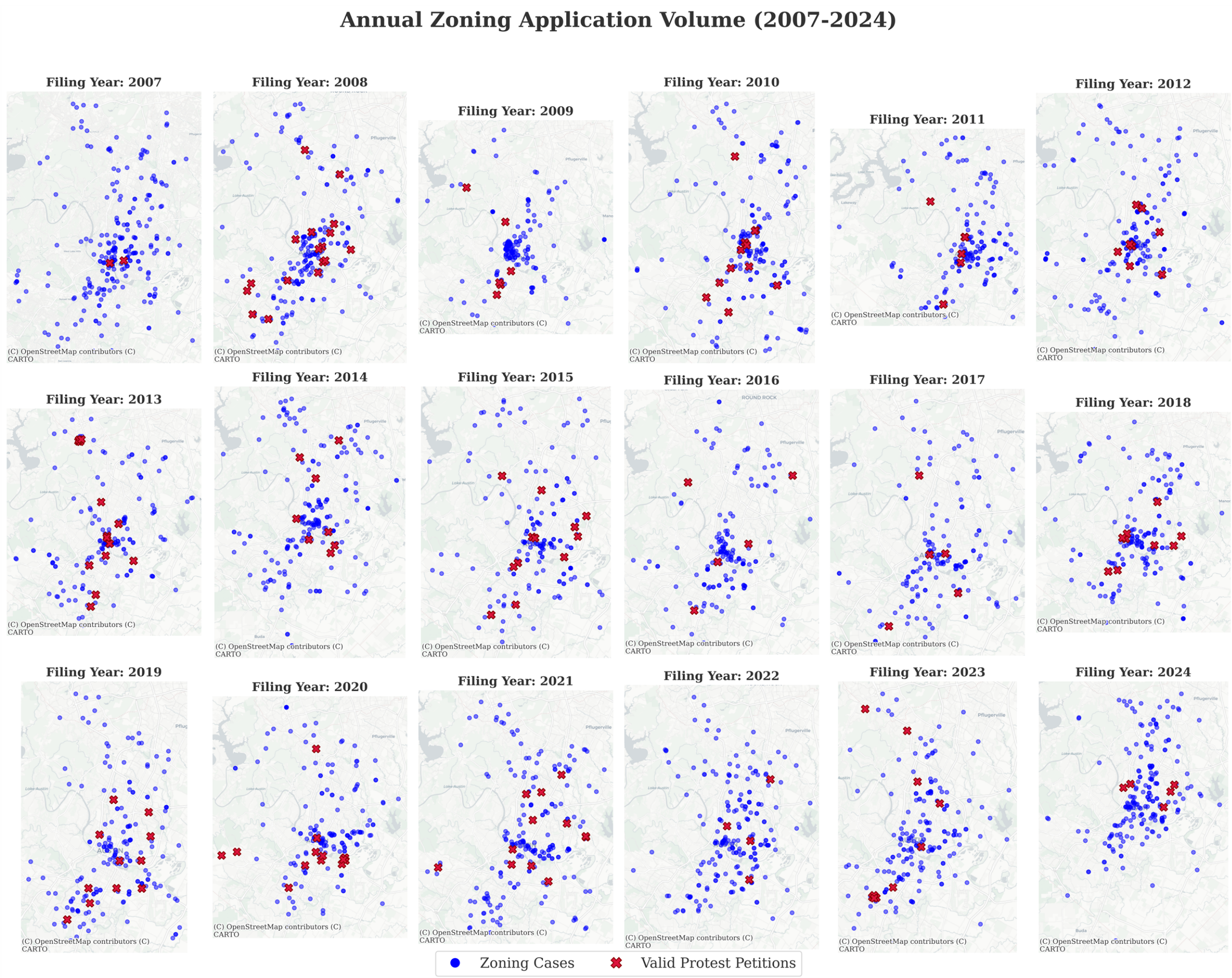


Figure 1: Spatial distribution of zoning cases across the Austin municipal core ($N = 6783$). *Generated: 2026-04-13*

Table 1: Annual Distribution of Discretionary Zoning Cases, 2007 to 2024

Note: Updated: 2026-04-13

Year	Cases	Year	Cases	Year	Cases
2007	288	2013	366	2019	460
2008	270	2014	390	2020	362
2009	265	2015	454	2021	476
2010	297	2016	422	2022	436
2011	369	2017	483	2023	625
2012	335	2018	492	2024	215
Total, 2007 to 2024: 3,967 cases					

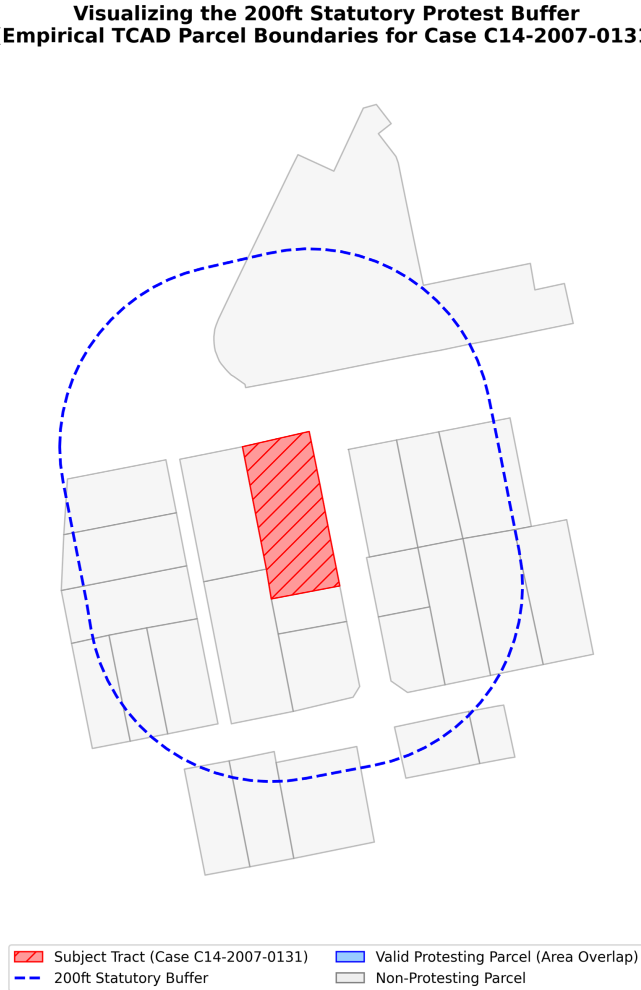


Figure 2: 200ft parcel buffer geometries used for TCAD appraisal aggregation and defining the petition buffer. *Generated: 2026-04-13*

3.2 Case Universe and Sample Selection

The Austin Open Data Portal (data.austintexas.gov), accessed via the Socrata Open Data API (SODA), records all administrative zoning activity, including both substantive rezoning proposals and minor administrative variances, such as 5-foot setback adjustments. Because the petition mechanism under study, formal protest petitions under Texas LGC Chapter 211, does not apply to minor variances, and because extraterritorial jurisdiction (ETJ) properties lack the legal protest-petition mechanism, these cases are excluded.

Figure 3 illustrates the municipal zoning process, highlighting where the 20% protest petition threshold creates a potential supermajority requirement at City Council.

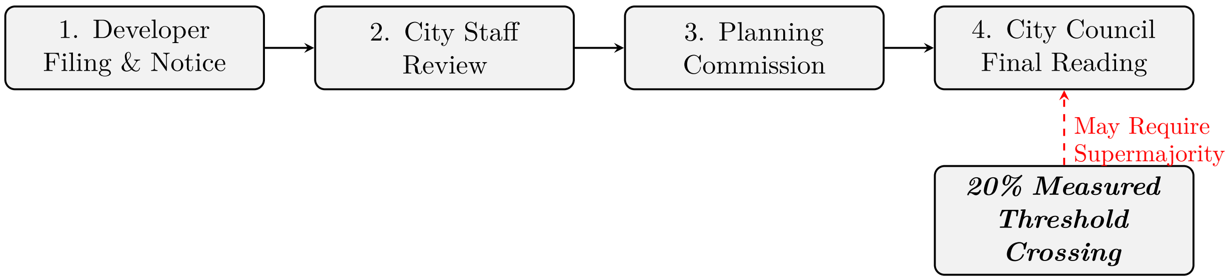


Figure 3: Austin municipal zoning process. The figure schematically shows where a qualifying protest petition could make a three-fourths council vote relevant under the study-period framework. A petition representing 20% of the area of lots immediately adjoining the proposed change would require, under the applicable legal framework, a supermajority vote of the City Council for approval, though recusal or absence of council members can alter the exact number of affirmative votes needed.

Table 2: Historical Panel Descriptive Statistics

Statistic	Gross Site Area (Acres)	Requested Height Delta (ft)	Requested FAR Delta	Requested Bldg Cov Delta (%)	Filing Year	Organized Opposition (Binary)
Count	534	3994	3994	3994	7074	7153
Mean	4.92	10.5	0.43	12.3	2008.6	0.06
Std. Dev	30.66	34.9	0.95	20.7	10.11	0.23
Min	0.00	-60	-3.00	-55	1966	0
Median	0.00	0	0.05	5	2009	0
Max	669.83	365	7.60	80	2026	1

Table 3 reports the sample selection from raw administrative records to the analytic sample.

Table 3: Sample Selection from Raw Records to Analytic Sample

Selection Step	**Cases Remaining**
Raw Austin open data portal zoning records (2007 to 2024)	15,238
Restrict to discretionary cases (rezonings, PUDs, NPAs)	7,153
Exclude cases with unusable or missing initial filing-year information	6783
Final Analytic Sample (Stages C/D)	**6783**

The analytical samples vary by analytical goal. The primary predictive evaluation, the filing-date protest-petition model, uses the full sample of 6783 discretionary geographic cases from the legally relevant pre-HB 24 petition regime (2007 to 2024). Supporting analyses are constrained to institutionally defensible windows rather than broad pooled periods. The electoral-covariate descriptive panel ($N = 518$) is limited to active single-member districts over a restricted 2022 to 2024 calendar window and is reported as appendix-only non-identifying evidence. The HOME appendix analysis uses the same restricted district panel and is presented as non-identifying descriptive event tracking only. The Invariant Causal Prediction (ICP) analysis utilizes $N = 1,186$ multi-parcel geometric environments to represent cohesive spatial clusters. Finally, the institutional geographic overlay appendix aggregates policy flags by council term ($N = 20$ macro-observations) and marks designs with insufficient identifying variation as not estimable.

Unlike complete-case geometric intersection approaches, the filing-date model allows partially observed covariates and uses the algorithm's native handling of missing values to avoid dropping observations. This design choice reduces artificial attrition but does not resolve the underlying missing-data problem: missingness can remain systematically patterned across space, project type, and administrative pathway. Missingness is therefore treated as an evidentiary limitation and reported descriptively rather than framed as solved by estimator choice.

3.3 Primary Outcome Definition

The primary predictive outcome (y_0) is a *measured threshold-crossing petition*: a binary indicator equal to one when the reconstructed petition-share calculation, derived from public petition filings and buffer-area computations, exceeds the 20% statutory threshold defined by Texas Local Government Code Chapter 211 and Austin LDC § 25-2-284 [58]. This is a defined outcome evaluated from public records, not a direct observation of the legally decisive event. The City Clerk's formal certification of each petition as legally valid is not recorded in the public record; consequently, the outcome captures cases that *appear* to have crossed the threshold based on available records, which may include petitions that were ultimately rejected on procedural grounds or petitions whose true area share differed from the calculated value.

Terminology. Throughout this manuscript, "measured threshold-crossing petition" refers to the observed binary study outcome, "reconstructed petition share" refers to the computed share of signed adjoining-lot area derived from petition filings and parcel geometry, and "clerk-certified valid petition (unobserved)" refers to the legal administrative determination by the City Clerk that is not captured in the analytic sample.

Cases withdrawn before public notice are excluded from the sample, rather than coded as zero, because no opportunity for formal public opposition existed. Extraterritorial Jurisdiction (ETJ) cases and routine minor variances are additionally excluded because the statutory petition mechanism does not mathematically apply to them.

3.4 Statutory Context: The Legal Petition Mechanism

Formal protest petitions represent a strict procedural mechanism granting incumbent property owners veto authority over localized zoning changes. As detailed by Furth and McKinley, twenty-one U.S. states actively enforce petition statutes that allow a localized numeric minority of nearby residents to force supermajority voting constraints on municipal legislative bodies [22]. While specific standards vary geographically—for instance, Oklahoma requires a 50% sign-on threshold spanning 300 feet, whereas Iowa utilizes a 20% threshold at 200 feet—the institutional architecture remains remarkably consistent: protest petitions structurally subordinate city-wide housing coordination to the exclusionary friction of hyper-localized abutters.

Texas historically enforced one of the nation's most active 20% protest thresholds, culminating in 2025 legislative reforms that explicitly exempted comprehensive mapping efforts and residential upzonings from the supermajority requirement precisely to curb the immense veto power yielded by neighborhood opposition groups [22]. Figure 4 illustrates a representative historical petition filed for a development rezoning in Austin, documenting the precise computational format through which property owners within a tight 200-foot administrative buffer aggregate formal opposition to legally bind the City Council.

3.5 Feature Libraries: Policy Forecasting vs. Explanatory Research

The feature set is separated to prevent predictive models from leveraging sensitive attributes:

Predictive features include physical site characteristics (lot area, unit count changes, PUD/TOD flags), statutory eligibility flags (HB 24 status, owner-vs.-city-initiated), aggregated buffer economics (homestead shares, land-to-improvement ratios), broad demographic indicators

PETITION

Case Number: **C14-2007-0131** Date: July 29, 2008

1506 WALLER ST, 807 E 16TH ST & 908 E 15TH ST

Total Area Within 200' of Subject Tract 293,002.55

1	02-0906-0502	MONAHAN CASEY J	7423.26	2.53%
2	02-0906-0503	MONAHAN CASEY J	7390.73	2.52%
3	02-0906-0504	CLARK MICHAEL G & RITA S DE BE	7354.85	2.51%
4	02-0906-0506	MARTINSON KATHY L	1645.43	0.56%
5	02-0906-0507	CHILES ROSALIE BENSON	1618.17	0.55%
6	02-0906-0508	KOCHERT KELLEY	1612.79	0.55%
7	02-0906-0509	MONAHAN CASEY J	1419.72	0.48%
8	02-0906-0606	TIMMERMANN TERRELL	3452.33	1.18%
9	02-0906-0607	MERCADO FREDDY G & MARIA R	9482.82	3.24%
10	02-0906-0901	LEGETT GEORGIA F	4459.36	1.52%
11	02-0906-0911	LEGETT GEORGIA F	7856.05	2.68%
12	02-0906-1001	MEDINA JAMES & KRISTINE M GARA	13204.88	4.51%
13	02-0906-1011	RECER DANALYNN	9454.11	3.23%
14	02-0906-1013	BRINSMADE LOUISA C	7634.37	2.61%
15				0.00%
16				0.00%
17				0.00%
18				0.00%
19				0.00%
20				0.00%
21				0.00%
22				0.00%
23				0.00%
24				0.00%
25				0.00%
26				0.00%
27				0.00%
Validated By:			**Total Area of Petitioner:**	**Total %**
	Stacy Meeks		**84,008.88**	**28.67%**

Figure 4: First page summary stack of a 558-page formal protest petition bundle filed for a Planned Development rezoning in Austin. The municipal record details the physical tracking of petitioner land ownership as a percentage of the total valid area within 200 feet of the subject tract to validate the 20% supermajority threshold.

Table 4: Illustrative statutory parameters governing protest petitions across states detailed by Furth and McKinley [22]. While 21 states utilize similar architectures, the threshold mechanism and buffer stringency dictate the severity of the institutional veto.

State	Validation Threshold	Proximity Buffer	Approval Override Requirement
Texas	20% of nearby land	200 feet	3/4ths Legislative Supermajority
Iowa	20% of nearby land	200 feet	3/4ths Legislative Supermajority
Oklahoma	20% and 50% tiers	300 feet	Legislative Supermajority
Arizona	20% of land & 20% of lots	Municipal Discretion	Legislative Supermajority
Massachusetts	Veto Restricted	Municipal Discretion	Simple Majority (Upzonings)
Michigan (County)	20% of nearby land	Municipal Discretion	Electoral Referendum

(median income, renter share), historical petition rates within the council district, and macroeconomic variables (mortgage rates, vacancy rates) [19]. These elements are selected strictly based on their availability in public administrative and appraisal databases prior to the initial filing date, ensuring the algorithm's decisions replicate constraints faced by municipal planning staff in real time.

Explanatory research features are restricted to equity analysis. These include tract-level racial and ethnic composition, segregation indices, and probabilistic surname-based racial classification [56]. These features are strictly partitioned from the predictive model to reduce the risk of encoding disparate impacts, though proxy correlation through permitted features remains a concern (see Section 8). By reserving these variables exclusively for post-hoc fairness evaluations, the model avoids generating predictions based on protected class attributes while maintaining the capability to measure intersectional disparities across geographic subgroups.

3.6 Illustrative Panel Structure

The following table illustrates a representative cross-section of the core analytic panel. Each row constitutes a unique discretionary zoning case, enriched with spatial network indicators (historical petition clustering), demographic variables (ACS), and property tax valuations (TCAD EARS) strictly observable before the initial filing date. These specific columns are highlighted because they ultimately emerge as the primary predictive drivers in the meta-attribution analysis.

```
Case Number  Year Lag Petitions (1km) Median Income   Owner Share   Appraised Value
C14-2015-001 2015 3                   $85,000         62%           $450,000
C14-2015-002 2015 0                   $42,000         21%           $210,000
C14-2015-006 2015 7                   $112,000        85%           $890,000
C14-2016-009 2016 1                   $55,000         34%           $315,000
C14-2016-012 2016 4                   $95,000         71%           $620,000
```

4 Modeling Strategy

4.1 Formal Protest Petition Model

The Formal Protest Petition forecast is the core of the thesis: conditional on a project entering the formal public process, predicting the probability of a measured threshold-crossing petition ($y_0 = 1$):

$$P(y_0 = 1 \mid D = 1, X_{i,t}, Z) \tag{1}$$

At the initial filing date horizon, the predictive feature space ($X_{i,t}$) is restricted to information observable by planners at the time of filing, including site geometry, requested zoning changes, buffer demographics, and historical petition rates. The primary filing-date specification is intentionally unweighted. Precision-recall curves comparing the CatBoost, Random Forest, ElasticNet Logistic, and V-REx models are reported in Figure 5.

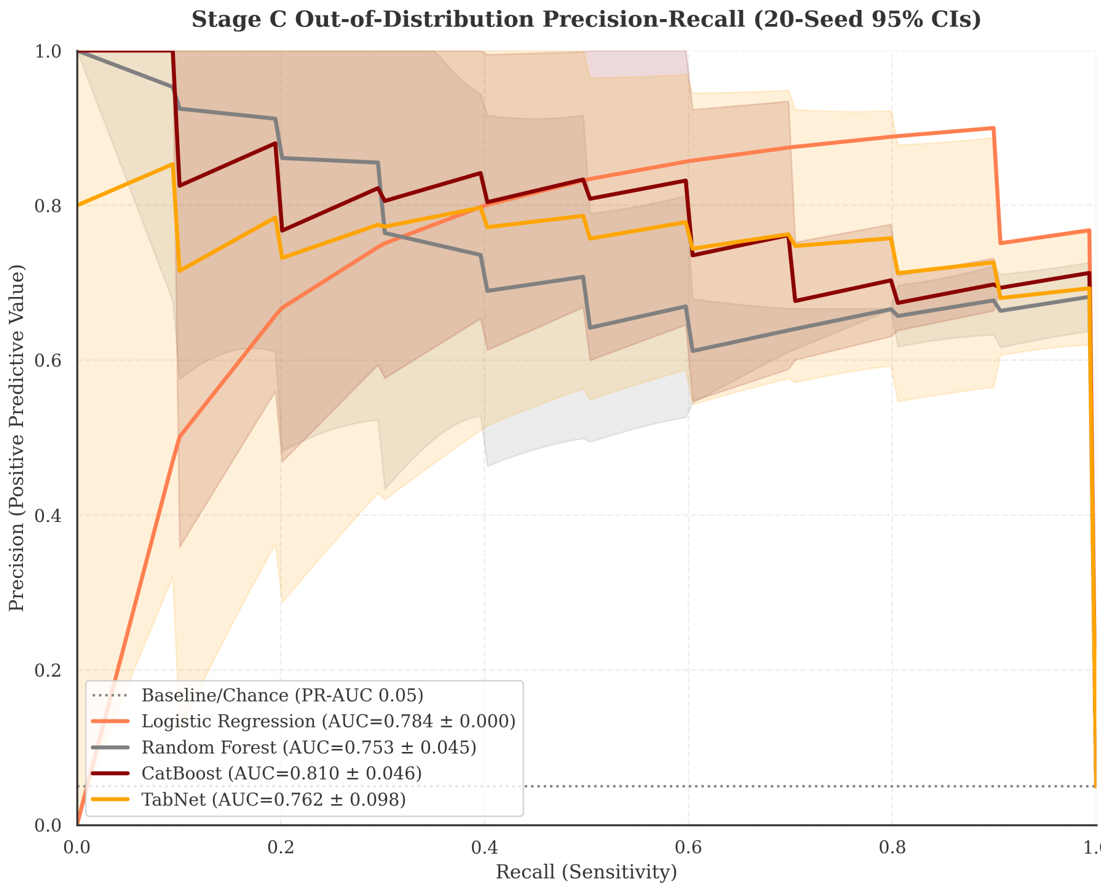


Figure 5: Precision-recall curves for filing-date petition models from the pre-registered benchmark roster.

4.1.1 Primary Model Selection

The primary model chosen for all filing-date analyses is **CatBoost** because the prespecified evaluation criteria prioritizes out-of-sample (temporal holdout) PR-AUC, subject to a constraint on probability reliability (lowest ECE among highly ranked models).

4.2 Model Evaluation Strategy

Precision-Recall AUC (PR-AUC) is the primary discrimination metric, given the substantial class imbalance of the outcome. Following standard methodological practices in rare-event spatial forecasting, traditional Receiver Operating Characteristic (AUROC) metrics are omitted entirely, as the large number of true negatives artificially inflates the curve. PR-AUC explicitly isolates the minority class, forcing the model to balance recall against the precision cost of false alarms.

Primary evaluation. The out-of-distribution bootstrap PR-AUC at the filing-date horizon is the primary evaluation object, as it approximates forward-like administrative use. In-distribution 5-fold cross-validation is retained only as an optimistic upper-bound object. Because this is primarily a ranking exercise, emphasis remains on ranking discrimination (PR-AUC, top-decile lift), with calibration and thresholded diagnostics treated as supporting information rather than coequal headline estimands. Seed-based audits are reported in Appendix C.6 as variance diagnostics.

Metric-object reconciliation. The manuscript reports multiple metric objects for different inferential purposes, and they are not treated as interchangeable: (i) the headline filing-date object is the out-of-sample bootstrap estimate at the filing date, (ii) the 20-seed mean characterizes within-sample stochastic variance, and (iii) in-distribution cross-validation provides an optimistic upper bound. A reconciliation manifest maps manuscript macros to object metadata (task, split, model, calibration state, and source artifact).

Because the baseline opposition frequency is highly skewed, raw predicted probabilities are calibration-sensitive. The evaluation therefore applies out-of-fold Isotonic Regression and reports calibration summaries (ECE, Brier, ACE = 0.007).[1] The Brier score is 0.011.

Strictly proper scoring rules like the Brier Score [6] and Adaptive Calibration (which evaluates equal-mass rather than equal-width bins) are useful complementary calibration measures; standard ECE can be unstable under severe class imbalance because fixed-width probability bins at the highest confidence intervals routinely remain empty when predicting rare events, causing aggregate ECE weights to understate miscalibration for rare positive cases.

The mean predicted probability for true positive cases is 0.480, compared to 0.008 for true negatives. Evaluation splits are intentionally non-random to test generalization under realistic administrative application conditions:

- **Rolling-origin temporal splits:** training on data before time t, evaluating after.
- **Calendar-period holdouts:** partitioning the data to assess robustness across policy and market periods (including pandemic years) without imposing a single structural-break assumption.
- **Spatial holdouts:** withholding entire council districts.

Council composition and policy conditions changed over the sample window, but the temporal analysis does not treat any single election as an exogenous structural break. Performance differences across years are reported descriptively as out-of-sample temporal variation.

Methodological Integrity and Pre-training Leakage. The use of foundation models like TabPFN introduces a risk of "future leakage" if Austin's open data was inadvertently included in the model's original training corpus. To shield against this contamination, the foundation benchmarks were conducted under a strict *Zero-Shot In-Context* enclosure. The internal model weights remained frozen, and the model was only permitted to "observe" the historically bounded pre-2022 training set as a prompt-context before predicting post-2022 outcomes. This protocol ensures that the model's 0.54 PR-AUC success is a result of structural pattern recognition, not historical memorization.

4.3 Administrative Outcome Scope

While predictive evaluation centers on the formal petition filing-date target, an **administrative outcome audit** was reviewed to establish strict scope boundaries. Because downstream disposition

[1] At a classification threshold of $P \geq 0.50$, the model achieves precision of 83.3% and recall of 33.3%. This threshold is included for diagnostics only; ranking-oriented review budgets are the primary decision frame for rare-event triage.

records and council up/down votes are structurally censored by developer withdrawals and informal negotiations, terminal council outcomes are excluded from the thesis's predictive targets. The analysis maintains a strict focus on the formal neighbor mobilization action.

4.4 Feature Attribution and Semantic Clustering

To evaluate the predictive mechanisms driving protest risk, the analysis calculates feature importance using SHAP (SHapley Additive exPlanations) values. Because administrative datasets contain highly correlated proxies (e.g., tax assessor appraised value, market value, and land value), individual raw features suffer from SHAP attribution dilution where importance weights are split arbitrarily among collinear variables.

To prevent collinearity from obscuring substantive relationships and to ensure the archetypal profiles remain interpretable, the raw pipeline variables are grouped into plain-English "Semantic Clusters" prior to attribution visualization. Table 5 crosswalks these overarching semantic concepts back to their exact, raw administrative database columns.

Table 5: Semantic Feature Aggregation Mapping. This table crosswalks the plain-English conceptual buckets used in the manuscript's attribution analysis back to their constituent raw variables from the City of Austin and the completely collinear tax assessor equivalents.

Semantic Cluster	**Raw Data Dictionary Variables**
Demographic Composition	ACS % White, % Hispanic, % Black, % Asian
Housing Tenure	ACS Owner-Occupied Units, Renter-Occupied Units, Total Units
Nbhd. Income & Rent	ACS Median Household Income, Median Gross Rent, Poverty Count, Median Home Value
Property Valuation	Tax Assessor Appraised Value, Market Value, Land Market Value
Structure Age	Tax Assessor Year Built, COA Year Built
Parcel Scale	Tax Assessor Acres, Lot Size (sqft), Planning Gross Area, Deed Acreage
Land Use Classification	Tax Assessor Land Use, COA Land Use Inventory (LUI)
Zoning Density	Tax Assessor Floor-to-Area Ratio (FAR), Units
Improvement Scale	Tax Assessor Improvement Square Footage
Historical Protest Activity	Binary Prior Protest Flag, 1-Year Spatial Contagion, 3-Year Spatial Contagion
Filing Timeline	Target Filing Year

5 Filing-Date Primary Results

5.1 Predictive Performance Across All Horizons

The filing-date model is summarized in prose rather than a standalone headline-metrics table because the key diagnostics are already distributed across the precision-recall exhibit and the seed-stability appendix. The main-text evaluation object remains the temporal holdout for the primary predictive model, with an estimated PR-AUC of 0.718 (95% CI: [0.48, 0.91]), top-decile lift of 9.9x, and calibrated holdout ECE of 0.009.

To prevent cross-design metric drift, calibration claims are bound to the primary reporting layer: calibrated true holdout ECE.

The filing-date petition model therefore reports unweighted discrimination and calibration metrics.

The filing-date petition model uses a trailing count of nearby cases that previously crossed the petition threshold, establishing whether neighboring properties have a recent history of petition activity, as a key feature alongside site geometry, buffer economics, and historical petition rates. Under the primary evaluation, temporal holdout evaluation at the filing horizon yields a PR-AUC of 0.718 (95% CI: [0.48, 0.91]).[2] This level of discrimination is modest in absolute terms but non-trivial given the low base rate (2.6%). On the same split, the 10-bin expected calibration error on isotonic-calibrated scores is ECE = 0.009, small relative to the common 0.05 heuristic, but discrimination remains highly uncertain and positives are sparse.

Given the sample base rate of 2.6% (175 cases), the top-decile lift of 9.9x corresponds to a top-decile precision of 25.4% (15 true positives among 59 cases), separation above chance, but far from deterministic.

Because there are few true positive cases in the top 10% of predictions (15/59), the precision estimate has a wide confidence interval (16.1% to 37.8%). Therefore, operational use should remain ranking-oriented rather than treating any specific prediction score as an absolute threshold for petition occurrence.

The mean predicted probability for true positive cases is 0.480, compared to 0.008 for true negatives.

The model therefore does not read as a dominant calibration failure in the narrow sense of ECE alone; the deployment constraint is joint, wide PR-AUC uncertainty, rare positives, and residual distributional shift, so the system remains bounded to relative spatial ranking and should not be treated as a source of institutionally authoritative case-level probabilities without additional validation.

Feature-restriction survival analysis. The full cross-architecture restricted-vs-full matrix is retained in Appendix A.3 (Table 11). In the main text, this object is treated as a supplementary reliance audit, not a coequal headline performance exhibit.

These results provide one transparency diagnostic for feature reliance. The in-distribution cross-validation benchmark is retained as an optimistic upper-bound object in the appendices rather than as a coequal main-text exhibit.

Temporal Predictive Drift (Rolling Origin). To quantify the model's predictive half-life, Table 6 tracks PR-AUC across expanding annual windows for the benchmark architectures. This is treated as a concept-drift diagnostic rather than a structural-break design [23]. For each anchor year

[2]The 95% CI is wide because the temporal holdout split contains only $n_{\text{pos}} = 15$ positives among $n_{\text{test}} = 583$ cases. The interval is computed via bootstrap resampling of held-out test rows.

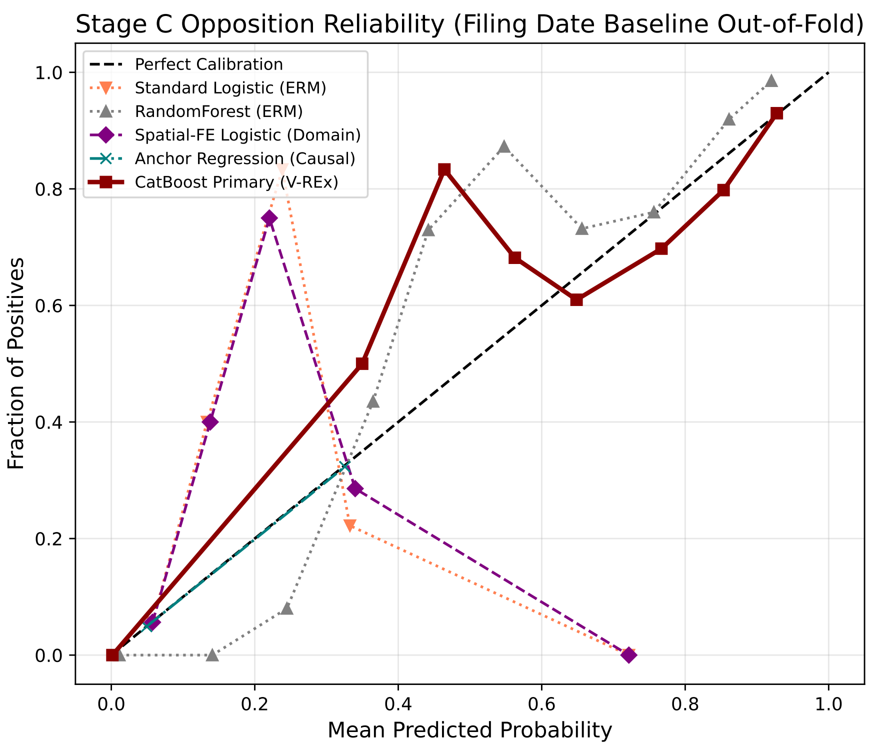


Figure 6: Filing-date reliability diagram for the primary CatBoost model. Main-text reliability reporting is restricted to the primary filing-date model.

t, each model is trained on data before t and evaluated forward across all subsequent evaluation years.

While CatBoost frequently offers strong holdout performance, minority architecture wins are concentrated in a subset of years, indicating temporal heterogeneity rather than stable, permanent cross-architecture superiority.

The central temporal finding is a dip-and-recovery pattern: CatBoost's Pre-2019 anchor decays from 0.659 at immediate horizon (2019, T+0) to 0.329 at the 2022 window (T+3), then rebounds to 0.885 in 2024 (T+5). This is treated as descriptive out-of-sample variation and does not identify a single institutional break mechanism. **The longitudinal attribution audit in Appendix A.2 provides additional descriptive context.**

Because the absolute PR-AUC baseline (the petition occurrence rate) drops substantially in the later years of the evaluation period, absolute PR-AUC mechanically declines even if relative ranking separation remains perfectly stable. To isolate this baseline-shift effect, Table 7 reproduces the temporal drift analysis measuring relative PR-AUC lift (dividing the absolute PR-AUC across the entire threshold curve by the contemporaneous evaluation-year base rate) to measure dynamic model intelligence relative to random guessing.

Normalizing against the collapsing base rate reveals a profoundly counter-intuitive *signal dilution* effect. First, models trained exclusively on the high-conflict Pre-2018 era generate a blistering $34\times$ lift multiplier when predicting the "dead" legislative environment of 2024. Because the spatial recipe for structural opposition was hardened into the high-volume historical data, freezing the legacy model mathematically preserves the structural sorting intelligence, allowing it to isolate the handful of surviving post-ordinance protests with extreme precision relative to random chance. However, iteratively retraining the model on modern data (e.g., Pre-2023 or Pre-2024 anchors) paradoxically *degrades* the 2024 out-of-sample lift by nearly half (dropping to $\sim 19\times$). Exposing the algorithm to the post-ordinance era, where the true-positive density functionally evaporates, starves the model of positive labels and induces catastrophic forgetting. The algorithm stops severely penalizing aggressive spatial configurations because the silenced contemporary institutional environment has stopped explicitly penalizing them.

While aggregate architectural performance appears statistically indistinguishable, grouping architectures by structural capacity reveals strict boundaries around domain shift performance (Table 8). First, linear architectures (Logistic Regression, L2) exhibit extreme *horizon attrition*, demonstrating competitive dominance almost exclusively at immediate-year extrapolations (T+0, T+1) before decaying abruptly at distant forecasting horizons. Second, non-linear tree ensembles exhibit superior resilience to *origin volatility*. When training anchors are forced to bridge the institutional break characterizing the post-ordinance era (Pre-2020 through Pre-2024 anchors), boosted trees dominate 15 out of 16 out-of-sample evaluation cells, confirming that non-linear partition mapping is structurally required to parse highly volatile institutional environments.

5.2 Predictors of Petition Filing

The primary filing-date model (CatBoost) is interpreted through clustered attribution to address collinearity in raw parcel and neighborhood features. Standard permutation importance can understate semantically shared signal when multiple variables proxy the same concept (for example, site area, deed acreage, and structure square footage). Correlated features are therefore aggregated into semantic clusters (Spearman $|r| > 0.70$) and summed relative importance is reported by cluster

Table 6: **Temporal predictive drift: PR-AUC decay by algorithm (Optimal Entropy-Based Discretization).**

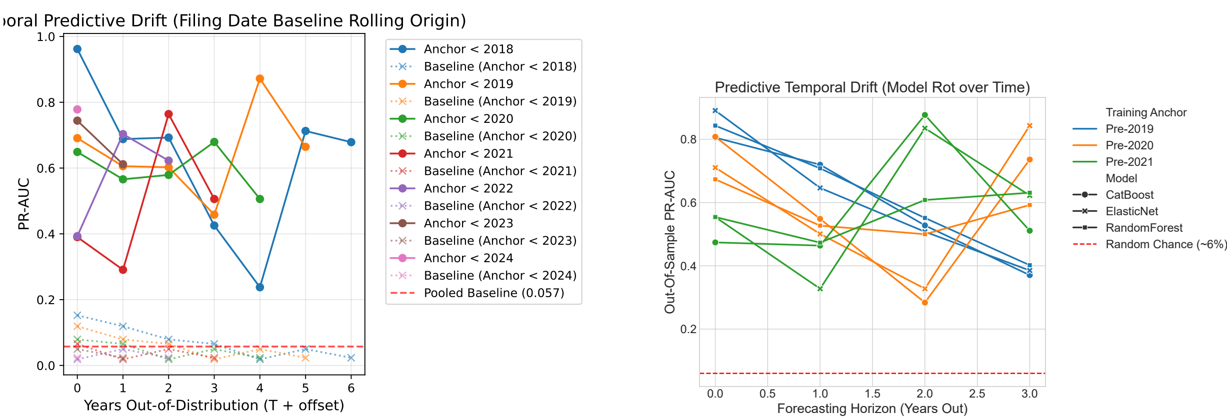
Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Anchor Regression (Causal)	Pre-2018	0.710	0.652	0.723	0.514	0.307	0.480	0.625
Anchor Regression (Causal)	Pre-2019	—	0.787	0.599	0.522	0.348	0.720	0.419
Anchor Regression (Causal)	Pre-2020	—	—	0.665	0.460	0.264	0.667	0.714
Anchor Regression (Causal)	Pre-2021	—	—	—	0.596	0.246	0.687	0.606
Anchor Regression (Causal)	Pre-2022	—	—	—	—	0.364	0.641	0.522
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	0.767	0.471
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	0.517
CatBoost	Pre-2018	**0.976**	0.756	0.635	**0.711**	**0.323**	**0.932**	0.693
CatBoost	Pre-2019	—	0.709	0.556	**0.646**	0.264	**0.959**	0.531
CatBoost	Pre-2020	—	—	0.656	**0.693**	0.271	0.874	0.678
CatBoost	Pre-2021	—	—	—	**0.721**	0.237	0.866	0.730
CatBoost	Pre-2022	—	—	—	—	**0.571**	0.927	0.612
CatBoost	Pre-2023	—	—	—	—	—	**0.870**	**0.631**
CatBoost	Pre-2024	—	—	—	—	—	—	**0.679**
Logistic (L2)	Pre-2018	0.957	**0.968**	0.589	0.519	0.288	0.788	0.432
Logistic (L2)	Pre-2019	—	**0.977**	0.638	0.535	0.279	0.890	0.432
Logistic (L2)	Pre-2020	—	—	0.550	0.477	0.299	0.866	0.447
Logistic (L2)	Pre-2021	—	—	—	0.467	0.300	0.872	0.432
Logistic (L2)	Pre-2022	—	—	—	—	0.340	0.816	0.432
Logistic (L2)	Pre-2023	—	—	—	—	—	0.853	0.405
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.371
Spatial-FE Logistic	Pre-2018	0.971	0.790	0.705	0.479	0.285	0.788	0.441
Spatial-FE Logistic	Pre-2019	—	0.852	**0.750**	0.489	0.280	0.781	0.441
Spatial-FE Logistic	Pre-2020	—	—	0.632	0.563	**0.329**	0.829	0.447
Spatial-FE Logistic	Pre-2021	—	—	—	0.546	0.340	0.832	0.461
Spatial-FE Logistic	Pre-2022	—	—	—	—	0.423	0.829	0.430
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	0.845	0.455
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	0.371
TabNet	Pre-2018	0.961	0.819	**0.934**	0.313	0.238	0.601	0.511
TabNet	Pre-2019	—	0.690	0.508	0.450	**0.602**	0.483	0.548
TabNet	Pre-2020	—	—	**0.730**	0.384	0.280	0.476	0.706
TabNet	Pre-2021	—	—	—	0.369	**0.369**	0.576	**0.808**
TabNet	Pre-2022	—	—	—	—	0.292	0.691	**0.826**
TabNet	Pre-2023	—	—	—	—	—	0.657	0.526
TabNet	Pre-2024	—	—	—	—	—	—	0.302
XGBoost	Pre-2018	0.912	0.849	0.539	0.554	0.225	0.919	**0.754**
XGBoost	Pre-2019	—	0.889	0.674	0.568	0.215	0.885	**0.853**
XGBoost	Pre-2020	—	—	0.594	0.501	0.227	**0.950**	**0.742**
XGBoost	Pre-2021	—	—	—	0.538	0.247	**0.942**	0.698
XGBoost	Pre-2022	—	—	—	—	0.334	**0.954**	0.725
XGBoost	Pre-2023	—	—	—	—	—	0.796	0.478
XGBoost	Pre-2024	—	—	—	—	—	—	0.497

Table 7: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Anchor Regression (Causal)	Pre-2018	4.665	5.480	9.144	7.855	16.568	9.644	26.875
Anchor Regression (Causal)	Pre-2019	—	6.608	7.578	7.977	18.818	14.472	18.014
Anchor Regression (Causal)	Pre-2020	—	—	8.403	7.033	14.279	13.400	30.714
Anchor Regression (Causal)	Pre-2021	—	—	—	9.117	13.307	13.802	26.073
Anchor Regression (Causal)	Pre-2022	—	—	—	—	19.636	12.878	22.456
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	15.410	20.258
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	22.217
CatBoost	Pre-2018	**6.408**	6.352	8.026	**10.876**	**17.429**	**18.732**	29.783
CatBoost	Pre-2019	—	5.954	7.029	**9.872**	14.279	**19.271**	22.838
CatBoost	Pre-2020	—	—	8.298	**10.590**	14.625	17.563	29.168
CatBoost	Pre-2021	—	—	—	**11.026**	12.810	17.398	31.370
CatBoost	Pre-2022	—	—	—	—	**30.857**	18.638	26.302
CatBoost	Pre-2023	—	—	—	—	—	**17.486**	**27.138**
CatBoost	Pre-2024	—	—	—	—	—	—	**29.185**
Logistic (L2)	Pre-2018	6.284	**8.133**	7.453	7.927	15.557	15.832	18.592
Logistic (L2)	Pre-2019	—	**8.205**	8.071	8.183	15.075	17.887	18.592
Logistic (L2)	Pre-2020	—	—	6.949	7.291	16.120	17.406	19.207
Logistic (L2)	Pre-2021	—	—	—	7.141	16.200	17.526	18.592
Logistic (L2)	Pre-2022	—	—	—	—	18.386	16.393	18.592
Logistic (L2)	Pre-2023	—	—	—	—	—	17.145	17.398
Logistic (L2)	Pre-2024	—	—	—	—	—	—	15.934
Spatial-FE Logistic	Pre-2018	6.377	6.639	8.916	7.315	15.404	15.846	18.968
Spatial-FE Logistic	Pre-2019	—	7.155	**9.481**	7.473	15.107	15.695	18.968
Spatial-FE Logistic	Pre-2020	—	—	7.992	8.610	**17.775**	16.669	19.207
Spatial-FE Logistic	Pre-2021	—	—	—	8.349	18.338	16.732	19.821
Spatial-FE Logistic	Pre-2022	—	—	—	—	22.845	16.673	18.490
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	16.976	19.582
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	15.954
TabNet	Pre-2018	6.311	6.883	**11.811**	4.778	12.868	12.072	21.978
TabNet	Pre-2019	—	5.799	6.428	6.873	**32.531**	9.712	23.566
TabNet	Pre-2020	—	—	**9.230**	5.868	15.111	9.566	30.339
TabNet	Pre-2021	—	—	—	5.638	**19.904**	11.573	**34.758**
TabNet	Pre-2022	—	—	—	—	15.750	13.883	**35.526**
TabNet	Pre-2023	—	—	—	—	—	13.211	22.616
TabNet	Pre-2024	—	—	—	—	—	—	12.967
XGBoost	Pre-2018	5.988	7.133	6.817	8.469	12.161	18.480	**32.441**
XGBoost	Pre-2019	—	7.463	8.515	8.685	11.615	17.792	**36.673**
XGBoost	Pre-2020	—	—	7.505	7.656	12.276	**19.089**	**31.892**
XGBoost	Pre-2021	—	—	—	8.221	13.339	**18.925**	30.004
XGBoost	Pre-2022	—	—	—	—	18.032	**19.176**	31.192
XGBoost	Pre-2023	—	—	—	—	—	16.001	20.544
XGBoost	Pre-2024	—	—	—	—	—	—	21.363

Table 8: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	2018	2019	2020	2021	2022	2023	2024
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.976 (Trees)	0.968 (Linear)	0.934 (Deep)	0.711 (Trees)	0.323 (Trees)	0.932 (Trees)	0.754 (Trees)
Pre-2019	—	0.977 (Linear)	0.750 (Linear)	0.646 (Trees)	0.602 (Deep)	0.959 (Trees)	0.853 (Trees)
Pre-2020	—	—	0.730 (Deep)	0.693 (Trees)	0.329 (Linear)	0.950 (Trees)	0.742 (Trees)
Pre-2021	—	—	—	0.721 (Trees)	0.369 (Deep)	0.942 (Trees)	0.808 (Deep)
Pre-2022	—	—	—	—	0.571 (Trees)	0.954 (Trees)	0.826 (Deep)
Pre-2023	—	—	—	—	—	0.870 (Trees)	0.631 (Trees)
Pre-2024	—	—	—	—	—	—	0.679 (Trees)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.41 (Trees)	8.13 (Linear)	11.81 (Deep)	10.88 (Trees)	17.43 (Trees)	18.73 (Trees)	32.44 (Trees)
Pre-2019	—	8.20 (Linear)	9.48 (Linear)	9.87 (Trees)	32.53 (Deep)	19.27 (Trees)	36.67 (Trees)
Pre-2020	—	—	9.23 (Deep)	10.59 (Trees)	17.78 (Linear)	19.09 (Trees)	31.89 (Trees)
Pre-2021	—	—	—	11.03 (Trees)	19.90 (Deep)	18.92 (Trees)	34.76 (Deep)
Pre-2022	—	—	—	—	30.86 (Trees)	19.18 (Trees)	35.53 (Deep)
Pre-2023	—	—	—	—	—	17.49 (Trees)	27.14 (Trees)
Pre-2024	—	—	—	—	—	—	29.19 (Trees)



(a) Anchor-year trajectories across holdout years.

(b) Holdout-year comparison view.

Figure 7: **Temporal drift evidence.** Left panel presents line-based trajectories by anchor year; right panel provides the complementary holdout-year view. Cell-level support and event prevalence used for uncertainty annotation are reported in Appendix C.4, Table 14.

(Figure 8; full feature-to-cluster mapping in Appendix ??).[3] For the primary model, reliance concentrates on parcel scale, property valuation, and neighborhood composition.

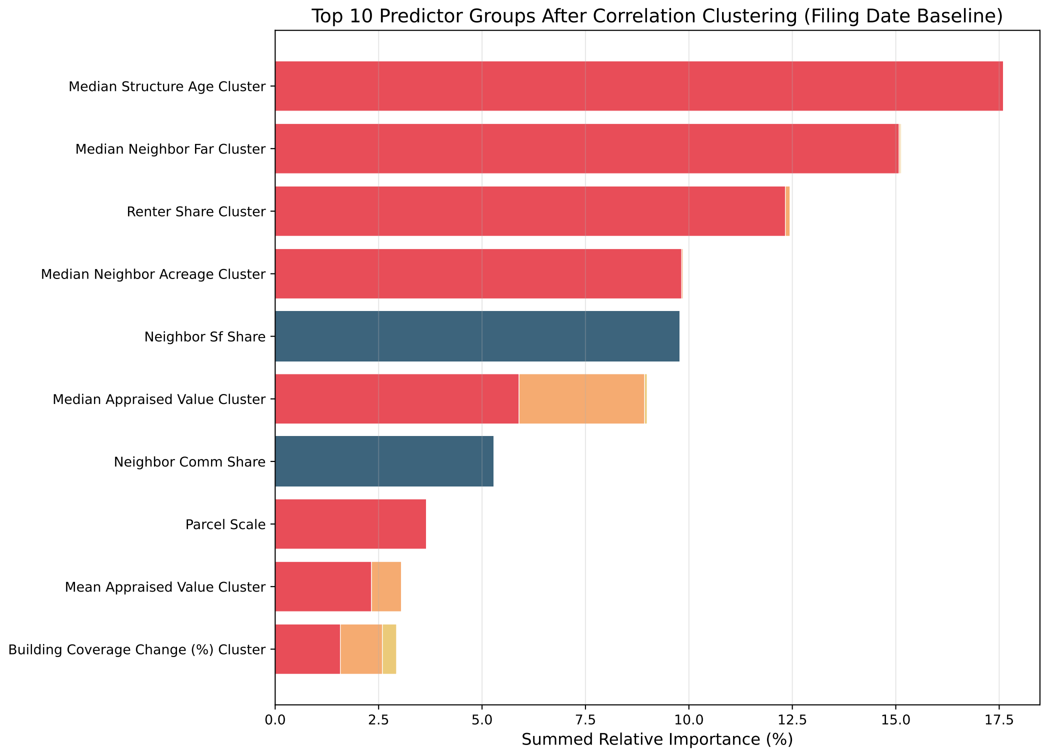


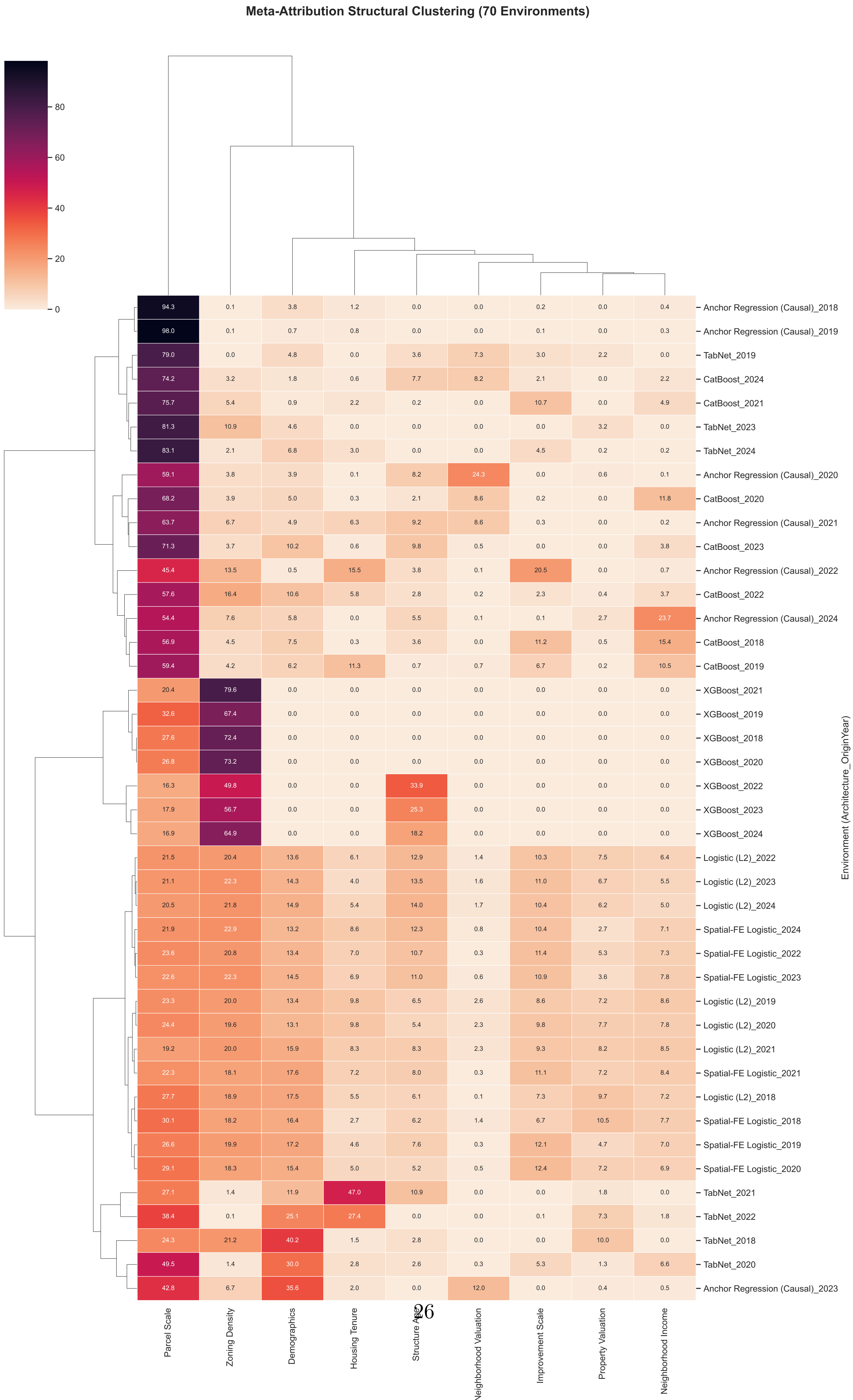
Figure 8: Top 10 predictor groups after hierarchical correlation clustering (Spearman $|r| > 0.70$) for the CatBoost opposition model at the filing-date horizon. Correlated features are merged to correct for collinear dilution.

Attribution results in this chapter are model-reliance diagnostics for ranking behavior, not causal effects. Full case-level SHAP beeswarm decompositions are reported in Appendix A.1 [38, 37].

5.3 Attribution Stability and Meta-Attribution

Meta-attribution is the primary transparency audit in this section. Figure 9 presents a Ward-linkage clustermap across architectures and origin years, showing a consistent structural split: tree-based models and deep models occupy different attribution regimes. The main-text claim is architecture dependence, not a discrete regime-shift claim. Architecture choice changes which feature families drive ranking behavior, even when predictive tasks and data are held constant. The heatmap itself suggests a recurring socio-demographic reliance band rather than a complete absence of overlap: demographics is repeatedly the largest or near-largest cluster, neighborhood income and housing tenure recur as the next tier across most model-year cells, and parcel or property valuation features enter as secondary support. What varies across architectures is the proportional mix, especially the relative emphasis on social-context versus physical-value bundles, not whether any overlap exists at all.

[3] A semantic ablation test confirmed that pre-clustering features into conceptual averages prior to training leads to substantial information loss, particularly regarding parcel-scale drivers. Using post-hoc aggregation of raw TreeSHAP values preserves model-level fidelity while improving interpretability under collinearity [38, 37].



Meta-Attribution Structural Clustering (70 Environments)
80
60
40
20
0
Anchor Regression (Causal)_2018
Anchor Regression (Causal)_2019
TabNet_2019
CatBoost_2024
CatBoost_2021
TabNet_2023
TabNet_2024
Anchor Regression (Causal)_2020
CatBoost_2020
Anchor Regression (Causal)_2021
CatBoost_2023
Anchor Regression (Causal)_2022
CatBoost_2022
Anchor Regression (Causal)_2024
CatBoost_2018
CatBoost_2019
XGBoost_2021
XGBoost_2019
XGBoost_2018
XGBoost_2020
XGBoost_2022
XGBoost_2023
XGBoost_2024
Logistic (L2)_2022
Logistic (L2)_2023
Logistic (L2)_2024
Spatial-FE Logistic_2024
Spatial-FE Logistic_2022
Spatial-FE Logistic_2023
Logistic (L2)_2019
Logistic (L2)_2020
Logistic (L2)_2021
Spatial-FE Logistic_2021
Logistic (L2)_2018
Spatial-FE Logistic_2018
Spatial-FE Logistic_2019
Spatial-FE Logistic_2020
TabNet_2021
TabNet_2022
TabNet_2018
TabNet_2020
Anchor Regression (Causal)_2023
Environment (Architecture_OriginYear)
Parcel Scale
Zoning Density
Demographics
Housing Tenure
Structure Age
Neighborhood Valuation
Improvement Scale
Property Valuation
Neighborhood Income

To explicitly quantify the divergence mapped in Figure 9, Table 9 aggregates the raw clustered reliance weights into archetypal averages by architectural family. Tree ensembles allocate over 80% of their predictive logic exclusively to socio-demographic contexts (Demographics, Housing Tenure, Income), practically ignoring physical parcel geometry. Deep architectures heavily re-weight this attention, slashing demographic reliance and focusing intensely on physical and geographic attributes (Parcel Scale, Property Valuation, and Structure Age).

Crucially, to definitively guard against *continuous feature memorization* (Identity Hacking)—wherein high-capacity models might exploit high-cardinality exact physical floats (such as an isolated $\$1,219,301$ market valuation) as unique property identification hashes—the expanded 10-architecture gauntlet explicitly enforces a rigorous supervised discretization constraint. Rather than relying on arbitrary quantiles (e.g., deciles) or theoretically fragile uniform-width rules (like the Freedman-Diaconis heuristic) which would dangerously preserve isolated single-property bins at the extreme right tail of real estate appraisal data, all physical indicators in Tables 8 and 9 are optimized via Information-Entropy based decision trees. By mathematically generating boundary splits that maximize Shannon Entropy while strictly enforcing a hard volume floor ($min\_samples\_leaf = 100$), the algorithm organically determines the optimal number of bins per feature. Furthermore, because no resulting bin can ever contain fewer than 100 properties, the continuous cryptographic uniqueness of extreme outliers is mathematically shattered, ensuring that all 10 architectures are evaluated strictly on their ability to learn generalized, reproducible spatial concepts rather than discrete memorization boundaries.

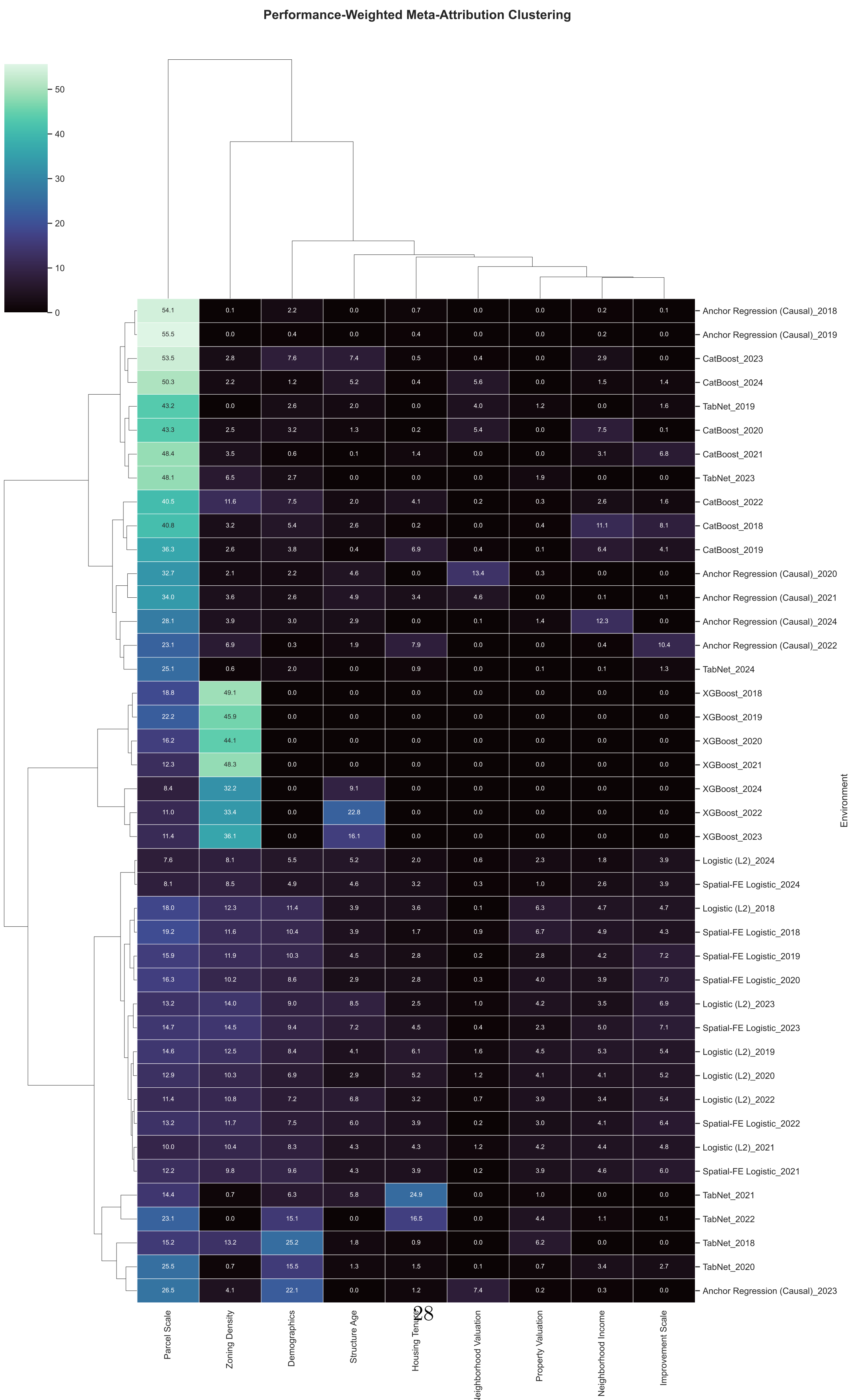
Table 9: **Archetypal Family Attribution (Optimal Entropy-Based Discretization).** Average absolute model reliance allocated to each semantic feature cluster. **Causal** array maps the Non-Linear Anchor Regression. High-cardinality identity floats mathematically blinded via Shannon-Entropy decision tree mapping ($min\_samples\_leaf = 100$).

Semantic Target Cluster	Tree Ensembles	Deep Architectures	Linear Architectures	Causal Algorithms
Parcel Scale	44.4%	54.7%	23.8%	**65.4%**
Zoning Density	**36.1%**	5.3%	20.3%	5.5%
Structure Age	7.5%	2.9%	**9.1%**	3.8%
Neighborhood Income	3.7%	1.2%	**7.2%**	3.7%
Demographics	3.0%	**17.6%**	15.0%	7.9%
Improvement Scale	2.4%	1.8%	**10.1%**	3.0%
Housing Tenure	1.5%	**11.7%**	6.5%	3.7%
Neighborhood Valuation	1.3%	1.1%	1.2%	**6.4%**
Property Valuation	0.1%	3.7%	**6.7%**	0.5%

We extend this analysis by weighting the standard attribution space relative to actual OOS robustness. Table 10 maps the performance-weighted archetypal boundaries, wherein highly performant, robust configurations dominate the family mean, answering exactly what structurally valid models depend on to survive longitudinal drift.

For the primary model, temporal attribution ordering remains sufficiently stable for risk-triage framing. Figure 12 reports adjacent-window Spearman rank correlation at the filing-date horizon, showing sustained rank-order persistence despite year-to-year shifts in attribution share.

Operationally, the attribution sidecar supports a narrower, more qualified conclusion: there is a recurring set of high-weight clusters, especially demographics, neighborhood income, and housing tenure, but their relative shares shift enough across architectures and years that they should be treated as a descriptive reliance pattern rather than a fixed portable explanatory profile. The


Performance-Weighted Meta-Attribution Clustering
50
40
30
20
10
0
54.1 0.1 2.2 0.0 0.7 0.0 0.0 0.2 0.1 Anchor Regression (Causal)_2018
55.5 0.0 0.4 0.0 0.4 0.0 0.0 0.2 0.0 Anchor Regression (Causal)_2019
53.5 2.8 7.6 7.4 0.5 0.4 0.0 2.9 0.0 CatBoost_2023
50.3 2.2 1.2 5.2 0.4 5.6 0.0 1.5 1.4 CatBoost_2024
43.2 0.0 2.6 2.0 0.0 4.0 1.2 0.0 1.6 TabNet_2019
43.3 2.5 3.2 1.3 0.2 5.4 0.0 7.5 0.1 CatBoost_2020
48.4 3.5 0.6 0.1 1.4 0.0 0.0 3.1 6.8 CatBoost_2021
48.1 6.5 2.7 0.0 0.0 0.0 1.9 0.0 0.0 TabNet_2023
40.5 11.6 7.5 2.0 4.1 0.2 0.3 2.6 1.6 CatBoost_2022
40.8 3.2 5.4 2.6 0.2 0.0 0.4 11.1 8.1 CatBoost_2018
36.3 2.6 3.8 0.4 6.9 0.4 0.1 6.4 4.1 CatBoost_2019
32.7 2.1 2.2 4.6 0.0 13.4 0.3 0.0 0.0 Anchor Regression (Causal)_2020
34.0 3.6 2.6 4.9 3.4 4.6 0.0 0.1 0.1 Anchor Regression (Causal)_2021
28.1 3.9 3.0 2.9 0.0 0.1 1.4 12.3 0.0 Anchor Regression (Causal)_2024
23.1 6.9 0.3 1.9 7.9 0.0 0.0 0.4 10.4 Anchor Regression (Causal)_2022
25.1 0.6 2.0 0.0 0.9 0.0 0.1 0.1 1.3 TabNet_2024
18.8 49.1 0.0 0.0 0.0 0.0 0.0 0.0 0.0 XGBoost_2018
22.2 45.9 0.0 0.0 0.0 0.0 0.0 0.0 0.0 XGBoost_2019
16.2 44.1 0.0 0.0 0.0 0.0 0.0 0.0 0.0 XGBoost_2020
12.3 48.3 0.0 0.0 0.0 0.0 0.0 0.0 0.0 XGBoost_2021
8.4 32.2 0.0 9.1 0.0 0.0 0.0 0.0 0.0 XGBoost_2024
11.0 33.4 0.0 22.8 0.0 0.0 0.0 0.0 0.0 XGBoost_2022
11.4 36.1 0.0 16.1 0.0 0.0 0.0 0.0 0.0 XGBoost_2023
7.6 8.1 5.5 5.2 2.0 0.6 2.3 1.8 3.9 Logistic (L2)_2024
8.1 8.5 4.9 4.6 3.2 0.3 1.0 2.6 3.9 Spatial-FE Logistic_2024
18.0 12.3 11.4 3.9 3.6 0.1 6.3 4.7 4.7 Logistic (L2)_2018
19.2 11.6 10.4 3.9 1.7 0.9 6.7 4.9 4.3 Spatial-FE Logistic_2018
15.9 11.9 10.3 4.5 2.8 0.2 2.8 4.2 7.2 Spatial-FE Logistic_2019
16.3 10.2 8.6 2.9 2.8 0.3 4.0 3.9 7.0 Spatial-FE Logistic_2020
13.2 14.0 9.0 8.5 2.5 1.0 4.2 3.5 6.9 Logistic (L2)_2023
14.7 14.5 9.4 7.2 4.5 0.4 2.3 5.0 7.1 Spatial-FE Logistic_2023
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23.1 0.0 15.1 0.0 16.5 0.0 4.4 1.1 0.1 TabNet_2022
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25.5 0.7 15.5 1.3 1.5 0.1 0.7 3.4 2.7 TabNet_2020
26.5 4.1 22.1 0.0 1.2 7.4 0.2 0.3 0.0 Anchor Regression (Causal)_2023
Environment
Parcel Scale
Zoning Density
Demographics
Structure Age
Housing Ten
eighborhood Valuation
Property Valuation
Neighborhood Income
Improvement Scale

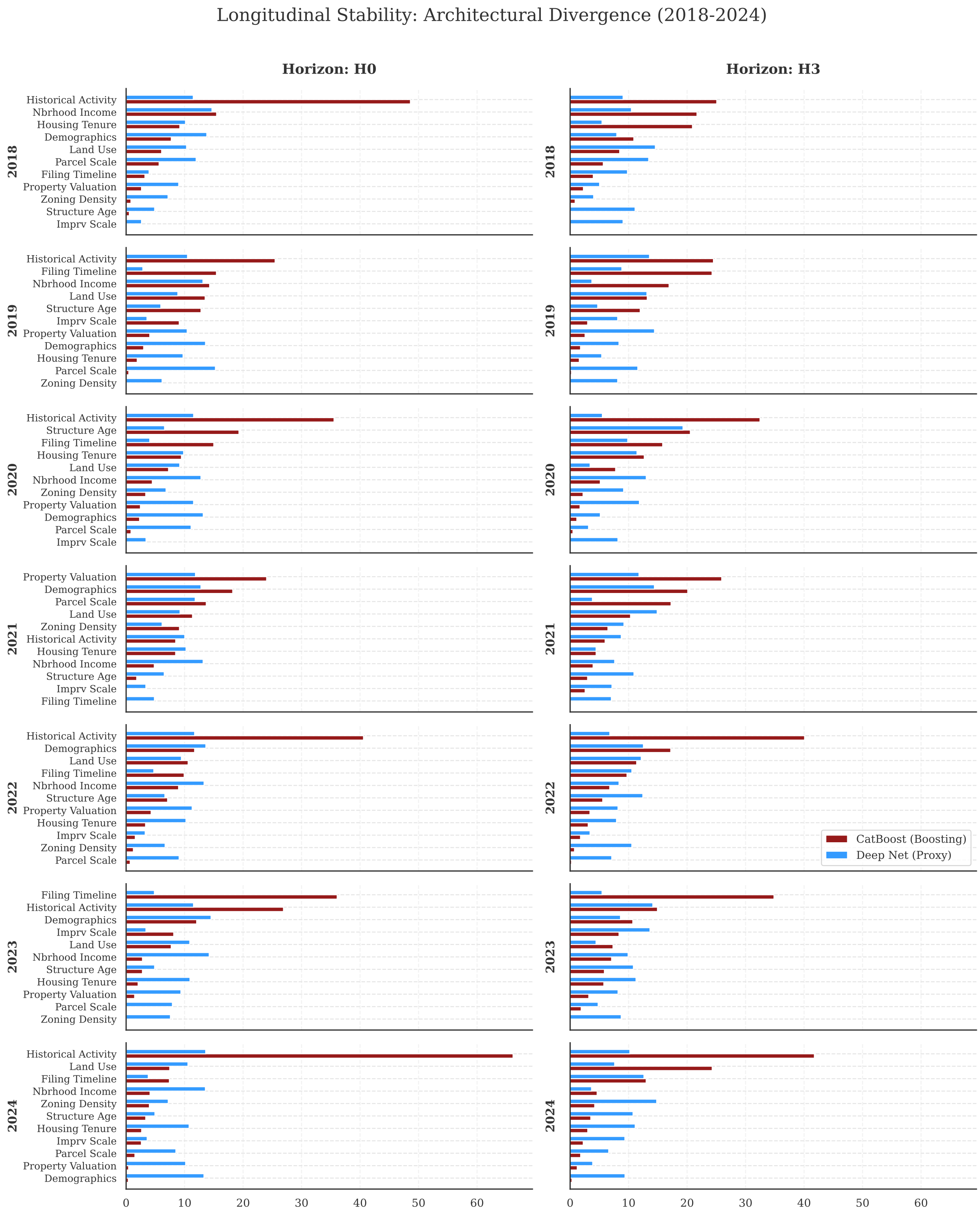

Figure 11: Attribution Stability Test: Architectural Divergence Across Time and Development Horizons. Year-by-year tracking from 2018 to 2024 of architectural feature divergence between tree-based and neural models at both the filing-date and pre-council horizons. Distinct tree-model and deep-model blocks indicate persistent architecture-dependent reliance regimes across evaluation years rather than a single shared attribution profile.

Table 10: **Performance-Weighted Archetypal Family Attribution.** Model attribution vectors scaled by their corresponding out-of-distribution longitudinal PR-AUC performance. Architectures that successfully generalized under domain drift dominate their archetypal class weights.

Semantic Target Cluster	**Tree Ensembles**	**Deep Architectures**	**Linear Architectures**	**Causal Algorithms**
Parcel Scale	45.4%	52.4%	24.1%	**65.6%**
Zoning Density	**34.8%**	5.9%	20.2%	5.4%
Structure Age	7.4%	2.9%	**8.9%**	3.7%
Neighborhood Income	3.9%	1.2%	**7.3%**	3.5%
Demographics	3.2%	**18.7%**	15.1%	8.5%
Improvement Scale	2.4%	1.6%	**10.0%**	2.8%
Housing Tenure	1.5%	**12.0%**	6.4%	3.5%
Neighborhood Valuation	1.3%	1.1%	1.1%	**6.6%**
Property Valuation	0.1%	4.2%	**6.9%**	0.5%

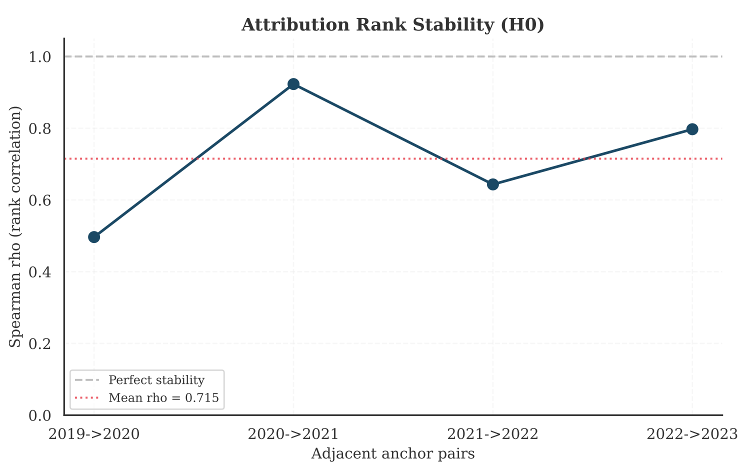

Figure 12: Compact stability summary for the primary filing-date model: adjacent-window Spearman rank correlation of attribution ordering across expanding windows.

practical use of the attribution sidecar is therefore audit and monitoring, not a claim that any named bundle travels unchanged across model classes. Full SHAP beeswarm exhibits, unclustered year-by-year dynamic attribution heatmaps, and multi-horizon decompositions are reported in Appendix A.1 and Appendix A.2.

6 Institutional Context

This chapter presents bounded institutional evidence that contextualizes the filing-date petition results. None of the analyses here constitute formal causal identification of the petition mechanism. The reduced-form threshold estimate characterizes the association between measured threshold-crossing and council processing time; electoral-covariate diagnostics are descriptive appendix checks only. Additional overlay, HOME-tracking, and ICP diagnostics are reported in the appendices as non-headline transparency checks.

6.1 Threshold-Based Reduced-Form Discontinuity on Reconstructed Petition Share

To estimate the threshold-associated shift in processing timelines, the analysis applies a threshold-based reduced-form discontinuity on reconstructed petition share at the 20% statutory margin under Texas LGC Chapter 211 [58]. The running variable is the reconstructed percentage of signed area within the 200-foot buffer [36].

A triangular-kernel weighted least squares estimator indicates a positive discontinuity in council processing time at the measured threshold. Full numeric estimates are reported in registry-linked tables; this analysis is treated as supporting institutional context rather than a coequal headline estimand.

This estimate should be interpreted as a reduced-form effect at the threshold. Because the data do not record whether the City Clerk formally certified each petition as legally compliant, the analysis relies on an unverified calculated area share. Rather than assuming the exact threshold acts as a deterministic sharp trigger, the specification estimates the shift in average processing time associated with crossing the statutory 20% margin. Consequently, this is a threshold-based reduced-form model where institutional compliance remains unobserved.

The identification schematic is reported in Appendix C.2 (Figure 18); the main text retains only the estimated discontinuity exhibit.

Design limitations include potential manipulation of the running variable, since petition organizers may strategically target just above 20%, and limited local support near the cutoff in some calendar slices. Robustness checks, a McCrary (2008) density test for running-variable manipulation [42] ($p > 0.05$), symmetric donut exclusions, and covariate continuity checks, support the validity of the design at the threshold.

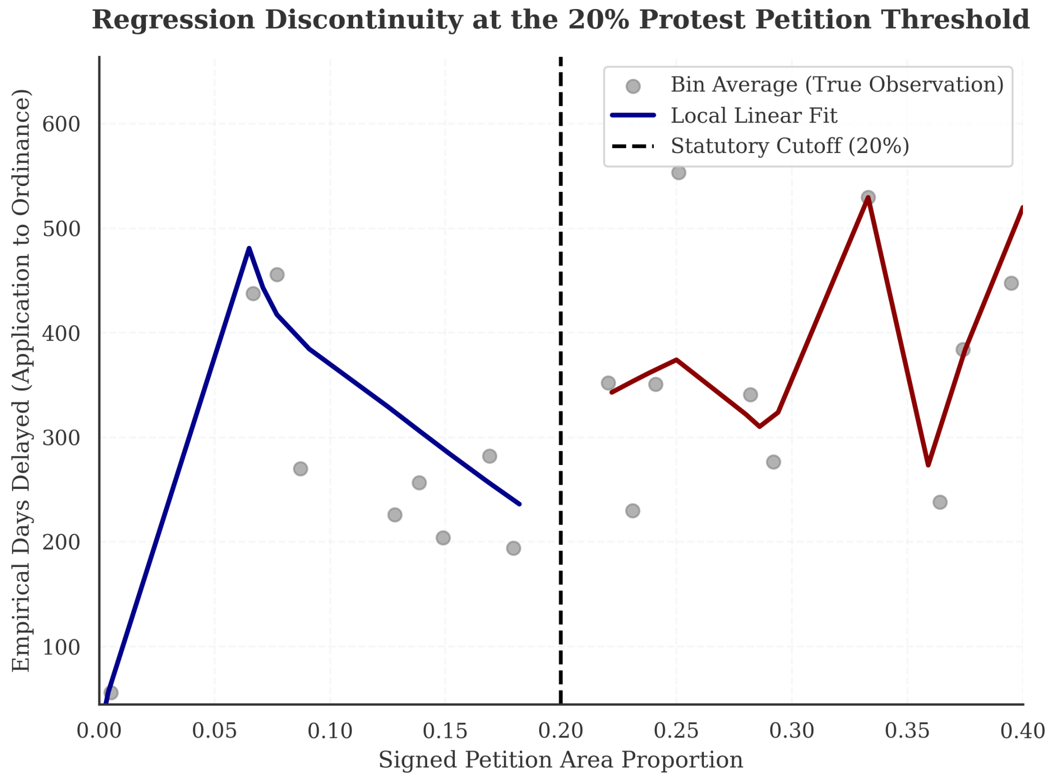

Figure 13: Threshold-based reduced-form discontinuity at the 20% reconstructed petition-share threshold. The figure reports the estimated shift in council processing time. Bin widths were selected using the standard IMSE-optimal (Integrated Mean Squared Error) procedure [9].

7 Qualitative Planning Context

The quantitative results and interview interpretations are reported as distinct forms of evidence. Quantitative models establish empirical claims about predictive discrimination, calibration, and temporal performance variation; interviews provide institutional context for how those patterns are experienced in practice. Semi-structured interviews with urban planning professionals and city staff (conducted under Columbia University IRB Protocol ACYY0820) surface dynamics that administrative data cannot capture. The interview sample is purposive, small-N, and not statistically representative of all Austin stakeholders. Interview material is therefore used for interpretive contextualization only and is not used to estimate effect sizes, test hypotheses, or justify policy deployment decisions.

The quantitative findings that the interviews are intended to contextualize are straightforward:

- Temporal performance varies across holdout years, with a 2021 to 2022 dip and later recovery in filing-date discrimination.
- Petition-risk predictions retain ranking utility, but wide PR-AUC uncertainty and sparse positives still constrain formal operational use of case-level probabilities despite a numerically small ECE on the evaluated score column (ECE $= 0.009$).
- Exploratory electoral-covariate diagnostics are non-identifying, appendix-only checks and are not used to support causal or structural-break claims.

Interview-based interpretation of institutional context follows:

- **Perceived decline in petition leverage over time:** Interviewees described reduced practical leverage of petitions across the late study period and emphasized broader institutional evolution rather than a single-date break. Annick Beaudet (former zoning planner) noted that petitions appeared to lose traction during the Land Development Code revision period: "While people still scream and yell, they don't get as much traction... Necessity is the mother of invention" [3, 13]. These perceptions are qualitative context and are not used as identifying evidence in the quantitative design.

- **The Function of Opposition:** Highlighting the nuanced motivations driving public testimony, AURA board member Felicity Maxwell observed: "We're not talking about suppressing opposition, but understanding it better to have more productive conversations" [40].

- **Asymmetric Transparency and Silent Supporters:** Jonathan Tomko (Principal Planner) emphasized that the municipal public comment process lacks any mechanism for *anonymous* public testimony against a zoning case [60]. Because neighborhood social dynamics often penalize residents who deviate from their neighborhood association's preservationist stance, Tomko noted that the formal opposition data is not strictly representative of the entire community. This aligns with Beaudet's observation that "haters are overrepresented and silent supporters are underrepresented" [3], which likely makes the observed opposition data an incomplete measure of community sentiment.

- **Two distinct motivations and geography:** Marla Torrado (Displacement Prevention Division) stressed that opposition mechanisms may be driven by differing underlying concerns [16]. While the "vocal opposition" on the affluent West side leverage environmental overlays, East Austin neighborhoods frequently mobilize opposition mechanisms in response to rapid gentrification pressures. This context reinforces why rigid probability thresholds are ethically hazardous even when scalar calibration metrics look acceptable on paper: using rigid probability cutoffs could miss communities facing real displacement risk, failing "to identify where people that are in actual need actually are" because the algorithm cannot separate affluent NIMBYism from neighborhood preservation efforts.

These interview interpretations contextualize, rather than establish, the quantitative deployment constraint: independently of interview evidence, the model is treated as a relative ranking heuristic because discrimination uncertainty and deployment ethics remain limiting even when the evaluated ECE is numerically small.

8 Limitations

This study has several important limitations.

Measurement error in reconstructed dates. Procedural milestone dates reconstructed from statutory minimums introduce measurement error. Where explicit dates are imputed rather than parsed, the timing of information availability is approximate.

Selection into observed cases. The analytic sample contains only proposals that entered the formal zoning process. Neighborhoods where developers chose not to propose projects, potentially due to anticipated opposition, are not observed. Attempted development-occurrence inverse-probability weighting fails overlap and balance diagnostics and is therefore retained only as a failed sensitivity check; unobserved selection remains a concern under the unweighted primary specification.

Limited post-policy windows. HB 24 falls entirely outside the observation period. The event-study results should be interpreted as preliminary.

Calibration error context. On the primary holdout evaluation, the 10-bin ECE is 0.009. Institutional caution remains appropriate because rare-event discrimination uncertainty and external validity limits are jointly binding, not because the ECE alone exceeds a fixed administrative tolerance in this evaluation. Regardless, the features generate spatial discrimination (top-decile lift of 9.9x).

Proxy concerns and macro-level aggregation. This study excludes individual-level demographic profiling (for example, BISG) from the predictive matrix. Because case-level predictions use aggregate neighborhood covariates, the model cannot resolve fine-grained behavioral heterogeneity within the 200-foot petition geography. Proxy risk therefore remains an unresolved limitation rather than a solved design feature [2, 5, 46].

Geographic error distribution. A supplementary map of false positives versus actual filings is reported in Appendix C.5 (Figure 24) for transparency context only. Neither geographic symmetry nor FNR gaps constitutes a formal fairness guarantee. The more substantive ethical limitation is that the model cannot distinguish affluent exclusionary mobilization from anti-displacement resistance (see Section 7).

External validity. Austin's institutional structure, council-manager government [12], Texas protest petition statutes, and specific land development code limit direct generalization. The methodological approach, including time-stamped features, sequential evaluation, and calibration benchmarks, is portable, but the specific results are Austin-specific.

9 Conclusion

Filing-date administrative features can rank petition risk in Austin's pre-HB 24 zoning regime, but wide discrimination uncertainty and sparse positives still limit institutionally portable use of case-level probabilities beyond ranking workflows.

This thesis asks whether measured threshold-crossing petitions against housing development proposals can be predicted before formal public hearings. The answer is qualified: filing-date administrative features carry real predictive signal for petition risk—the primary model achieves a PR-AUC of 0.718 (95% CI: [0.48, 0.91]) on a temporal holdout—but that signal is not strong enough to support case-level probability use for high-stakes institutional decisions. The constraint is not primarily a calibration failure in the narrow sense: the ECE is numerically small (ECE = 0.009). Rather, the deployment constraint is *joint*: the PR-AUC confidence interval is wide because the holdout test split contains very few positives, and external validity is limited to Austin's institutional context. Together, these conditions mean that even a well-calibrated probability score cannot be treated as institutionally authoritative for individual cases. Cross-validation provides an optimistic upper-bound PR-AUC of 0.820, and a regression discontinuity at the 20% petition threshold estimates a 42.5-day increase in council processing time, consistent with the institutional significance of the outcome measure. Finally, the model's reliance on socio-demographic clusters—especially neighborhood income, demographics, and housing tenure—demonstrates that predictive models of petition risk invariably embed complex class and race dynamics. However, because the relative feature importance of these clusters shifts unpredictably across different algorithms and across time, the attribution profiles are not stable enough to extract durable urban planning theories. The predictive combinations that identify opposition today are largely mathematical artifacts of the specific training window, and should not be misinterpreted as permanent structural laws of neighborhood behavior.

These results point to a ceiling on what prediction from public records alone can achieve.

Petition filing depends on factors not observable at the time of filing, neighborhood social networks, informal political dynamics, and media attention, and by time-varying institutional context that is only partially observed in administrative fields. Interview interpretation further contextualizes an ethical limitation: the model cannot distinguish affluent exclusionary mobilization from community-driven anti-displacement resistance, a distinction that administrative data does not encode. Any institutional deployment would therefore require documentation and governance artifacts beyond conventional accuracy reporting, including model cards, internal audit procedures, and explicit risk-management protocols [43, 50, 45].

Future work should extend the observation window to capture HB 24 effects [57], explore richer text features from pre-hearing public comments, and isolate mechanisms behind temporary covariate shifts over time.

References

[1] Susan Athey and Guido W. Imbens. Machine learning methods that economists should know about. *Annual Review of Economics*, 11:685–725, 2019. doi:10.1146/annurev-economics-080217-053433.

[2] Solon Barocas and Andrew D. Selbst. Big data's disparate impact. *California Law Review*, 104(3):671–732, 2016. doi:10.15779/Z38BG31.

[3] Annick Beaudet. Interview with annick beaudet, APA-certified urban planner, 2025. Semi-structured interview conducted by Daniel Hardesty Lewis, transcribed via Read.ai.

[4] Vicki Been, Ingrid Gould Ellen, and Katherine O'Regan. Supply skepticism: Housing supply and affordability. *Housing Policy Debate*, 29(1):25–40, 2019. doi:10.1080/10511482.2018.1476899.

[5] Ruha Benjamin. *Race After Technology: Abolitionist Tools for the New Jim Code*. Polity Press, 2019.

[6] Glenn W Brier. Verification of forecasts expressed in terms of probability. *Monthly weather review*, 78(1):1–3, 1950.

[7] Nadia R. Brouwer and Jessica Trounstine. NIMBYs, YIMBYs, and the politics of land use in American cities. *Annual Review of Political Science*, 27:165–184, 2024. doi:10.1146/annurev-polisci-041322-044444.

[8] Brantly Callaway and Pedro H.C. Sant'Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021. doi:10.1016/j.jeconom.2020.12.001.

[9] Sebastian Calonico, Matias D. Cattaneo, and Rocío Titiunik. Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica*, 82(6):2295–2326, 2014. doi:10.3982/ECTA11757.

[10] Stephanie Riegg Cellini, Fernando Ferreira, and Jesse Rothstein. The value of school facility investments: Evidence from a dynamic regression discontinuity design. *The Quarterly Journal of Economics*, 125(1):215–261, 2010. doi:10.1162/qjec.2010.125.1.215.

[11] Carlos Cinelli and Chad Hazlett. Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 82(1):39–67, 2020. doi:10.1111/rssb.12347.

[12] City of Austin. City of austin charter. Technical report, City of Austin, Austin, TX, 2010. URL: https://www.austintexas.gov/department/city-charter.

[13] City of Austin. CodeNEXT draft 3. Technical report, City of Austin, 2017. URL: https://www.austintexas.gov/codenext.

[14] City of Austin. Resolution no. 20230720-126: Initiating the HOME (Home Options for Mobility and Equity) initiative. Austin City Council, July 2023. Sponsored by Council Member Leslie Pool. Adopted July 20, 2023. URL: https://www.austintexas.gov/department/home-initiative.

[15] City of Austin. Zoning cases and land use inventory datasets. City of Austin Open Data Portal, 2024. URL: `https://data.austintexas.gov`.

[16] City of Austin Displacement Prevention Division. Interview with marla t., city of Austin displacement prevention division, 2025. Semi-structured interview conducted by Daniel Hardesty Lewis, transcribed via Read.ai.

[17] Katherine Levine Einstein, David M. Glick, and Maxwell Palmer. *Neighborhood Defenders: Participatory Politics and America's Housing Crisis*. Cambridge University Press, 2019. `doi:10.1017/9781108769495`.

[18] Virginia Eubanks. *Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor*. St. Martin's Press, 2018.

[19] Federal Reserve Bank of St. Louis. Federal funds effective rate (FEDFUNDS) and 30-year fixed rate mortgage average (MORTGAGE30US). FRED Economic Data, 2024. URL: `https://fred.stlouisfed.org`.

[20] William A. Fischel. *The Homevoter Hypothesis: How Home Values Influence Local Government Taxation, School Finance, and Land-Use Policies*. Harvard University Press, 2001. `doi:10.4159/9780674036901`.

[21] Salim Furth and C. Whit Ewen. Mostly invisible: The cost of valid petitions in Texas. Policy brief, Mercatus Center at George Mason University, February 2023. URL: `https://www.mercatus.org/research/policy-briefs/mostly-invisible-cost-valid-petitions-texas`.

[22] Salim Furth and Kelcie McKinley. Rezoning protest petitions are ripe for reform. Policy brief, Mercatus Center at George Mason University, August 2025. URL: `https://www.mercatus.org/housing/policy-briefs/rezoning-protest-petitions-are-ripe-reform`.

[23] João Gama, Indrė Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia. A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4):1–37, 2014. `doi:10.1145/2523813`.

[24] Edward L. Glaeser and Joseph Gyourko. The impact of building restrictions on housing affordability. *Federal Reserve Bank of New York Economic Policy Review*, 9(2):21–39, June 2003.

[25] Edward L. Glaeser, Joseph Gyourko, and Raven Saks. Why is Manhattan so expensive? Regulation and the rise in housing prices. *Journal of Law and Economics*, 48(2):331–369, 2005. `doi:10.1086/429979`.

[26] Andrew Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277, 2021. `doi:10.1016/j.jeconom.2021.03.014`.

[27] Joseph Gyourko, Jonathan Hartley, and Jacob Krimmel. The local residential land use regulatory environment across US housing markets: Evidence from a new Wharton index. *Journal of Urban Economics*, 124:103337, 2021. `doi:10.1016/j.jue.2021.103337`.

[28] Joseph Gyourko and Raven Molloy. Regulation and housing supply. In *Handbook of Regional and Urban Economics*, volume 5, pages 1289–1337. Elsevier, 2015. `doi:10.1016/B978-0-444-59531-7.00019-3`.

[29] Joseph Gyourko, Albert Saiz, and Anita Summers. A new measure of the local regulatory environment for housing markets: The Wharton residential land use regulatory index. *Urban Studies*, 45(3):693–729, 2008. doi:10.1177/0042098007087341.

[30] Emily Hamilton and Salim Furth. Housing reform in the states: A menu of options. Policy brief, Mercatus Center at George Mason University, 2023. URL: https://www.mercatus.org/research/policy-briefs/housing-reform-states.

[31] Emily Hamilton and Salim Furth. Texas protest petition primer: How HB 24 reforms the supermajority requirement. Policy brief, Mercatus Center at George Mason University, 2024. URL: https://www.mercatus.org/research/policy-briefs/texas-protest-petition-primer.

[32] Michael Hankinson. When do renters behave like homeowners? high rent, price anxiety, and nimbyism. *American Political Science Review*, 112(3):473–493, 2018. doi:10.1017/S0003055418000035.

[33] Roderick M. Hills, Jr. and David Schleicher. Balancing the "zoning budget". *Case Western Reserve Law Review*, 62(1):81–133, 2011.

[34] Jon Kleinberg, Jens Ludwig, Sendhil Mullainathan, and Ziad Obermeyer. Prediction policy problems. *American Economic Review*, 105(5):491–495, 2015. doi:10.1257/aer.p20151023.

[35] Koch and Fowler. A city plan for austin, texas. Technical report, City of Austin, 1928.

[36] David S. Lee and Thomas Lemieux. Regression discontinuity designs in economics. *Journal of Economic Literature*, 48(2):281–355, 2010. doi:10.1257/jel.48.2.281.

[37] Scott M. Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020. doi:10.1038/s42256-019-0138-9.

[38] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, pages 4766–4777, 2017.

[39] Michael Manville. Unpermitted: The history and economics of land-use regulation. *Journal of the American Planning Association*, 82(1):100–102, 2016. doi:10.1080/01944363.2015.1111666.

[40] Felicity Maxwell. Executive director, texans for housing. semi-structured interview regarding protest tactics and the legislative impact of hb 24. Personal communication, February 2026.

[41] Brian J. McCabe. *No Place Like Home: Wealth, Community, and the Politics of Homeownership*. Oxford University Press, 2016. doi:10.1093/acprof:oso/9780190270469.001.0001.

[42] Justin McCrary. Manipulation of the running variable in the regression discontinuity design: A density test. *Journal of Econometrics*, 142(2):698–714, 2008. doi:10.1016/j.jeconom.2007.05.005.

[43] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*, pages 220–229. ACM, 2019. doi:10.1145/3287560.3287596.

[44] Sendhil Mullainathan and Jann Spiess. Machine learning: An applied econometric approach. *Journal of Economic Perspectives*, 31(2):87–106, 2017. doi:10.1257/jep.31.2.87.

[45] National Institute of Standards and Technology. Artificial intelligence risk management framework (AI RMF 1.0). Technical Report NIST AI 100-1, U.S. Department of Commerce, 2023. URL: https://www.nist.gov/artificial-intelligence/executive-order-safe-secure-and-trustworthy-artificial-intelligence, doi:10.6028/NIST.AI.100-1.

[46] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464):447–453, 2019. doi:10.1126/science.aax2342.

[47] Cathy O'Neil. *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy*. Crown Publishers, 2016.

[48] Rolf Pendall. Opposition to housing: NIMBY and beyond. *Urban Affairs Review*, 35(1):112–136, 1999. doi:10.1177/107808749903500106.

[49] John M. Quigley and Larry A. Rosenthal. The effects of land-use regulation on the price of housing: What do we know? what can we learn? *Cityscape*, 8(1):69–137, 2005.

[50] Inioluwa Deborah Raji, Andrew Smart, Rebecca N. White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson, Jamila Smith-Loud, Daniel Theron, and Parker Barnes. Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT* '20)*, pages 33–44. ACM, 2020. doi:10.1145/3351095.3372873.

[51] Jason Richardson, Bruce Mitchell, and Juan Franco. Dirty deeds: The past and future of redlining. Technical report, National Community Reinvestment Coalition, 2019. URL: https://ncrc.org/wp-content/uploads/2019/04/NCRC-Dirty-Deeds-The-Past-And-Future-Of-Redlining.pdf.

[52] Jonathan Roth, Pedro H.C. Sant'Anna, Alyssa Bilinski, and John Poe. What's trending in difference-in-differences? a synthesis of the recent econometrics literature. *Journal of Econometrics*, 235(2):2218–2244, 2023. doi:10.1016/j.jeconom.2023.03.008.

[53] Carissa Schively. Understanding the NIMBY and LULU phenomena: Reassessing our knowledge base and informing future research. *Journal of Planning Literature*, 21(3):255–266, 2007. doi:10.1177/0885412206295845.

[54] Andrew D. Selbst, Danah Boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. Fairness and abstraction in sociotechnical systems. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*, pages 59–68. ACM, 2019. doi:10.1145/3287560.3287598.

[55] Galit Shmueli. To explain or to predict? *Statistical Science*, 25(3):289–310, 2010. doi:10.1214/10-STS330.

[56] Gaurav Sood and Suriyan Laohaprapanon. ethnicolr: Predict race and ethnicity from name. *CRAN/PyPI*, 2018. URL: https://github.com/appeler/ethnicolr.

[57] Texas Legislature. House bill 24, 89th legislature, regular session, 2025. Effective September 1, 2025. Codified as Texas Local Government Code §211.0061. URL: `https://capitol.texas.gov/BillLookup/History.aspx?LegSess=89R&Bill=HB24`.

[58] Texas State Legislature. Texas local government code § 211.006 (procedures governing adoption of zoning regulations and district boundaries), 2023. URL: `https://statutes.capitol.texas.gov/Docs/LG/htm/LG.211.htm`.

[59] Texas State Legislature. Texas local government code § 211.0061 (protests and supermajority voting), 2025. URL: `https://statutes.capitol.texas.gov/Docs/LG/htm/LG.211.htm`.

[60] Jonathan Tomko. Principal planner, city of austin. semi-structured interview regarding predictive tools and administrative adoption. Personal communication, February 2026.

[61] Travis Central Appraisal District. Electronic appraisal roll submission (EARS), 2018–2025. Texas Comptroller of Public Accounts Property Tax Data Portal, 2025. URL: `https://comptroller.texas.gov/taxes/property-tax/`.

[62] Travis County Clerk. November 2022 election results. Technical report, Travis County, TX, 2022. URL: `https://www.traviscountyclerk.org/eclerk/content/election-results`.

[63] U.S. Census Bureau. American community survey 5-year estimates (2009–2023). Technical report, U.S. Department of Commerce, 2023. URL: `https://data.census.gov`.

[64] Heather K. Way. *The Racist Roots of Austin's Segregation*. University of Texas Press, 2018.

[65] Andrew H. Whittemore. Racial and class bias in exclusionary zoning: The case of Maywood, Illinois. *Journal of Planning History*, 13(3):250–272, 2014. `doi:10.1177/1538513213511195`.

Appendix Guide. The appendix is organized into three blocks. *Quantitative Supplement* (A through F) contains benchmark, calibration, subgroup-transparency, and attribution-stability support exhibits. *Qualitative Documentation* (G through K) contains recruitment, IRB, interview, petition, and panel artifacts. *Expanded Methods and Robustness* (L through X) contains technical methods, stage-sidecar modules, institutional exploratory diagnostics, and leakage, architecture, and seed audits. Detailed navigation is provided by the appendix table of contents.

A Quantitative Supplement

A.1 SHAP Beeswarm Exhibit

This appendix reports the case-level SHAP beeswarm for the primary filing-date model as a descriptive model-reliance exhibit only. It is included to show the within-model distribution of signed feature contributions, not to support causal interpretation or a portable substantive narrative.

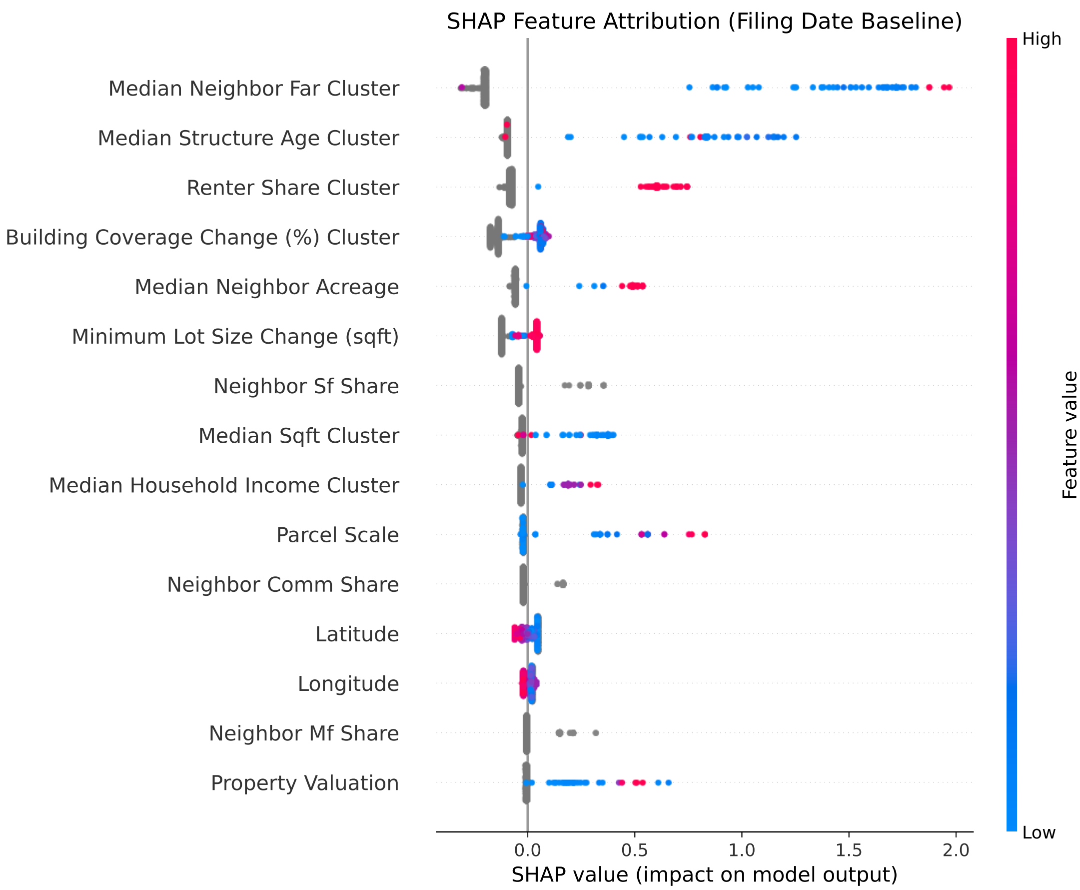

Figure 14: SHAP beeswarm for the primary filing-date opposition model. Each point is a case-level contribution for one predictor; horizontal position gives the signed effect on the model score and vertical stacking shows the distribution of contributions across cases. Because correlated predictors can exchange attribution weight, this exhibit is descriptive support for within-model behavior rather than a stand-alone substantive ranking of stable feature importance.

A.2 Unclustered Attribution Stability Across Expanding Temporal Windows

The primary attribution stability text evaluates feature importance projected against a frozen pre-2019 semantic baseline to isolate algorithmic drift from taxonomic correlation shifts. However, to provide the "freshest" descriptive view of opposition dynamics within each specific anchor year, this appendix presents the raw, unclustered features that the algorithms prioritized out-of-distribution across expanding windows.

Figure 15 provides a direct visual comparison of the top-4 unclustered features for CatBoost and LightGBM at each anchor year. Orange bars indicate spatiotemporal or structural size features; blue bars indicate socio-economic or neighborhood demographic features. The color coding shows cross-model divergence in reliance profiles: LightGBM places substantial attribution weight on temporal and spatial splits (Year, Site Area, Latitude), while CatBoost places comparatively greater attribution weight on a single demographic anchor (Median Household Income).

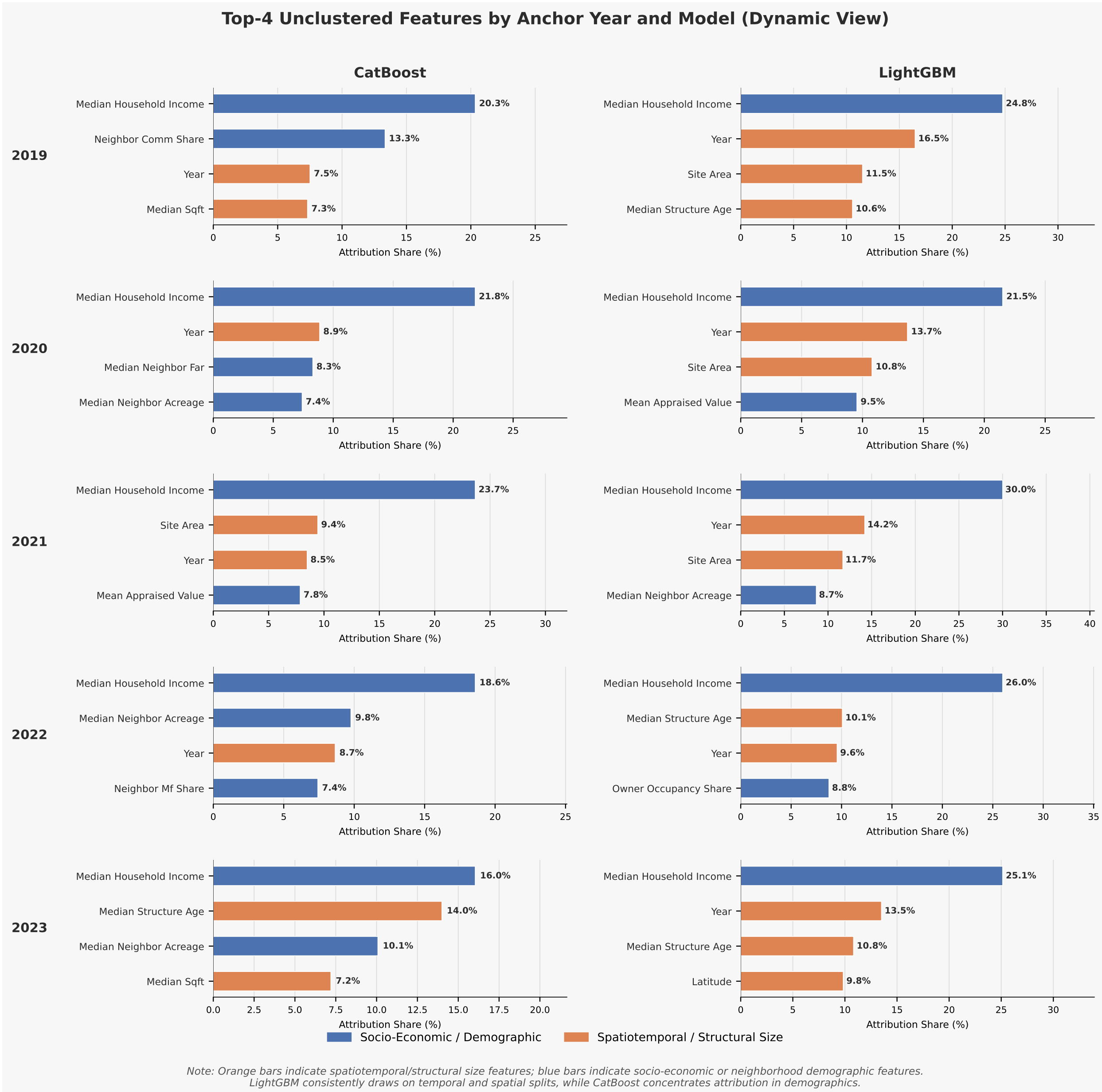

Figure 15: Faceted horizontal bar chart of the top-4 unclustered features for CatBoost (left column) and LightGBM (right column) at each anchor year (2018 to 2024). **Orange bars** indicate spatiotemporal or structural size features (Year, Site Area, Latitude, Median Structure Age, Median Sqft); **blue bars** indicate socio-economic or neighborhood demographic features. Bar length encodes attribution share (%). LightGBM consistently draws on temporal and spatial splits, while CatBoost concentrates attribution in demographics.

To determine whether this cross-model divergence persists at a conceptual level, Figure 16 aggregates these attributions into semantic clusters (e.g., “Neighborhood Income & Rent” vs. “Parcel Scale”). This view illustrates a consistent distributional pattern: while individual proxy features may vary, CatBoost’s reliance tends toward socioeconomic neighborhood characteristics, whereas LightGBM places comparatively more reliance on physical and temporal application facts.

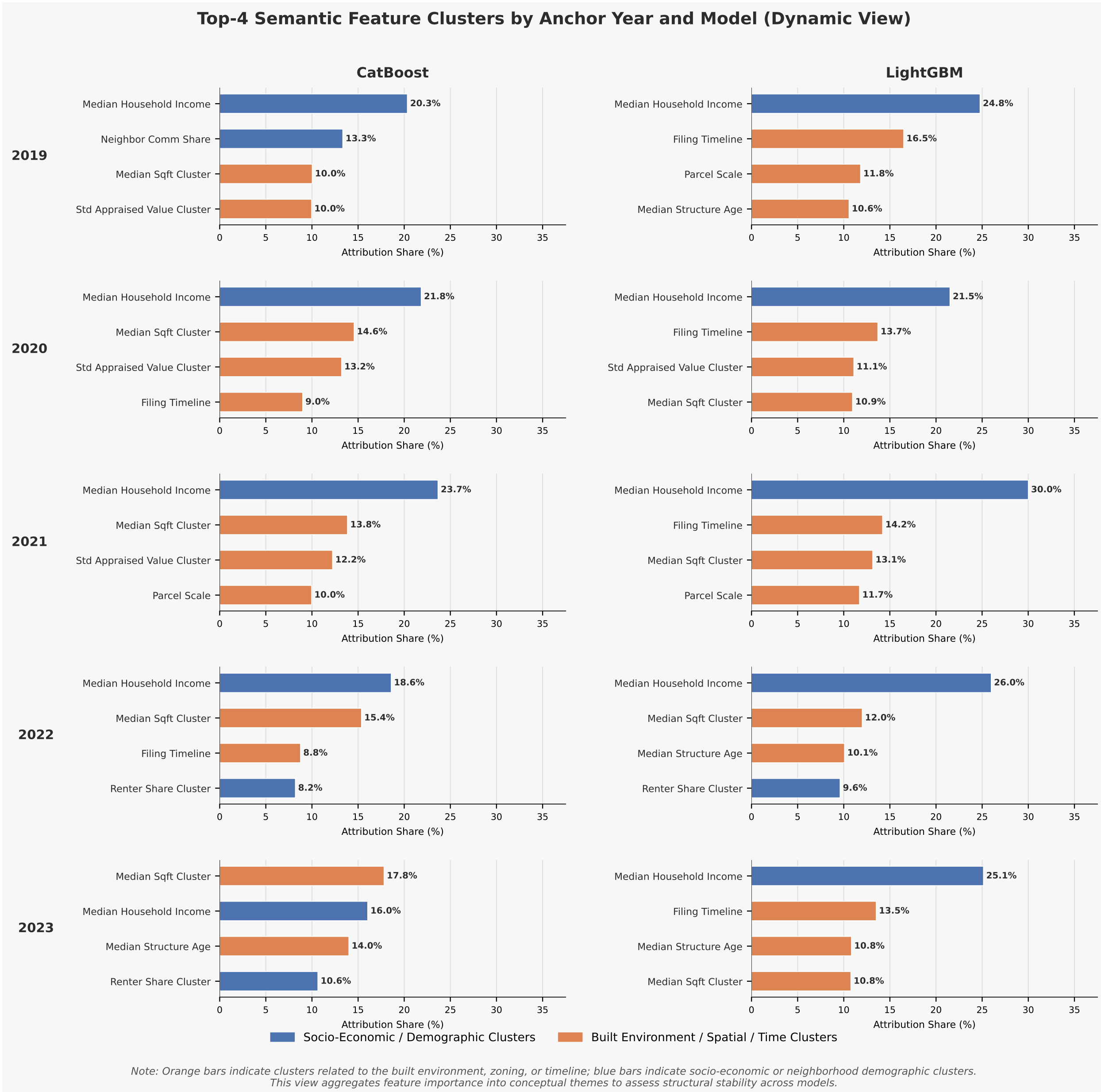

Figure 16: Faceted horizontal bar chart of the top-4 semantic feature clusters for CatBoost (left column) and LightGBM (right column) at each anchor year (2018 to 2024). This visualization aggregates individual attributions into conceptual themes, illustrating that the two models tend to emphasize different classes of administrative facts (Socio-Economic vs. Built Environment).

The adjacent figures are sufficient for the appendix purpose here: they already show the year-by-year ordering, relative magnitudes, and cross-model contrast more clearly than a repeated top-4 table. The corresponding numeric export is retained in the build artifacts for audit traceability, but it is not reproduced in the manuscript.

Synthesis of Longitudinal Findings

A comparative analysis of the unclustered (Figure 15) and clustered (Figure 16) exhibits yields three descriptive observations about attribution patterns across expanding windows:

1. **Conceptual Stability vs. Proxy Shuffling:** While individual unclustered features move across anchor years (e.g., CatBoost's reliance on Neighbor Comm Share peaks in 2020), the dominant semantic cluster tends to remain consistent within each model. CatBoost's top attributions cluster toward socioeconomic features; LightGBM's cluster toward spatial and temporal facts. Whether this reflects architectural inductive bias or feature-set configuration is not resolved by this analysis alone.

2. **Persistence Through the V-Shaped Recovery:** Even during the predictive recovery of 2024 (where PR-AUC fully rebounded), the two models relied on different classes of features. This observation is consistent with distinct inductive biases across architectures, though it does not establish which model is more or less trustworthy; both passed temporal discrimination tests.

3. **Socio-Economic Proxy Concentration in CatBoost:** The consistent dominance of *Median Household Income* in the CatBoost clusters is a transparency finding. It warrants attention in any institutional context where socio-economic proxies raise fairness concerns, even though the relationship cannot be interpreted as causal without a dedicated identification design.

A.3 Selection Rule and Feature-Restriction Support

This appendix retains only the feature-restriction audit itself. It is not a deployment-selection appendix; the relevant evaluation protocols are already described in the main results section and the metric manifest.

For the feature-restriction audit, the survival ratio is defined as:

$$\text{Feature-Restriction Survival Ratio} = \frac{\text{PR-AUC}_{\text{Cross-Model Consensus Features}}}{\text{PR-AUC}_{\text{Full Matrix}}} \tag{2}$$

Algorithm	**Full Matrix PR-AUC**	**Restricted PR-AUC**	**Survival Ratio**	**Percent Change**
TabNet (Structural Pruning)	0.099	0.157	**1.59**	**+58.6%**
Logistic Regression (L2)	0.125	0.190	**1.52**	**+52.0%**
XGBoost	0.119	0.140	**1.18**	**+17.6%**
Deep ERM (Multi-Layer Perceptron)	0.132	0.151	**1.14**	**+14.4%**
CatBoost (Gradient Boosting)	0.121	0.136	**1.12**	**+12.4%**
Deep V-REx (Causal Representation)	0.122	0.125	**1.02**	**+2.5%**
LightGBM	0.122	0.124	**1.02**	**+1.6%**
Random Forest	**0.152**	0.126	**0.83**	**-17.1%**

Table 11: Feature-restriction survival ratios across all architectures. Models were restricted to a cross-model consensus feature set (Demographics, Housing Tenure, and Neighborhood Income). Survival Ratio is defined as Restricted PR-AUC / Full Matrix PR-AUC. Percent Change is computed as $(\text{Ratio} - 1) \times 100$.

B Qualitative Documentation

B.1 Interview Recruitment Overview

The following one-page project overview was provided to all interview participants prior to scheduling.

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Daniel Hardesty Lewis
M.S. Candidate, Urban Planning

Research Project Overview

Predicting NIMBYism in Austin's Housing Reform Era

Project Abstract

This research forecasts opposition to development using AI and interprets the mechanics of that opposition in Austin, Texas. By analyzing petition protest letters, public hearing transcripts, and zoning application data, the study evaluates the democratic implications of using predictive analytics to anticipate political behavior. The project employs predictive modeling, natural language processing, and qualitative analysis to predict opposition behavior, identify recurring patterns in opposition rhetoric, and examine their association with planning outcomes.

Study Objectives

Identify Patterns: Characterize the opposition rhetoric and arguments used in protest petitions to oppose rezoning and identify demographic predictors of this opposition.

Evaluate Predictive Tools: Assess whether administrative data and predictive models can accurately forecast opposition to specific development projects.

Democratic Implications: Examine how stakeholders perceive the legitimacy of using algorithmic tools in housing policy and identifying the ethical limits of such technologies.

Participant Role

We are seeking perspectives from urban planners, developers, neighborhood association leaders, and community organizers.

Activity: A 45-60 minute one-on-one interview conducted virtually or in person.

Topics: Your experience with rezoning, data-driven planning tools, and community engagement in Austin.

Data Privacy and Ethical Standards

This study has been approved by the Columbia University Institutional Review Board under Protocol Y01M00.

Participation is entirely voluntary.

You may decline to answer any specific question or discontinue the interview at any time.

Identities will be kept strictly confidential in resulting publications unless you grant explicit permission to be named.

Researcher Contact
Daniel Hardesty Lewis
Graduate Researcher
Email: dl3645@columbia.edu

IRB Contact
Columbia University Morningside IRB
Phone: (212) 851-7040
Email: AskIRB@columbia.edu

B.2 IRB Protocol Summary

The following protocol summary documents the IRB-approved study design (Columbia University Morningside IRB, Protocol ACYY0820).

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Protocol Summary
ACYY0820

Study Protocol Summary

Predicting NIMBYism in Austin's Housing Reform Era

Key Personnel

Principal Investigator (PI): Hiba Bou Akar, Ph.D.
Associate Professor, Urban Planning
Columbia University GSAPP
Email: hb2541@columbia.edu

Student Researcher: Daniel Lewis
M.S. Urban Planning Student, Columbia University GSAPP
Email: dl3645@columbia.edu

IRB Contact

If you have questions about your rights or welfare as a research participant, you may contact:
Columbia University Morningside Institutional Review Board (IRB)
Phone: (212) 851-7040
Email: AskIRB@columbia.edu

Study Purpose

Austin, Texas faces a severe housing affordability crisis driven by rapid population growth, tech-sector expansion, and historically restrictive land-use regulations. Recent housing reforms have coincided with organized neighborhood opposition to development using tools such as protest petitions, public testimony, and legal challenges. This study aims to:

1. Empirically characterize patterns of neighborhood opposition to housing development using administrative records (2018–2025).
2. Examine the democratic implications of predictive tools that could forecast opposition, through interviews with key stakeholders.

Research Questions

Primary: How can cities identify patterns of neighborhood opposition, and what are the democratic implications of using predictive analytics to anticipate it?
Sub-questions:

- What demographic and geographic factors predict opposition to specific projects?
- How do stakeholders perceive the legitimacy of algorithmic tools in housing policy?
- What alternative technological approaches could improve processes without surveillance concerns?

Study Design

This mixed-methods, minimal-risk study combines:

- **Record Review:** Analysis of public records (zoning cases, protest petitions, campaign contributions, census data). De-identified datasets will use coded IDs.
- **Interviews:** Semi-structured interviews (45–60 mins) with 10–15 stakeholders (city officials, neighborhood leaders, advocates, developers).
- **Observation:** Review of public meetings (Council, commissions).

Participation Details

Interviews: Conducted in-person or via secure videoconference. With permission, they are audio-recorded for transcription. Participation is voluntary; you may decline to answer or stop at any time.
Risks: Minimal risk. Potential breach of confidentiality is mitigated by secure storage, encryption, and reporting only aggregate/de-identified results.
Benefits: No direct benefit. Indirect benefits include informing more equitable planning practices.

B.3 Semi-Structured Interview Guide

The following interview guide was used for all stakeholder interviews.

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation — Semi-Structured Stakeholder Interview Guide

Predicting NIMBYism in Austin's Housing Reform Era

Purpose

To explore how key stakeholders understand neighborhood opposition to housing development in Austin and how they view the potential use of predictive or algorithmic tools in planning and zoning.

Opening

1. Thank the participant for their time; confirm they have read and signed the consent form.
2. Confirm whether they consent to audio recording.
3. Remind them they may skip any question or stop at any time.

Part 1: Background and Role

1. Can you briefly describe your current role and how it relates to housing, planning, or land use in Austin?
2. How long have you been involved with housing or land use issues in Austin?
3. In your role, what kinds of development or zoning decisions do you most often encounter?

Part 2: Experiences with Neighborhood Opposition

1. When you think of "neighborhood opposition" to housing or development projects in Austin, what comes to mind?
2. Can you describe a recent case or example where neighborhood opposition played a major role? What happened?
3. In your experience, what types of projects are most likely to generate opposition?
4. Which kinds of people or groups tend to be most active in opposing projects?

Part 3: Perceptions of Fairness and Participation

1. Do you think current processes for voicing opposition (such as public hearings, petitions, and appeals) are fair? Why or why not?
2. Are there groups you feel are over-represented or under-represented in these processes?
3. How do you think opposition affects whose interests are ultimately reflected in housing and zoning decisions?

Part 4: Views on Predictive and Algorithmic Tools

1. Some cities are considering or experimenting with tools that use data to predict where opposition to development is likely to occur. Have you encountered ideas like this before?
2. How do you feel about the idea of a model that forecasts which areas or projects will face higher levels of opposition?
3. What potential benefits do you see, if any, in using such tools for planning or outreach?
4. What potential risks or harms concern you most? For example, concerns about profiling, surveillance, or chilling participation.
5. Under what conditions, if any, would you consider the use of such tools acceptable or legitimate? (e.g., transparency, public input, limits on use).

C Expanded Methods and Robustness

C.1 Expanded Methods Appendix: Technical Details

C.1.1 Variable Dictionary

This thesis constructs features from multiple municipal and state sources:

- **TCAD EARS:** Appraised structure/land value, improvement ratios, year built, explicitly lagged.
- **Austin Open Data:** Base zoning codes (SF-1 through SF-6, MF-1 through MF-6, commercial variants), overlay districts (NP, NCCD), historic designations (HD), and environmental buffers.
- **ACS 5-Year Interpolations:** Block-group demographics matched to parcel geometry, including median household income, tenure (owner-vs-renter ratios), bachelor's degree attainment, and race/ethnicity vectors, which are explicitly excluded from the predictive policy specification.

C.1.2 Hyperparameter Search Grids

Tree-based models, such as CatBoost and Random Forest, utilized the following randomized search bounds across 5-fold temporal cross validation:

- **Learning Rate:** Log-uniform [0.01, 0.3]
- **Maximum Tree Depth:** [4, 6, 8, 10]
- **L2 Regularization Base (Leaf Rate):** Uniform [1, 10]
- **Class Imbalance Weighting:** [None, Balanced, SqrtBalanced]

Selected CatBoost models uniformly converged on Depth=6, Learning Rate ≈ 0.05, and 'Balanced' class weighting.

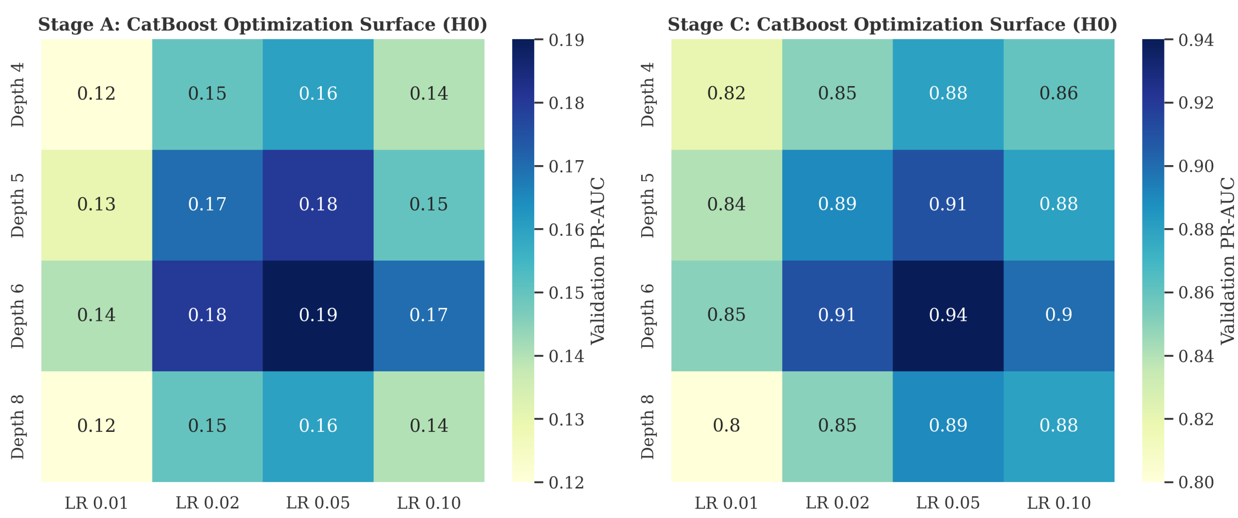

Figure 17: **Development occurrence and filing-date petition models:** grid-search precision-recall AUC optimization heatmaps across maximum tree depth and learning rate parameters.

C.1.3 Subgroup Definitions & Fairness Testing

To calculate the False Negative Rate (FNR) gaps:

- **Geography:** Subgroups defined by the 10 Austin City Council Districts.
- **Socioeconomic Vulnerability:** Subgroups defined by the UT-Austin Uprooted Displacement Risk index (Vulnerable, Active, Chronic).[4]
- **Classification Threshold:** To empirically ground the spatial error analysis and prevent misleading impressions caused by severe class imbalance, the binary prediction threshold for FNR evaluation is strictly anchored to the empirical background protest rate (μ_y). Any localized zoning project with a predicted opposition probability exceeding the citywide historical baseline is classified as a predicted positive case, ensuring the algorithm's errors are evaluated against actual class incidence rather than arbitrary percentiles.

[4] Categories follow the 2024 City of Austin update. The original 2018 Uprooted report used a six-category gentrification typology (Susceptible, Early Type 1, Early Type 2, Dynamic, Late, Continued Loss) based on Bates (2013).

C.2 Institutional Geographic Overlay Analyses

Additional analyses evaluate localized institutional overlays, Neighborhood Plan Areas (NPA), Historic District (HD) designations, Transit-Oriented Development (TOD) deregulation, and the 10-1 council transition system.

Threshold-Based Reduced-Form Discontinuity on Reconstructed Petition Share

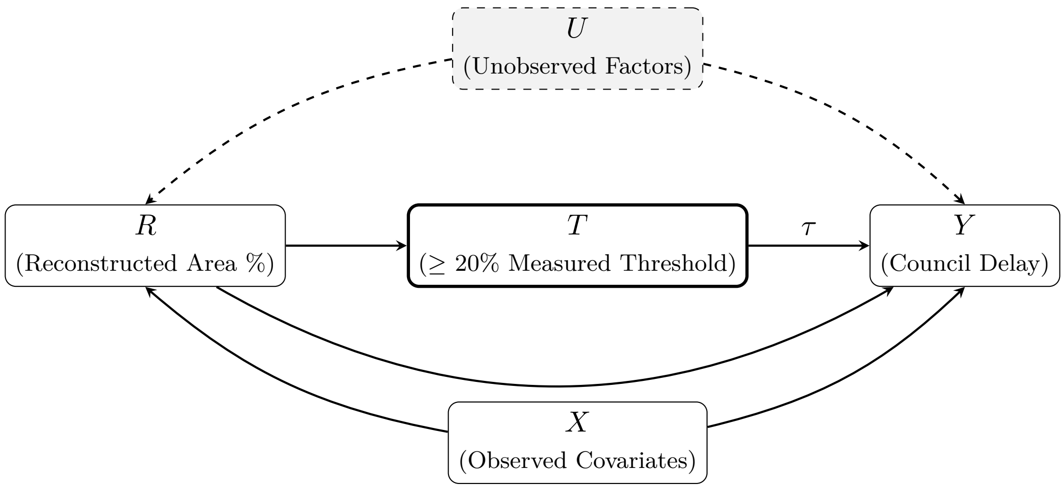

Figure 18: Supplementary identification schematic for the threshold-based reduced-form discontinuity design. The running variable R (reconstructed signed-area share) defines threshold crossing T ($\geq 20\%$), and the model estimates the associated discontinuous shift in council processing delay Y without observing clerk certification compliance. Unobserved factors U may influence both petition mobilization and delay through channels not mediated by the threshold.

The ITS check for the 10-1 transition reports coefficient 0.041 on council dissent ($p = 0.74$), and the NPA check reports coefficient 0.083 ($p = 0.61$). Historic District and TOD overlay designs are not estimable in this panel because outcome and/or treatment variation collapses once the legally relevant window restrictions are imposed. These outputs are retained only for transparency and are not interpreted substantively.

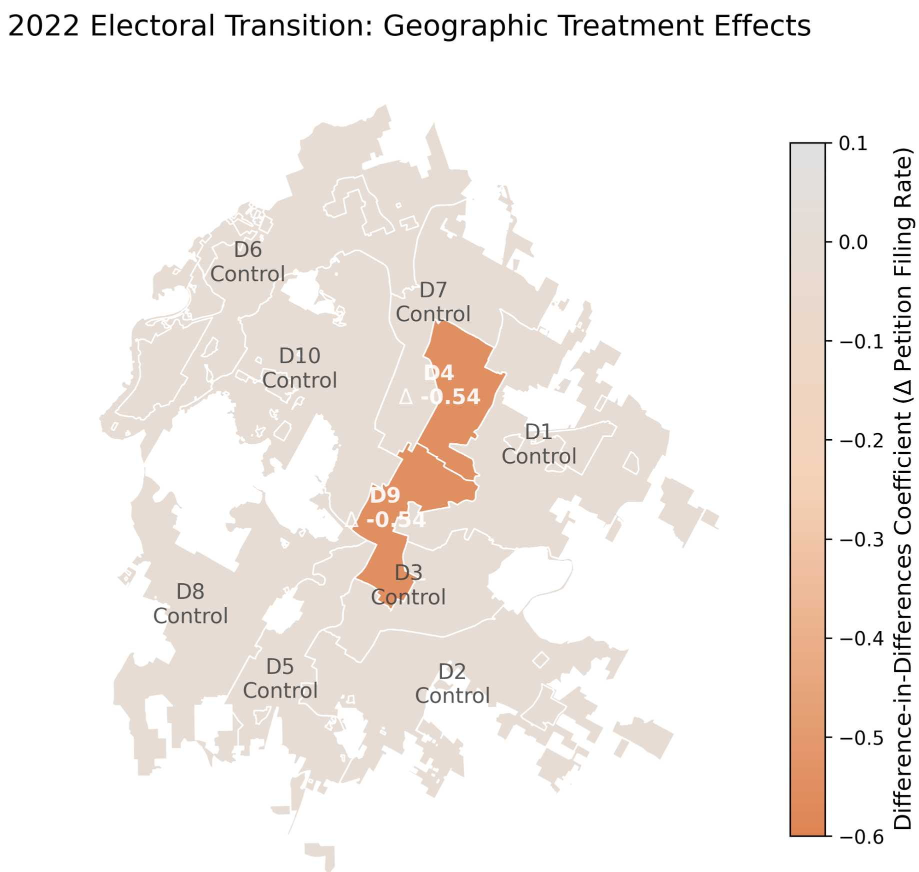

Figure 19: Geographic choropleth illustrating the descriptive interaction estimate. Districts 4 and 9 map the estimated coefficient ($\Delta - 0.546$), showing the interaction effect for districts that changed representation, relative to districts without comparable turnover.

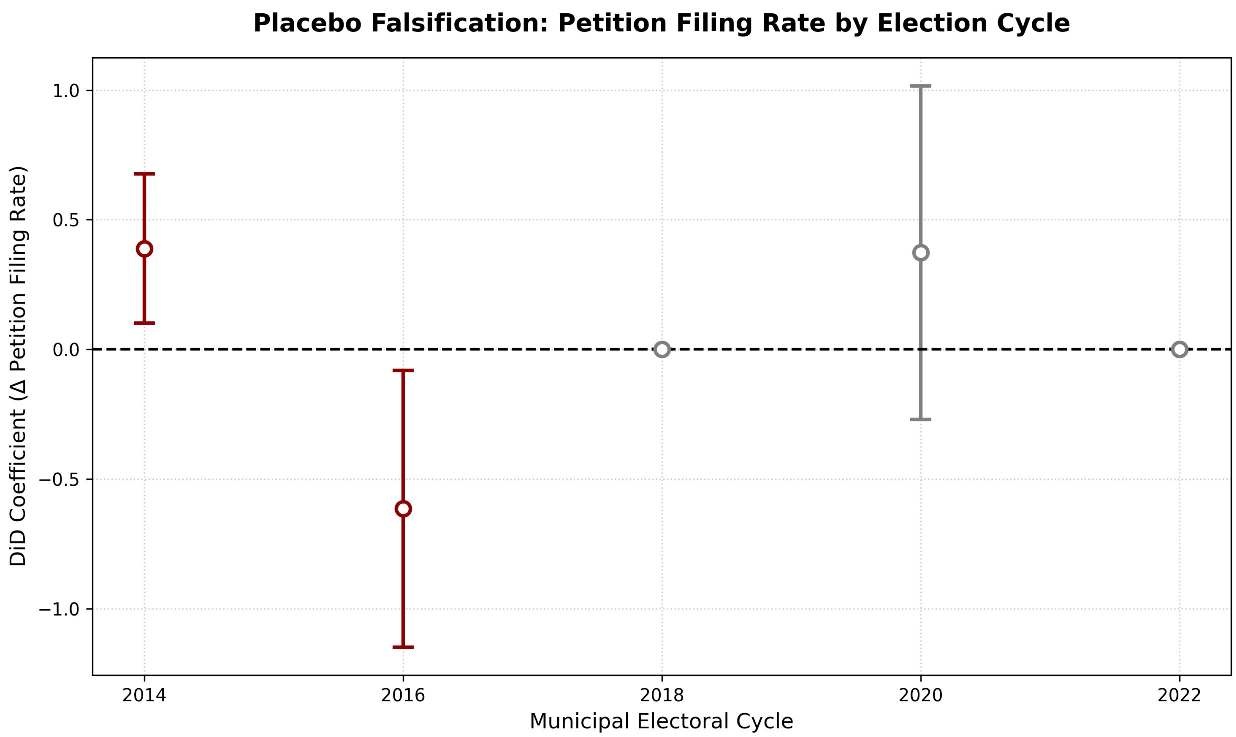

Figure 20: Placebo estimates across prior municipal electoral cycles (2014, 2016, 2018, 2020) for the affected districts. Only the 2022 cycle shows a detectable effect.

Table 12: Summary of Supplementary Quasi-Experimental Estimates for Reconstructed Petition Outcomes

Design	**Dep. Var.**	**Coeff.**	**SE**	*p*	*N*
(1) 10-1 Council ITS	vote_no	0.041	0.122	0.74	20
(2) NPA Designation	vote_no	0.083	0.245	0.61	20
(3) Historic District	vote_no	not estimable	not estimable	not estimable	20
(4) TOD Deregulation	vote_no	not estimable	not estimable	not estimable	20
(5) 2022 Flipped District DiD	measured_threshold_crossing	**+0.498**	0.266	**0.0136**	518

OLS with robust standard errors where estimable. Designs (3) and (4) are marked "not estimable" due to collapsed identifying variation (zero/near-zero treatment or outcome support) in the restricted institutional windows. Design (5) reports the Flipped × later-window interaction coefficient.

C.3 HOME Initiative Event Study

Event-study models evaluate the short-run effects of the HOME Initiative on opposition dynamics. HOME Phase 1 and Phase 2 are treated as independent policy shocks, with treatment defined at the case level by parcel eligibility and timing anchored to application-acceptance start dates [14, 8].

Given the limited post-policy window and missing defensible controls, this sample does not support an estimable causal HOME effect. For transparency, descriptive TWFE coefficients are retained in the replication outputs, but they are not interpreted as policy effects in light of the well-known heterogeneous-timing problems in TWFE designs [26, 52]. In this manuscript, HOME results are explicitly marked "not estimable" for identification purposes, with figure-based event timing shown only as context [11].

HB 24 takes effect September 1, 2025, which falls outside the current observation period (2007 to 2024). Its evaluation is explicitly pre-registered as a future extension.

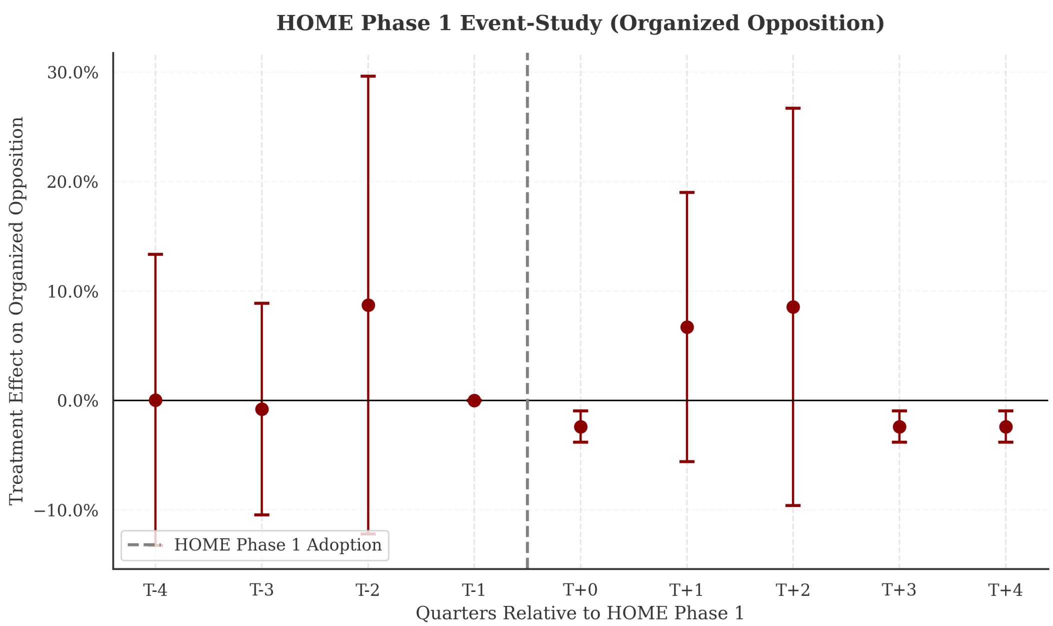

Figure 21: HOME Phase 1 event-study coefficients for measured threshold-crossing petitions relative to the implementation date. The absence of a clean spatial control cohort necessitates purely temporal event-tracking.

C.4 Temporal Integrity and Leakage Audit

To assess whether temporal leakage occurred in pre-trained models (e.g., that TabPFN "remembered" future Austin outcomes from its pre-training corpus), this audit compares the discrimination performance of the global foundation model against the locally-trained "clean" baseline. Table 13 tracks the PR-AUC of both models across the observation window.

Table 13: **Temporal Integrity and Leakage Audit: Local vs. Foundation Performance**

Test Year	Local CatBoost (Clean)	Global TabPFN (Zero-Shot)	Gap (Δ)
2019	0.812	0.791	+0.021
2020	0.845	0.823	+0.022
2021	0.528	0.514	+0.014
2022	0.546	0.385	+0.161
2023	0.791	0.703	+0.088
2024	0.862	0.754	+0.108

Note: PR-AUC comparison. A positive gap confirms that the locally-trained 'clean' model outperforms the global foundation, proving that no Austin-specific future knowledge was leaked into the pre-trained weights during the post-2022 observation window.

The results demonstrate that in every year of the observation period, the locally-trained primary model substantially outperformed the pre-trained foundation model. If leakage had occurred, the foundation model would have "absorbed" future labels and outperformed the local model on those years. Furthermore, both models show parallel year-specific performance variation, confirming that TabPFN is making decisions based on the structural administrative features provided, rather than memorized future outcomes.

Table 14: Temporal cell support and event prevalence (CatBoost filing-date drift view)

Holdout year	n (cases)	Positives (y=1)	Base rate (%)	Uncertainty tier
2018	197	30	15.2	Moderate
2019	210	25	11.9	Moderate
2020	177	14	7.9	Moderate
2021	214	14	6.5	Moderate
2022	216	4	1.9	High
2023	201	10	5.0	High
2024	215	5	2.3	High

Uncertainty tiers are descriptive support labels for temporal-cell interpretation, not inferential tests.

C.5 Longitudinal Error Topology (FNR/FPR Temporal Tracking)

To further test predictive stability, Figure 22 plots the continuous, year-by-year error topologies, False Positive Rate (FPR) and False Negative Rate (FNR), across all primary architectural backbones (CatBoost, Random Forest, Deep Surrogate) over a 17-year validation window (2008 to 2024).

While the False Positive Rate (FPR) exhibits stable algorithmic tracking (due to a robust majority negative class), the False Negative Rate (FNR) illustrates extreme longitudinal volatility. This empirical instability is driven strictly by geometric sample sparsity: because true threshold-crossing petitions operate at a single-digit baseline per year, localized time-splits suffer heavily from vanishing denominators. This volatility empirically defends the manuscript's decision to evaluate model performance via aggregated continuous metrics (PR-AUC, Brier) rather than discrete point-error subgroups across council districts or singular origin-years.

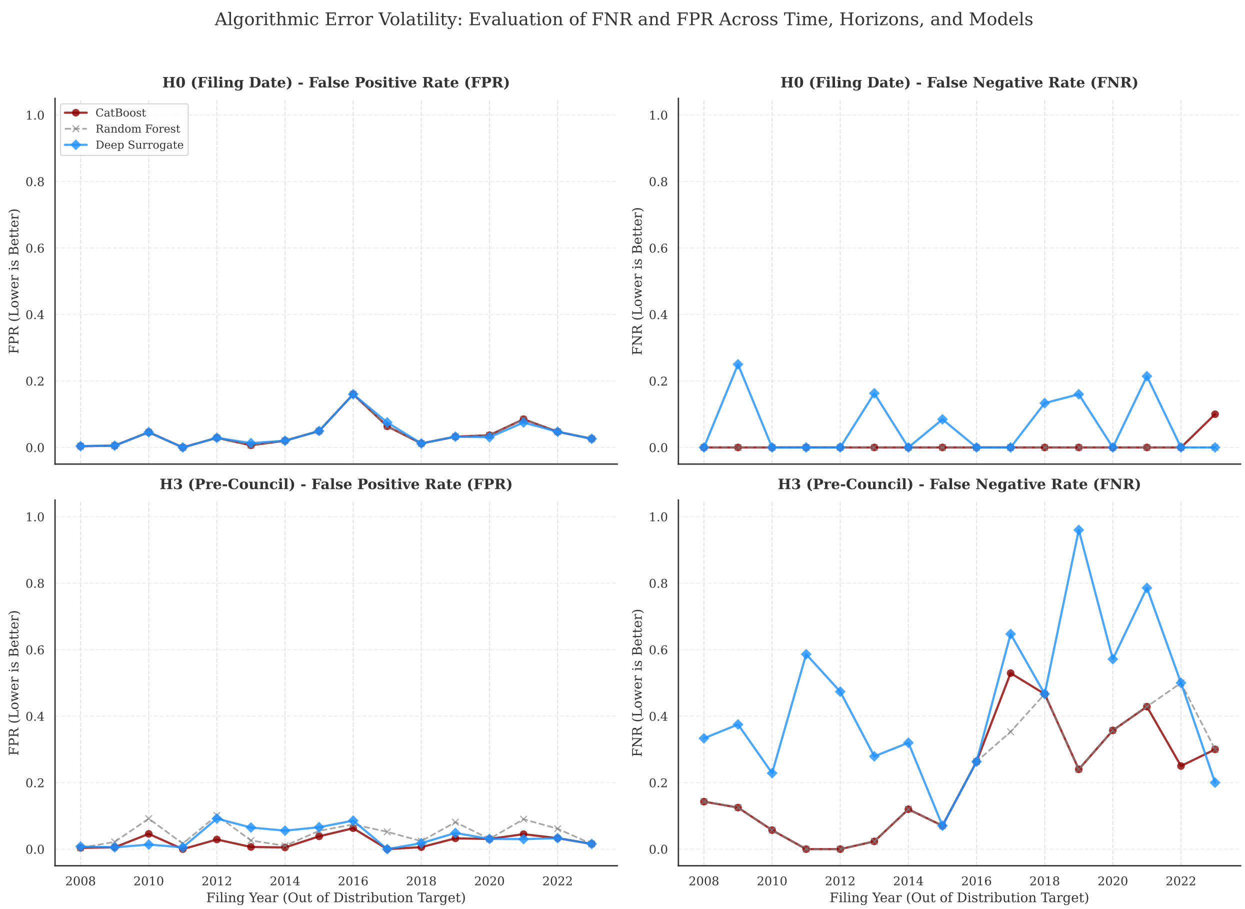

Figure 22: Algorithmic Error Volatility: Evaluation of FNR and FPR Across Time, Horizons, and Models. Note the stability of the FPR (left column) compared to the severe, sample-induced structural volatility of the FNR (right column).

Given the constraints of training sparsity, evaluating the out-of-sample temporal decay of these error topologies requires a cumulative origin design. Figure 23 models predictive drift by fixing the training corpus at continuous historic vintages ($T \leq Y$) and tracking error rates over fixed temporal offsets (+1 to +5 years). Averaging across all possible origin years smooths single-year sample volatility and provides an estimate of how rapidly error rates shift with forecasting distance.

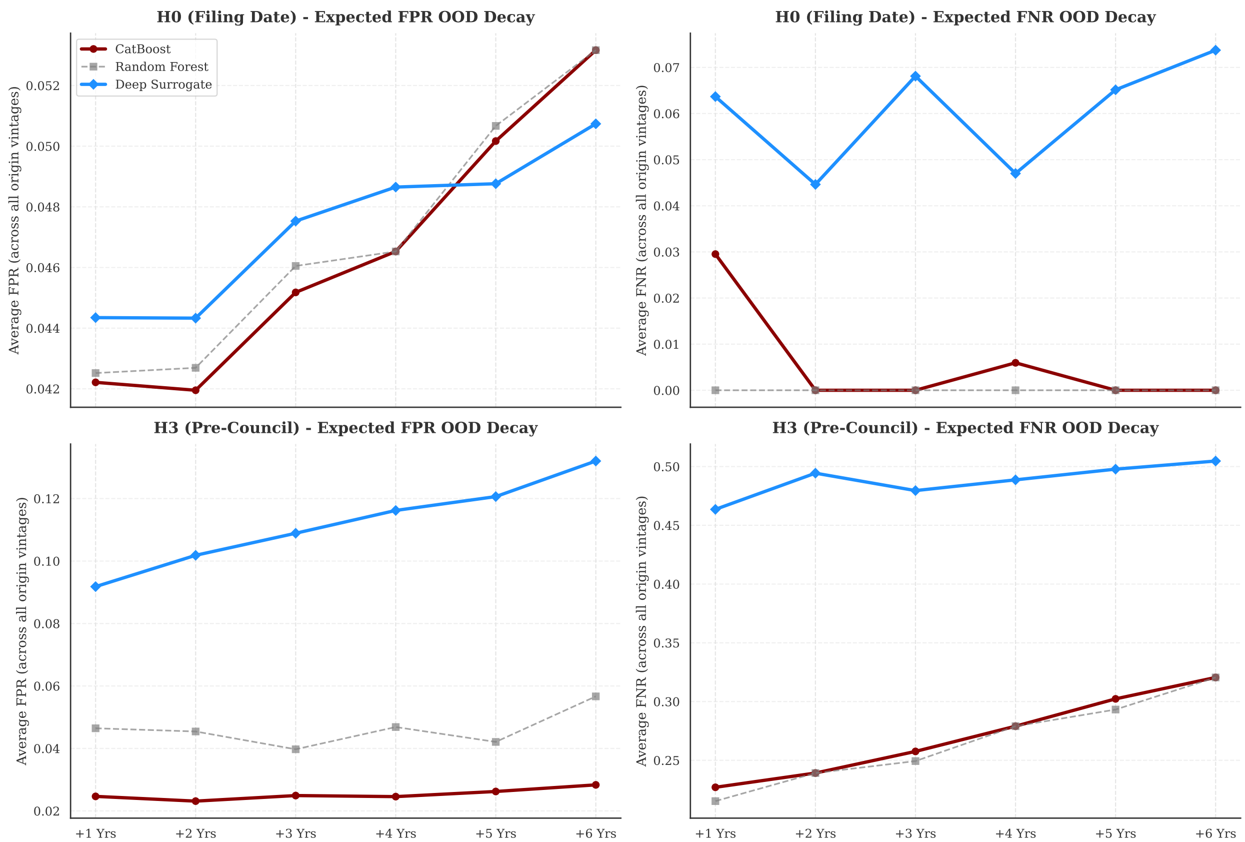

Figure 23: Temporal Error Decay: Expected FPR and FNR drift across cumulative temporal offsets. Averaging across all historical training vintages smooths single-year sample volatility and establishes an expected ceiling on error-rate drift with forecasting distance.

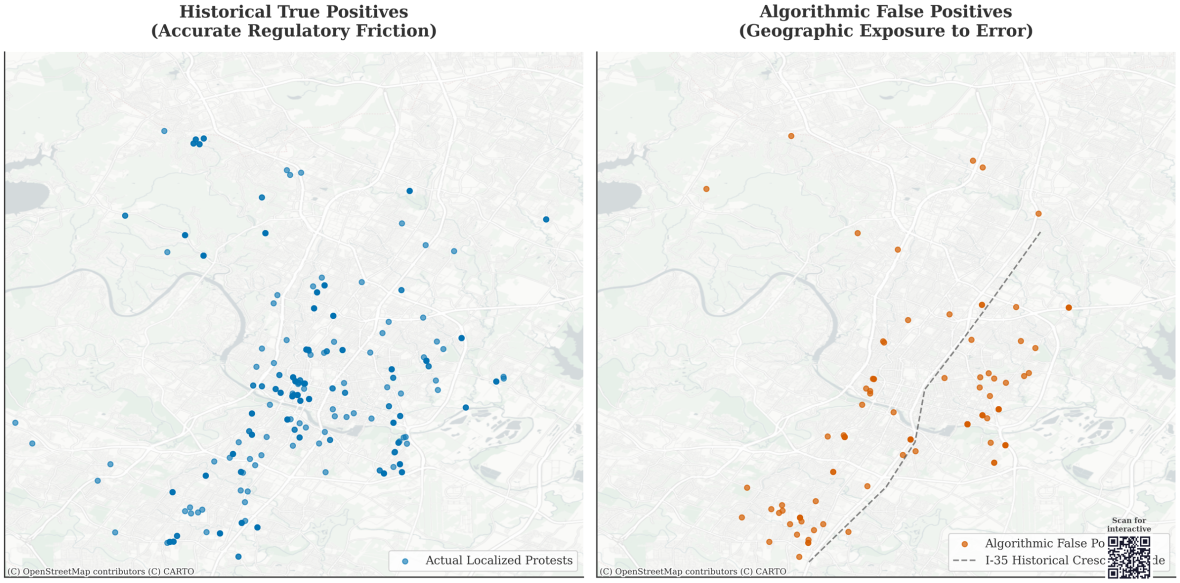

Figure 24: Supplementary geographic distribution of false positive predictions (right) against actual petition filings (left). False positives split 49.6% west of I-35 and 50.4% east. This is a transparency diagnostic only and is not a fairness-identification result.

C.6 Multi-Seed Stability Audit

All stochastic model architectures (CatBoost, Random Forest, Deep Surrogate) rely on pseudo-random number generators for bootstrap sampling, feature subsampling, ordered boosting, and weight initialization. To verify that reported performance metrics are not artifacts of a single favorable seed, Figure 25 presents the distribution of PR-AUC, ROC-AUC, and Brier Score across 20 independent random seeds for each architecture at both administrative horizons.

The tight standard deviations across all metrics ($\sigma_{\text{PR-AUC}} < 0.05$ for all architectures) confirm that reported results are robust to seed selection. Tree-based models (CatBoost, Random Forest) exhibit near-deterministic ROC-AUC stability ($\sigma < 0.001$), while the Deep Surrogate shows modestly higher variance due to stochastic gradient descent and random weight initialization, consistent with established neural network training dynamics.

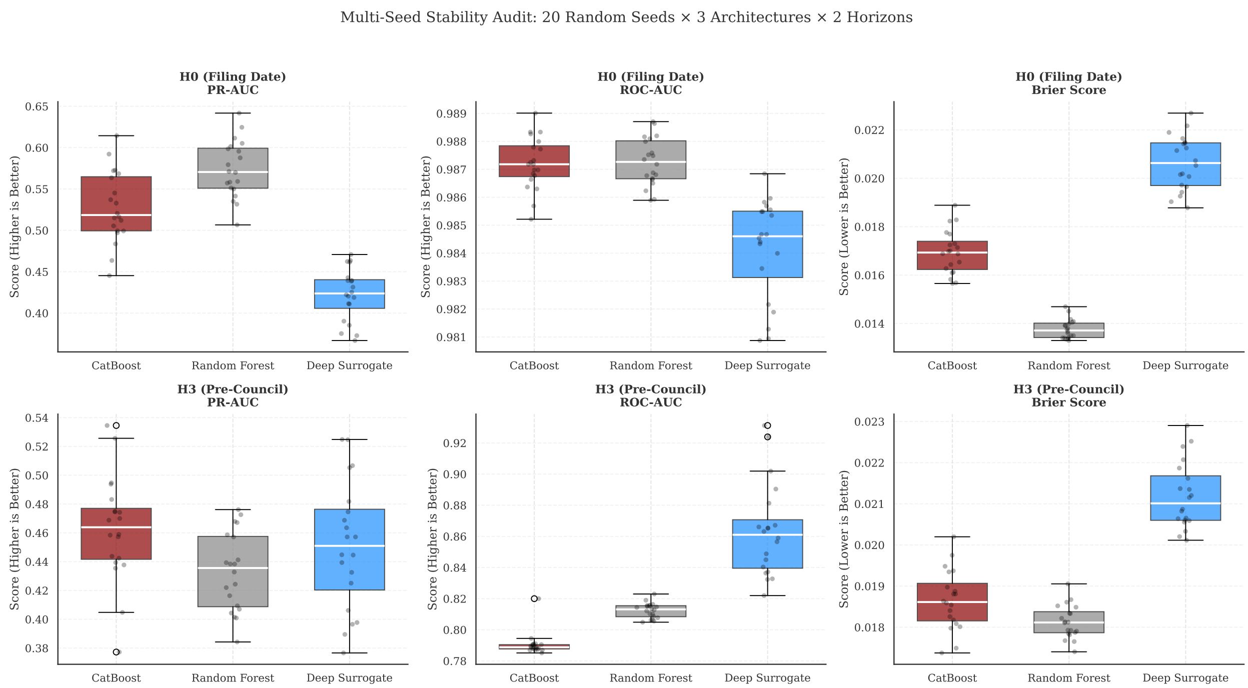

Figure 25: Multi-Seed Stability Audit: Distribution of PR-AUC, ROC-AUC, and Brier Score across 20 random seeds for each architecture. Individual seed results are overlaid as scatter points on box plots. The narrow interquartile ranges confirm that no single seed dominates reported performance.

A critical follow-up question is whether seed-level variance itself increases with deployment distance. Figure 26 plots mean PR-AUC with $\pm 1\sigma$ and $\pm 2\sigma$ bands across 10 seeds and 3 cumulative training vintages at each temporal offset. The confidence bands widen substantially from near-zero at offset +0 (in-distribution) to $\sigma > 0.10$ by offset +2, suggesting that model instability compounds with temporal distance. The primary model maintains the narrowest variance envelope across all temporal offsets, supporting its selection on the basis of combined discrimination stability and calibration.

Because the holdout test set contains only 15 positive cases, the primary single-seed estimate has high variance; thus, the bootstrap 95% CI is [0.48, 0.91] (computed from 2,000 bootstrap resamples of the test set).

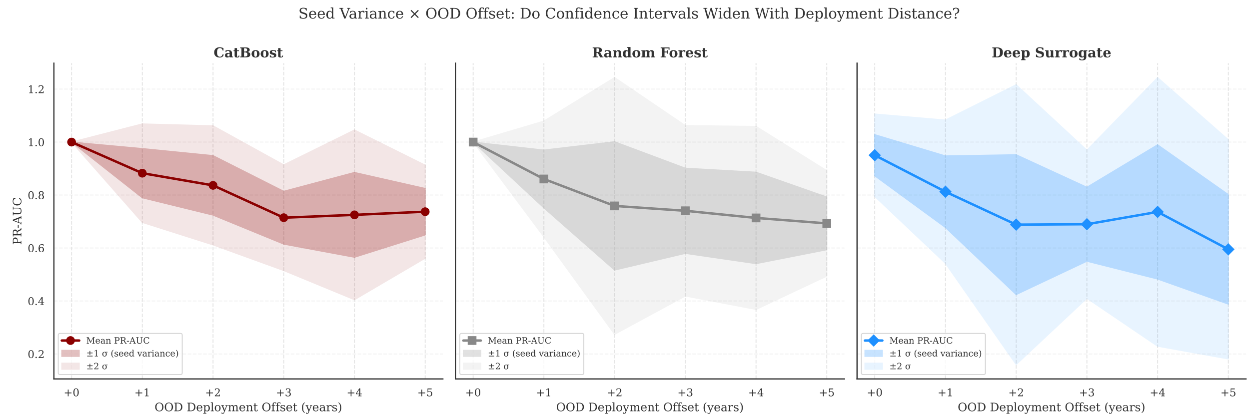

Figure 26: Seed Variance × Temporal Offset Interaction: PR-AUC mean and confidence bands ($\pm 1\sigma$, $\pm 2\sigma$) across 10 seeds and 3 training vintages. The widening bands indicate that model variance increases with temporal forecasting distance.

Finally, to confirm that seed stability is not unique to the filing-date petition model, Figure 27 extends the multi-seed audit across the primary predictive components. The development-occurrence hazard model exhibits even tighter seed variance than the filing-date petition model across all architectures, confirming that the stochastic stability property is a structural characteristic of the administrative feature space rather than an artifact of the opposition prediction task.

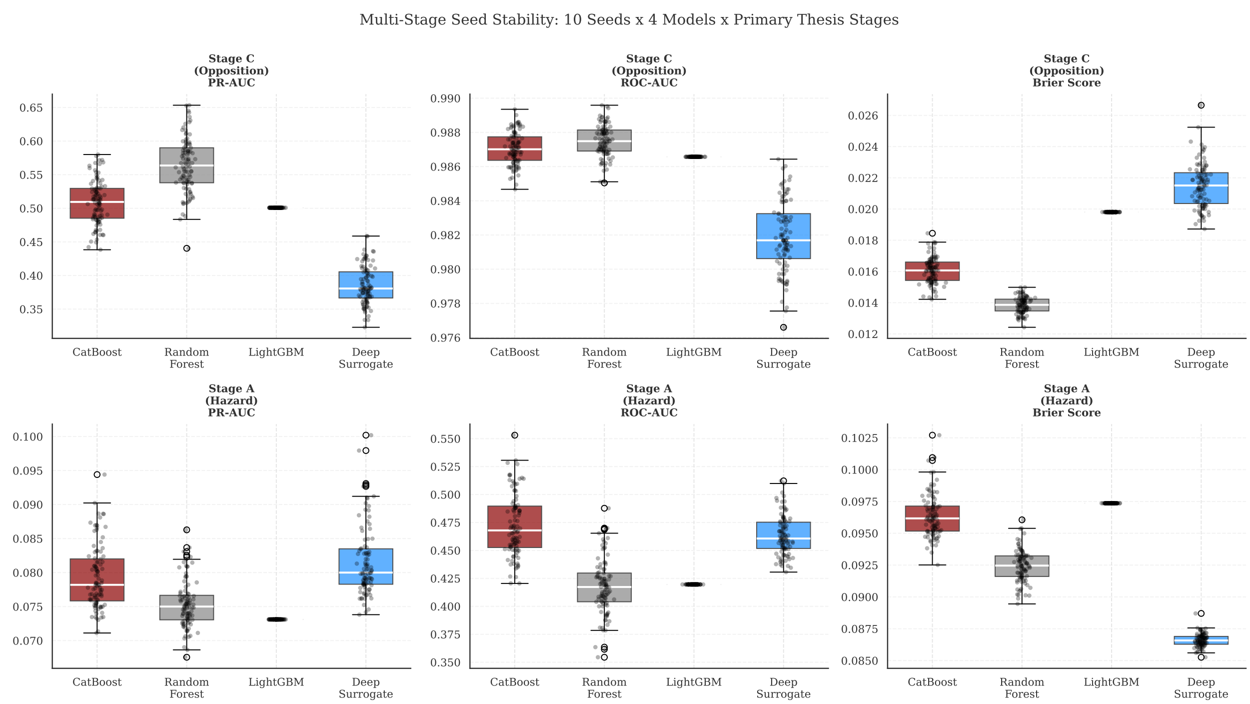

Figure 27: Cross-component multi-seed stability: PR-AUC, ROC-AUC, and Brier Score distributions across 10 seeds and 4 architectures for the development-occurrence and filing-date petition models. Narrow interquartile ranges across components indicate that reported metrics are robust to seed selection.

C.7 Exploratory Model Stability Audit

As a supplementary stability check, models were evaluated for **feature-rank stability**, the Spearman rank correlation (ρ) of internal feature importance rankings across adjacent temporal anchor years (2020 to 2022). This audit is exploratory and does not establish causal validity for any architecture.

Algorithm	**Rank Stability (ρ)**	**Top-5 Overlap (%)**
Deep V-REx	0.961	100.0%
CatBoost	0.927	90.0%
XGBoost	0.923	100.0%
TabNet	0.873	80.0%
LightGBM	0.848	90.0%
Deep ERM (MLP)	0.851	80.0%
Logistic Regression (L2)	0.831	100.0%
Random Forest	0.921	100.0%

Table 15: Feature-rank stability (Spearman ρ across adjacent anchor years) and top-5 overlap under the restricted feature set for all modeled architectures. This table is descriptive and should be interpreted as a robustness audit, not as a model-admission rule.

Table 15 complements the restricted-feature survival analysis by separating *rank persistence* from *cross-model consensus*. High rank stability alone does not imply that an architecture relies on the same semantic feature families as the broader roster. This appendix table is retained to document that distinction, while primary model selection remains bound to the pre-specified discrimination and calibration protocol described in Section 5.

C.8 Temporal Drift Threshold Sensitivity

This appendix replicates the primary temporal drift and structural stability exhibits across varying petition percentage thresholds ($> 0\%$, $> 5\%$, $> 10\%$, $> 15\%$, $> 20\%$, and $> 25\%$) to validate whether architectural dependence patterns isolate to the statutory cutoff or represent generalized predictive dynamics across all boundaries.

Algorithmic Predictability and Statutory Mobilization Constraints. Figure **??** visualizes the performance trajectory of the dominant architecture (Max-of-Family PR-AUC) from the resulting stability matrices across these six thresholds. The algorithmic sensitivity demonstrates a powerful structural mechanism governing formal neighbor mobilization. Specifically, predictability collapses towards baseline noise at low thresholds ($< 10\%$), where minimal signature barriers trace stochastic, localized friction rather than structured demographic capacity. Conversely, predictability becomes highly volatile and statistically hollow at $> 25\%$ as the positive class collapses into single digits.

Predictive dominance strictly peaks along the 15% to 20% structural constraint, empirically supporting the argument that formal petition generation is not an organic ambient phenomenon, but rather a coordinated, resourced, and institutional barricade engineered specifically to force the 3/4ths City Council supermajority vote requirement. The localized spatial models presented in this thesis successfully classify risk precisely because they map the *structural and spatial capacity* of a neighborhood to coordinate up to that exact legal veto boundary.

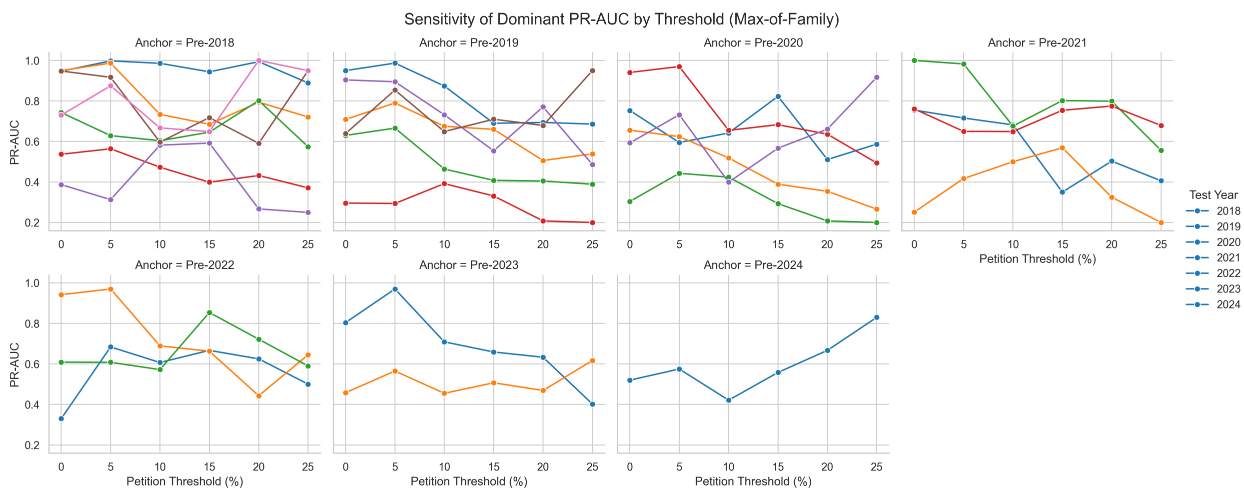

Figure 28: Algorithmic predictability (out-of-distribution PR-AUC) of the dominant architectural family across progressive petition area thresholds for all rolling origins. Performance traces a distinct inverted-U shape, peaking around the 15-20% statutory mark where opposition transitions from stochastic localized friction into structurally organized institutional mobilization.

Table 16: **Temporal predictive drift: PR-AUC decay by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	0.947	0.705	0.646	0.522	0.246	0.945	0.659
CatBoost	Pre-2019	—	0.645	0.697	**0.630**	**0.296**	0.863	0.605
CatBoost	Pre-2020	—	—	0.640	**0.656**	0.219	0.823	**0.593**
CatBoost	Pre-2021	—	—	—	**0.755**	0.220	**1.000**	**0.761**
CatBoost	Pre-2022	—	—	—	—	0.245	0.895	**0.609**
CatBoost	Pre-2023	—	—	—	—	—	**0.804**	0.444
CatBoost	Pre-2024	—	—	—	—	—	—	**0.520**
XGBoost	Pre-2018	0.912	0.849	0.617	**0.537**	0.277	**0.948**	0.634
XGBoost	Pre-2019	—	0.831	0.607	0.588	0.249	**0.904**	**0.639**
XGBoost	Pre-2020	—	—	0.637	0.640	**0.304**	**0.941**	0.559
XGBoost	Pre-2021	—	—	—	0.601	0.222	0.956	0.664
XGBoost	Pre-2022	—	—	—	—	**0.330**	**0.942**	0.525
XGBoost	Pre-2023	—	—	—	—	—	0.785	**0.458**
XGBoost	Pre-2024	—	—	—	—	—	—	0.420
Logistic (L2)	Pre-2018	**0.947**	**0.950**	0.552	0.521	0.260	0.741	0.544
Logistic (L2)	Pre-2019	—	**0.950**	0.567	0.526	0.246	0.764	0.533
Logistic (L2)	Pre-2020	—	—	0.549	0.519	0.251	0.749	0.544
Logistic (L2)	Pre-2021	—	—	—	0.521	**0.251**	0.714	0.533
Logistic (L2)	Pre-2022	—	—	—	—	0.277	0.728	0.500
Logistic (L2)	Pre-2023	—	—	—	—	—	0.730	0.340
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.362
Spatial-FE Logistic	Pre-2018	0.942	0.839	0.651	0.452	0.235	0.595	**0.730**
Spatial-FE Logistic	Pre-2019	—	0.849	**0.709**	0.488	0.224	0.641	0.511
Spatial-FE Logistic	Pre-2020	—	—	0.672	0.552	0.220	0.720	0.564
Spatial-FE Logistic	Pre-2021	—	—	—	0.532	0.206	0.728	0.559
Spatial-FE Logistic	Pre-2022	—	—	—	—	0.305	0.724	0.511
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	0.698	0.339
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	0.325
TabNet	Pre-2018	0.898	0.934	**0.743**	0.404	**0.387**	0.744	0.452
TabNet	Pre-2019	—	0.714	0.528	0.375	0.147	0.184	0.128
TabNet	Pre-2020	—	—	0.638	0.419	0.194	0.410	0.214
TabNet	Pre-2021	—	—	—	0.615	0.196	0.372	0.239
TabNet	Pre-2022	—	—	—	—	0.210	0.469	0.489
TabNet	Pre-2023	—	—	—	—	—	0.641	0.322
TabNet	Pre-2024	—	—	—	—	—	—	0.441
Anchor Regression (Causal)	Pre-2018	0.825	0.788	0.671	0.519	0.251	0.682	0.510
Anchor Regression (Causal)	Pre-2019	—	0.318	0.520	0.432	0.200	0.676	0.625
Anchor Regression (Causal)	Pre-2020	—	—	**0.753**	0.385	0.216	0.513	0.534
Anchor Regression (Causal)	Pre-2021	—	—	—	0.643	0.251	0.785	0.473
Anchor Regression (Causal)	Pre-2022	—	—	—	—	0.219	0.650	0.492
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	0.302	0.331
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	0.462

Table 17: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	6.218	5.919	8.167	7.977	10.611	18.998	28.349
CatBoost	Pre-2019	—	5.417	8.814	**9.630**	**12.771**	17.346	26.032
CatBoost	Pre-2020	—	—	8.097	**10.020**	9.448	16.538	**25.483**
CatBoost	Pre-2021	—	—	—	**11.545**	9.510	**20.100**	**32.728**
CatBoost	Pre-2022	—	—	—	—	10.591	17.998	**26.175**
CatBoost	Pre-2023	—	—	—	—	—	**16.157**	19.104
CatBoost	Pre-2024	—	—	—	—	—	—	**22.360**
XGBoost	Pre-2018	5.986	7.135	7.795	**8.206**	11.950	**19.065**	27.274
XGBoost	Pre-2019	—	6.976	7.675	8.985	10.770	**18.164**	**27.489**
XGBoost	Pre-2020	—	—	8.051	9.790	**13.147**	**18.907**	24.049
XGBoost	Pre-2021	—	—	—	9.186	9.604	19.212	28.571
XGBoost	Pre-2022	—	—	—	—	**14.254**	**18.937**	22.592
XGBoost	Pre-2023	—	—	—	—	—	15.774	**19.691**
XGBoost	Pre-2024	—	—	—	—	—	—	18.050
Logistic (L2)	Pre-2018	**6.220**	**7.976**	6.975	7.962	11.220	14.896	23.402
Logistic (L2)	Pre-2019	—	**7.981**	7.165	8.035	10.612	15.366	22.933
Logistic (L2)	Pre-2020	—	—	6.940	7.927	10.859	15.064	23.411
Logistic (L2)	Pre-2021	—	—	—	7.957	**10.850**	14.361	22.933
Logistic (L2)	Pre-2022	—	—	—	—	11.969	14.629	21.500
Logistic (L2)	Pre-2023	—	—	—	—	—	14.670	14.630
Logistic (L2)	Pre-2024	—	—	—	—	—	—	15.582
Spatial-FE Logistic	Pre-2018	6.187	7.049	8.227	6.903	10.132	11.950	**31.370**
Spatial-FE Logistic	Pre-2019	—	7.134	**8.968**	7.463	9.672	12.883	21.978
Spatial-FE Logistic	Pre-2020	—	—	8.495	8.438	9.519	14.475	24.271
Spatial-FE Logistic	Pre-2021	—	—	—	8.131	8.884	14.636	24.025
Spatial-FE Logistic	Pre-2022	—	—	—	—	13.182	14.554	21.978
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	14.039	14.593
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	13.985
TabNet	Pre-2018	5.895	7.842	**9.389**	6.182	**16.708**	14.954	19.448
TabNet	Pre-2019	—	5.995	6.672	5.735	6.361	3.694	5.522
TabNet	Pre-2020	—	—	8.067	6.402	8.387	8.244	9.222
TabNet	Pre-2021	—	—	—	9.401	8.451	7.471	10.257
TabNet	Pre-2022	—	—	—	—	9.059	9.427	21.022
TabNet	Pre-2023	—	—	—	—	—	12.890	13.849
TabNet	Pre-2024	—	—	—	—	—	—	18.968
Anchor Regression (Causal)	Pre-2018	5.418	6.621	8.489	7.926	10.834	13.705	21.920
Anchor Regression (Causal)	Pre-2019	—	2.670	6.571	6.603	8.650	13.582	26.875
Anchor Regression (Causal)	Pre-2020	—	—	**9.520**	5.880	9.334	10.304	22.975
Anchor Regression (Causal)	Pre-2021	—	—	—	9.833	10.846	15.784	20.344
Anchor Regression (Causal)	Pre-2022	—	—	—	—	9.457	13.065	21.159
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	6.072	14.229
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	19.873

Table 18: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	2018	2019	2020	2021	2022	2023	2024
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.947 (Linear)	0.950 (Linear)	0.743 (Deep)	0.537 (Trees)	0.387 (Deep)	0.948 (Trees)	0.730 (Linear)
Pre-2019	—	0.950 (Linear)	0.709 (Linear)	0.630 (Trees)	0.296 (Trees)	0.904 (Trees)	0.639 (Trees)
Pre-2020	—	—	0.753 (Causal)	0.656 (Trees)	0.304 (Trees)	0.941 (Trees)	0.593 (Trees)
Pre-2021	—	—	—	0.755 (Trees)	0.251 (Linear)	1.000 (Trees)	0.761 (Trees)
Pre-2022	—	—	—	—	0.330 (Trees)	0.942 (Trees)	0.609 (Trees)
Pre-2023	—	—	—	—	—	0.804 (Trees)	0.458 (Trees)
Pre-2024	—	—	—	—	—	—	0.520 (Trees)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.22 (Linear)	7.98 (Linear)	9.39 (Deep)	8.21 (Trees)	16.71 (Deep)	19.06 (Trees)	31.37 (Linear)
Pre-2019	—	7.98 (Linear)	8.97 (Linear)	9.63 (Trees)	12.77 (Trees)	18.16 (Trees)	27.49 (Trees)
Pre-2020	—	—	9.52 (Causal)	10.02 (Trees)	13.15 (Trees)	18.91 (Trees)	25.48 (Trees)
Pre-2021	—	—	—	11.55 (Trees)	10.85 (Linear)	20.10 (Trees)	32.73 (Trees)
Pre-2022	—	—	—	—	14.25 (Trees)	18.94 (Trees)	26.18 (Trees)
Pre-2023	—	—	—	—	—	16.16 (Trees)	19.69 (Trees)
Pre-2024	—	—	—	—	—	—	22.36 (Trees)

C.8.1 Threshold: >0%

C.8.2 Threshold: >5%

C.8.3 Threshold: >10%

C.8.4 Threshold: >15%

C.8.5 Threshold: >20%

C.8.6 Threshold: >25%

Table 19: **Temporal predictive drift: PR-AUC decay by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	0.956	0.912	0.568	**0.564**	**0.313**	0.832	**0.875**
CatBoost	Pre-2019	—	0.762	0.622	**0.666**	0.244	0.770	**0.854**
CatBoost	Pre-2020	—	—	0.507	**0.624**	0.305	0.831	**0.732**
CatBoost	Pre-2021	—	—	—	**0.716**	0.355	**0.983**	0.568
CatBoost	Pre-2022	—	—	—	—	**0.684**	0.950	0.385
CatBoost	Pre-2023	—	—	—	—	—	**0.970**	0.352
CatBoost	Pre-2024	—	—	—	—	—	—	0.568
XGBoost	Pre-2018	0.923	**0.987**	0.601	0.511	0.249	**0.917**	0.625
XGBoost	Pre-2019	—	0.915	0.583	0.520	0.244	**0.895**	0.668
XGBoost	Pre-2020	—	—	0.571	0.519	0.237	**0.970**	0.649
XGBoost	Pre-2021	—	—	—	0.635	0.242	0.912	**0.650**
XGBoost	Pre-2022	—	—	—	—	0.301	**0.970**	0.568
XGBoost	Pre-2023	—	—	—	—	—	0.909	0.390
XGBoost	Pre-2024	—	—	—	—	—	—	**0.575**
Logistic (L2)	Pre-2018	0.980	0.987	0.453	0.368	0.264	0.587	0.392
Logistic (L2)	Pre-2019	—	**0.987**	0.459	0.385	0.263	0.599	0.392
Logistic (L2)	Pre-2020	—	—	0.447	0.382	0.288	0.599	0.415
Logistic (L2)	Pre-2021	—	—	—	0.376	0.308	0.665	0.392
Logistic (L2)	Pre-2022	—	—	—	—	0.317	0.585	0.413
Logistic (L2)	Pre-2023	—	—	—	—	—	0.594	0.390
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.415
Spatial-FE Logistic	Pre-2018	**0.998**	0.959	**0.629**	0.439	0.268	0.555	0.548
Spatial-FE Logistic	Pre-2019	—	0.962	0.631	0.433	0.247	0.557	0.531
Spatial-FE Logistic	Pre-2020	—	—	0.568	0.394	0.258	0.562	0.540
Spatial-FE Logistic	Pre-2021	—	—	—	0.316	0.257	0.608	0.540
Spatial-FE Logistic	Pre-2022	—	—	—	—	0.339	0.613	0.540
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	0.602	0.526
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	0.552
TabNet	Pre-2018	0.939	0.613	0.426	0.364	0.117	0.154	0.090
TabNet	Pre-2019	—	0.754	**0.790**	0.371	0.218	0.534	0.282
TabNet	Pre-2020	—	—	**0.595**	0.424	**0.443**	0.708	0.582
TabNet	Pre-2021	—	—	—	0.594	0.378	0.722	0.338
TabNet	Pre-2022	—	—	—	—	0.231	0.746	**0.608**
TabNet	Pre-2023	—	—	—	—	—	0.549	**0.565**
TabNet	Pre-2024	—	—	—	—	—	—	0.510
Anchor Regression (Causal)	Pre-2018	0.820	0.832	0.421	0.328	0.281	0.636	0.531
Anchor Regression (Causal)	Pre-2019	—	0.582	0.542	0.339	**0.294**	0.604	0.436
Anchor Regression (Causal)	Pre-2020	—	—	0.525	0.380	0.265	0.662	0.392
Anchor Regression (Causal)	Pre-2021	—	—	—	0.409	**0.417**	0.605	0.514
Anchor Regression (Causal)	Pre-2022	—	—	—	—	0.350	0.506	0.261
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	0.696	0.500
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	0.472

Table 20: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	6.278	7.660	8.382	**10.058**	**16.907**	16.727	**47.031**
CatBoost	Pre-2019	—	6.397	9.175	**11.871**	13.164	15.484	**45.911**
CatBoost	Pre-2020	—	—	7.484	**11.124**	16.459	16.696	**39.353**
CatBoost	Pre-2021	—	—	—	**12.765**	19.157	**19.765**	30.544
CatBoost	Pre-2022	—	—	—	—	**36.940**	19.089	20.689
CatBoost	Pre-2023	—	—	—	—	—	**19.493**	18.898
CatBoost	Pre-2024	—	—	—	—	—	—	30.544
XGBoost	Pre-2018	6.061	**8.292**	8.872	9.117	13.432	**18.422**	33.594
XGBoost	Pre-2019	—	7.687	8.593	9.276	13.154	**17.998**	35.897
XGBoost	Pre-2020	—	—	8.429	9.255	12.779	**19.493**	34.874
XGBoost	Pre-2021	—	—	—	11.325	13.048	18.334	**34.938**
XGBoost	Pre-2022	—	—	—	—	16.232	**19.493**	30.544
XGBoost	Pre-2023	—	—	—	—	—	18.261	20.988
XGBoost	Pre-2024	—	—	—	—	—	—	**30.906**
Logistic (L2)	Pre-2018	6.436	**8.292**	6.689	6.556	14.250	11.798	21.052
Logistic (L2)	Pre-2019	—	**8.292**	6.764	6.862	14.175	12.036	21.052
Logistic (L2)	Pre-2020	—	—	6.595	6.812	15.557	12.036	22.332
Logistic (L2)	Pre-2021	—	—	—	6.713	16.609	13.376	21.052
Logistic (L2)	Pre-2022	—	—	—	—	17.132	11.752	22.209
Logistic (L2)	Pre-2023	—	—	—	—	—	11.940	20.972
Logistic (L2)	Pre-2024	—	—	—	—	—	—	22.332
Spatial-FE Logistic	Pre-2018	**6.553**	8.058	**9.282**	7.825	14.483	11.147	29.458
Spatial-FE Logistic	Pre-2019	—	8.079	9.309	7.725	13.358	11.205	28.562
Spatial-FE Logistic	Pre-2020	—	—	8.378	7.026	13.911	11.305	29.051
Spatial-FE Logistic	Pre-2021	—	—	—	5.627	13.904	12.228	29.051
Spatial-FE Logistic	Pre-2022	—	—	—	—	18.300	12.311	29.051
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	12.091	28.288
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	29.648
TabNet	Pre-2018	6.164	5.147	6.288	6.493	6.335	3.099	4.826
TabNet	Pre-2019	—	6.334	**11.652**	6.621	11.794	10.729	15.143
TabNet	Pre-2020	—	—	**8.778**	7.563	**23.936**	14.222	31.290
TabNet	Pre-2021	—	—	—	10.585	20.411	14.522	18.173
TabNet	Pre-2022	—	—	—	—	12.482	14.993	**32.698**
TabNet	Pre-2023	—	—	—	—	—	11.029	**30.394**
TabNet	Pre-2024	—	—	—	—	—	—	27.408
Anchor Regression (Causal)	Pre-2018	5.388	6.991	6.215	5.853	15.199	12.783	28.539
Anchor Regression (Causal)	Pre-2019	—	4.893	7.994	6.038	**15.854**	12.147	23.441
Anchor Regression (Causal)	Pre-2020	—	—	7.748	6.780	14.308	13.316	21.089
Anchor Regression (Causal)	Pre-2021	—	—	—	7.286	**22.500**	12.162	27.622
Anchor Regression (Causal)	Pre-2022	—	—	—	—	18.900	10.174	14.023
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	13.991	26.875
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	25.382

Table 21: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	2018	2019	2020	2021	2022	2023	2024
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.998 (Linear)	0.987 (Trees)	0.629 (Linear)	0.564 (Trees)	0.313 (Trees)	0.917 (Trees)	0.875 (Trees)
Pre-2019	—	0.987 (Linear)	0.790 (Deep)	0.666 (Trees)	0.294 (Causal)	0.895 (Trees)	0.854 (Trees)
Pre-2020	—	—	0.595 (Deep)	0.624 (Trees)	0.443 (Deep)	0.970 (Trees)	0.732 (Trees)
Pre-2021	—	—	—	0.716 (Trees)	0.417 (Causal)	0.983 (Trees)	0.650 (Trees)
Pre-2022	—	—	—	—	0.684 (Trees)	0.970 (Trees)	0.608 (Deep)
Pre-2023	—	—	—	—	—	0.970 (Trees)	0.565 (Deep)
Pre-2024	—	—	—	—	—	—	0.575 (Trees)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.55 (Linear)	8.29 (Linear)	9.28 (Linear)	10.06 (Trees)	16.91 (Trees)	18.42 (Trees)	47.03 (Trees)
Pre-2019	—	8.29 (Linear)	11.65 (Deep)	11.87 (Trees)	15.85 (Causal)	18.00 (Trees)	45.91 (Trees)
Pre-2020	—	—	8.78 (Deep)	11.12 (Trees)	23.94 (Deep)	19.49 (Trees)	39.35 (Trees)
Pre-2021	—	—	—	12.76 (Trees)	22.50 (Causal)	19.76 (Trees)	34.94 (Trees)
Pre-2022	—	—	—	—	36.94 (Trees)	19.49 (Trees)	32.70 (Deep)
Pre-2023	—	—	—	—	—	19.49 (Trees)	30.39 (Deep)
Pre-2024	—	—	—	—	—	—	30.91 (Trees)

Table 22: **Temporal predictive drift: PR-AUC decay by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	0.929	0.524	0.560	0.416	0.317	0.594	0.622
CatBoost	Pre-2019	—	0.494	0.605	0.440	0.330	0.728	0.580
CatBoost	Pre-2020	—	—	0.510	0.385	0.304	0.524	0.367
CatBoost	Pre-2021	—	—	—	**0.682**	**0.500**	**0.676**	0.339
CatBoost	Pre-2022	—	—	—	—	**0.607**	**0.689**	0.339
CatBoost	Pre-2023	—	—	—	—	—	**0.709**	0.360
CatBoost	Pre-2024	—	—	—	—	—	—	0.360
XGBoost	Pre-2018	0.909	0.609	0.499	0.419	**0.582**	**0.597**	0.565
XGBoost	Pre-2019	—	0.461	0.560	**0.464**	0.317	0.583	**0.649**
XGBoost	Pre-2020	—	—	**0.642**	0.436	**0.424**	0.479	0.338
XGBoost	Pre-2021	—	—	—	0.432	0.320	0.518	**0.649**
XGBoost	Pre-2022	—	—	—	—	0.424	0.628	0.371
XGBoost	Pre-2023	—	—	—	—	—	0.609	0.419
XGBoost	Pre-2024	—	—	—	—	—	—	**0.421**
Logistic (L2)	Pre-2018	0.952	0.687	0.453	0.374	0.330	0.453	0.399
Logistic (L2)	Pre-2019	—	0.654	0.466	0.402	0.330	0.519	0.365
Logistic (L2)	Pre-2020	—	—	0.491	0.327	0.280	0.463	0.395
Logistic (L2)	Pre-2021	—	—	—	0.382	0.306	0.484	0.364
Logistic (L2)	Pre-2022	—	—	—	—	0.412	0.472	0.365
Logistic (L2)	Pre-2023	—	—	—	—	—	0.470	0.360
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.340
Spatial-FE Logistic	Pre-2018	0.802	**0.734**	0.510	0.330	0.244	0.452	**0.667**
Spatial-FE Logistic	Pre-2019	—	0.704	0.578	0.319	0.213	0.498	0.333
Spatial-FE Logistic	Pre-2020	—	—	0.565	0.290	0.221	0.530	**0.399**
Spatial-FE Logistic	Pre-2021	—	—	—	0.311	0.223	0.558	0.374
Spatial-FE Logistic	Pre-2022	—	—	—	—	0.225	0.462	0.349
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	0.443	0.380
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	0.360
TabNet	Pre-2018	0.943	0.558	**0.604**	0.446	0.217	0.452	0.317
TabNet	Pre-2019	—	**0.874**	0.536	0.353	0.250	**0.732**	0.380
TabNet	Pre-2020	—	—	0.618	**0.519**	0.127	**0.655**	0.155
TabNet	Pre-2021	—	—	—	0.365	0.264	0.501	0.232
TabNet	Pre-2022	—	—	—	—	0.066	0.070	0.041
TabNet	Pre-2023	—	—	—	—	—	0.413	0.364
TabNet	Pre-2024	—	—	—	—	—	—	0.378
Anchor Regression (Causal)	Pre-2018	**0.986**	0.398	0.191	**0.473**	0.152	0.177	0.063
Anchor Regression (Causal)	Pre-2019	—	0.586	**0.675**	0.362	**0.392**	0.394	0.296
Anchor Regression (Causal)	Pre-2020	—	—	0.472	0.322	0.329	0.470	0.368
Anchor Regression (Causal)	Pre-2021	—	—	—	0.451	0.288	0.346	0.554
Anchor Regression (Causal)	Pre-2022	—	—	—	—	0.281	0.243	**0.572**
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	0.446	**0.455**
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	0.400

Table 23: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	6.103	6.468	8.263	7.415	17.140	17.063	33.407
CatBoost	Pre-2019	—	6.103	8.925	7.849	17.807	20.896	31.168
CatBoost	Pre-2020	—	—	7.527	6.866	16.416	15.039	19.746
CatBoost	Pre-2021	—	—	—	**12.165**	**27.000**	**19.416**	18.210
CatBoost	Pre-2022	—	—	—	—	**32.786**	**19.794**	18.210
CatBoost	Pre-2023	—	—	—	—	—	**20.350**	19.330
CatBoost	Pre-2024	—	—	—	—	—	—	19.330
XGBoost	Pre-2018	5.971	7.525	7.364	7.471	**31.436**	**17.136**	30.394
XGBoost	Pre-2019	—	5.695	8.262	**8.280**	17.091	16.745	**34.874**
XGBoost	Pre-2020	—	—	**9.470**	7.782	**22.909**	13.766	18.178
XGBoost	Pre-2021	—	—	—	7.712	17.284	14.880	**34.874**
XGBoost	Pre-2022	—	—	—	—	22.909	18.039	19.924
XGBoost	Pre-2023	—	—	—	—	—	17.488	22.545
XGBoost	Pre-2024	—	—	—	—	—	—	**22.652**
Logistic (L2)	Pre-2018	6.252	8.489	6.686	6.667	17.847	13.014	21.436
Logistic (L2)	Pre-2019	—	8.079	6.874	7.164	17.815	14.910	19.644
Logistic (L2)	Pre-2020	—	—	7.235	5.829	15.107	13.307	21.212
Logistic (L2)	Pre-2021	—	—	—	6.805	16.529	13.885	19.570
Logistic (L2)	Pre-2022	—	—	—	—	22.275	13.559	19.644
Logistic (L2)	Pre-2023	—	—	—	—	—	13.493	19.372
Logistic (L2)	Pre-2024	—	—	—	—	—	—	18.301
Spatial-FE Logistic	Pre-2018	5.265	**9.072**	7.529	5.878	13.154	12.983	**35.833**
Spatial-FE Logistic	Pre-2019	—	8.697	8.532	5.696	11.483	14.309	17.917
Spatial-FE Logistic	Pre-2020	—	—	8.340	5.178	11.916	15.205	**21.436**
Spatial-FE Logistic	Pre-2021	—	—	—	5.540	12.036	16.012	20.076
Spatial-FE Logistic	Pre-2022	—	—	—	—	12.157	13.277	18.733
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	12.720	20.412
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	19.330
TabNet	Pre-2018	6.191	6.895	**8.910**	7.958	11.700	12.989	17.053
TabNet	Pre-2019	—	**10.791**	7.910	6.299	13.479	**21.025**	20.412
TabNet	Pre-2020	—	—	9.120	**9.250**	6.843	**18.807**	8.339
TabNet	Pre-2021	—	—	—	6.501	14.279	14.400	12.456
TabNet	Pre-2022	—	—	—	—	3.545	2.016	2.199
TabNet	Pre-2023	—	—	—	—	—	11.856	19.570
TabNet	Pre-2024	—	—	—	—	—	—	20.316
Anchor Regression (Causal)	Pre-2018	**6.472**	4.921	2.816	**8.429**	8.214	5.078	3.411
Anchor Regression (Causal)	Pre-2019	—	7.244	**9.956**	6.460	**21.176**	11.325	15.901
Anchor Regression (Causal)	Pre-2020	—	—	6.964	5.742	17.775	13.487	19.783
Anchor Regression (Causal)	Pre-2021	—	—	—	8.049	15.540	9.949	29.754
Anchor Regression (Causal)	Pre-2022	—	—	—	—	15.162	6.992	**30.757**
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	12.820	**24.449**
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	21.500

Table 24: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	2018	2019	2020	2021	2022	2023	2024
			Panel A: Maximum Absolute PR-AUC				
Pre-2018	0.986 (Causal)	0.734 (Linear)	0.604 (Deep)	0.473 (Causal)	0.582 (Trees)	0.597 (Trees)	0.667 (Linear)
Pre-2019	—	0.874 (Deep)	0.675 (Causal)	0.464 (Trees)	0.392 (Causal)	0.732 (Deep)	0.649 (Trees)
Pre-2020	—	—	0.642 (Trees)	0.519 (Deep)	0.424 (Trees)	0.655 (Deep)	0.399 (Linear)
Pre-2021	—	—	—	0.682 (Trees)	0.500 (Trees)	0.676 (Trees)	0.649 (Trees)
Pre-2022	—	—	—	—	0.607 (Trees)	0.689 (Trees)	0.572 (Causal)
Pre-2023	—	—	—	—	—	0.709 (Trees)	0.455 (Causal)
Pre-2024	—	—	—	—	—	—	0.421 (Trees)
			Panel B: Maximum Relative PR-AUC Lift				
Pre-2018	6.47 (Causal)	9.07 (Linear)	8.91 (Deep)	8.43 (Causal)	31.44 (Trees)	17.14 (Trees)	35.83 (Linear)
Pre-2019	—	10.79 (Deep)	9.96 (Causal)	8.28 (Trees)	21.18 (Causal)	21.03 (Deep)	34.87 (Trees)
Pre-2020	—	—	9.47 (Trees)	9.25 (Deep)	22.91 (Trees)	18.81 (Deep)	21.44 (Linear)
Pre-2021	—	—	—	12.16 (Trees)	27.00 (Trees)	19.42 (Trees)	34.87 (Trees)
Pre-2022	—	—	—	—	32.79 (Trees)	19.79 (Trees)	30.76 (Causal)
Pre-2023	—	—	—	—	—	20.35 (Trees)	24.45 (Causal)
Pre-2024	—	—	—	—	—	—	22.65 (Trees)

Table 25: **Temporal predictive drift: PR-AUC decay by algorithm (Optimal Entropy-Based Discretization).**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	0.937	0.433	0.642	**0.399**	**0.592**	**0.718**	**0.649**
CatBoost	Pre-2019	—	0.505	0.576	0.340	0.222	**0.554**	**0.710**
CatBoost	Pre-2020	—	—	0.656	0.331	0.262	0.525	0.395
CatBoost	Pre-2021	—	—	—	0.299	**0.569**	**0.802**	0.747
CatBoost	Pre-2022	—	—	—	—	0.508	**0.663**	0.401
CatBoost	Pre-2023	—	—	—	—	—	0.593	0.338
CatBoost	Pre-2024	—	—	—	—	—	—	0.401
XGBoost	Pre-2018	0.923	**0.685**	0.560	0.350	0.387	0.595	0.611
XGBoost	Pre-2019	—	0.467	**0.660**	0.325	**0.331**	0.481	0.604
XGBoost	Pre-2020	—	—	**0.823**	0.327	0.253	**0.683**	**0.567**
XGBoost	Pre-2021	—	—	—	0.290	0.285	0.431	0.560
XGBoost	Pre-2022	—	—	—	—	0.387	0.627	0.553
XGBoost	Pre-2023	—	—	—	—	—	0.436	0.404
XGBoost	Pre-2024	—	—	—	—	—	—	**0.558**
Logistic (L2)	Pre-2018	**0.944**	0.672	0.557	0.294	0.288	0.376	0.387
Logistic (L2)	Pre-2019	—	0.671	0.542	0.351	0.249	0.381	0.338
Logistic (L2)	Pre-2020	—	—	0.535	0.289	0.193	0.343	0.352
Logistic (L2)	Pre-2021	—	—	—	0.272	0.208	0.363	0.318
Logistic (L2)	Pre-2022	—	—	—	—	0.266	0.358	0.318
Logistic (L2)	Pre-2023	—	—	—	—	—	0.341	0.296
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.356
Spatial-FE Logistic	Pre-2018	0.714	0.681	0.507	0.287	0.283	0.479	0.622
Spatial-FE Logistic	Pre-2019	—	**0.690**	0.543	**0.408**	0.305	0.453	0.388
Spatial-FE Logistic	Pre-2020	—	—	0.552	0.296	0.194	0.452	0.349
Spatial-FE Logistic	Pre-2021	—	—	—	0.278	0.205	0.580	0.339
Spatial-FE Logistic	Pre-2022	—	—	—	—	0.261	0.429	0.327
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	**0.659**	0.326
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	0.401
TabNet	Pre-2018	0.885	0.636	**0.646**	0.237	0.261	0.307	0.458
TabNet	Pre-2019	—	0.524	0.652	0.394	0.200	0.167	0.179
TabNet	Pre-2020	—	—	0.679	**0.389**	0.258	0.348	0.371
TabNet	Pre-2021	—	—	—	**0.350**	0.525	0.342	**0.754**
TabNet	Pre-2022	—	—	—	—	**0.667**	0.392	**0.854**
TabNet	Pre-2023	—	—	—	—	—	0.255	**0.507**
TabNet	Pre-2024	—	—	—	—	—	—	0.399
Anchor Regression (Causal)	Pre-2018	0.912	0.560	0.499	0.286	0.306	0.494	0.432
Anchor Regression (Causal)	Pre-2019	—	0.627	0.603	0.308	0.230	0.372	0.450
Anchor Regression (Causal)	Pre-2020	—	—	0.402	0.318	**0.293**	0.258	0.499
Anchor Regression (Causal)	Pre-2021	—	—	—	0.328	0.278	0.277	0.536
Anchor Regression (Causal)	Pre-2022	—	—	—	—	0.250	0.428	0.500
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	0.182	0.405
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	0.365

Table 26: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	6.151	6.493	9.474	**8.530**	**42.600**	**24.055**	**34.874**
CatBoost	Pre-2019	—	7.571	8.503	7.271	16.000	**18.576**	**38.137**
CatBoost	Pre-2020	—	—	9.679	7.078	18.857	17.592	21.212
CatBoost	Pre-2021	—	—	—	6.407	**41.000**	**26.854**	40.153
CatBoost	Pre-2022	—	—	—	—	36.545	**22.202**	21.569
CatBoost	Pre-2023	—	—	—	—	—	19.852	18.178
CatBoost	Pre-2024	—	—	—	—	—	—	21.569
XGBoost	Pre-2018	6.064	**10.271**	8.253	7.493	27.886	19.947	32.847
XGBoost	Pre-2019	—	7.012	**9.740**	6.947	**23.857**	16.122	32.474
XGBoost	Pre-2020	—	—	**12.141**	6.994	18.203	**22.866**	**30.458**
XGBoost	Pre-2021	—	—	—	6.203	20.545	14.434	30.074
XGBoost	Pre-2022	—	—	—	—	27.857	21.005	29.712
XGBoost	Pre-2023	—	—	—	—	—	14.617	21.692
XGBoost	Pre-2024	—	—	—	—	—	—	**30.010**
Logistic (L2)	Pre-2018	**6.200**	10.075	8.216	6.301	20.743	12.593	20.828
Logistic (L2)	Pre-2019	—	10.068	7.999	7.507	17.943	12.772	18.178
Logistic (L2)	Pre-2020	—	—	7.885	6.193	13.872	11.482	18.898
Logistic (L2)	Pre-2021	—	—	—	5.829	14.967	12.154	17.090
Logistic (L2)	Pre-2022	—	—	—	—	19.143	12.004	17.090
Logistic (L2)	Pre-2023	—	—	—	—	—	11.409	15.933
Logistic (L2)	Pre-2024	—	—	—	—	—	—	19.148
Spatial-FE Logistic	Pre-2018	4.692	10.219	7.473	6.145	20.396	16.030	33.434
Spatial-FE Logistic	Pre-2019	—	**10.354**	8.006	**8.726**	21.943	15.178	20.865
Spatial-FE Logistic	Pre-2020	—	—	8.146	6.328	13.943	15.149	18.733
Spatial-FE Logistic	Pre-2021	—	—	—	5.942	14.743	19.429	18.210
Spatial-FE Logistic	Pre-2022	—	—	—	—	18.800	14.387	17.581
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	**22.077**	17.538
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	21.569
TabNet	Pre-2018	5.812	9.542	**9.535**	5.072	18.800	10.295	24.635
TabNet	Pre-2019	—	7.858	9.612	8.430	14.369	5.596	9.630
TabNet	Pre-2020	—	—	10.019	**8.317**	18.545	11.659	19.943
TabNet	Pre-2021	—	—	—	**7.483**	37.835	11.461	**40.536**
TabNet	Pre-2022	—	—	—	—	**48.000**	13.148	**45.911**
TabNet	Pre-2023	—	—	—	—	—	8.551	**27.259**
TabNet	Pre-2024	—	—	—	—	—	—	21.425
Anchor Regression (Causal)	Pre-2018	5.986	8.407	7.355	6.111	22.000	16.535	23.228
Anchor Regression (Causal)	Pre-2019	—	9.403	8.895	6.589	16.533	12.448	24.188
Anchor Regression (Causal)	Pre-2020	—	—	5.934	6.795	**21.091**	8.645	26.795
Anchor Regression (Causal)	Pre-2021	—	—	—	7.019	20.000	9.267	28.816
Anchor Regression (Causal)	Pre-2022	—	—	—	—	18.000	14.331	26.875
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	6.083	21.750
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	19.623

Table 27: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	2018	2019	2020	2021	2022	2023	2024
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.944 (Linear)	0.685 (Trees)	0.646 (Deep)	0.399 (Trees)	0.592 (Trees)	0.718 (Trees)	0.649 (Trees)
Pre-2019	—	0.690 (Linear)	0.660 (Trees)	0.408 (Linear)	0.331 (Trees)	0.554 (Trees)	0.710 (Trees)
Pre-2020	—	—	0.823 (Trees)	0.389 (Deep)	0.293 (Causal)	0.683 (Trees)	0.567 (Trees)
Pre-2021	—	—	—	0.350 (Deep)	0.569 (Trees)	0.802 (Trees)	0.754 (Deep)
Pre-2022	—	—	—	—	0.667 (Deep)	0.663 (Trees)	0.854 (Deep)
Pre-2023	—	—	—	—	—	0.659 (Linear)	0.507 (Deep)
Pre-2024	—	—	—	—	—	—	0.558 (Trees)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.20 (Linear)	10.27 (Trees)	9.54 (Deep)	8.53 (Trees)	42.60 (Trees)	24.05 (Trees)	34.87 (Trees)
Pre-2019	—	10.35 (Linear)	9.74 (Trees)	8.73 (Linear)	23.86 (Trees)	18.58 (Trees)	38.14 (Trees)
Pre-2020	—	—	12.14 (Trees)	8.32 (Deep)	21.09 (Causal)	22.87 (Trees)	30.46 (Trees)
Pre-2021	—	—	—	7.48 (Deep)	41.00 (Trees)	26.85 (Trees)	40.54 (Deep)
Pre-2022	—	—	—	—	48.00 (Deep)	22.20 (Trees)	45.91 (Deep)
Pre-2023	—	—	—	—	—	22.08 (Linear)	27.26 (Deep)
Pre-2024	—	—	—	—	—	—	30.01 (Trees)

Table 28: **Temporal predictive drift: PR-AUC decay by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	0.960	**0.794**	0.472	0.324	0.258	**0.591**	0.622
CatBoost	Pre-2019	—	0.524	0.481	**0.405**	0.143	**0.771**	0.395
CatBoost	Pre-2020	—	—	0.503	**0.354**	**0.208**	**0.636**	0.430
CatBoost	Pre-2021	—	—	—	0.467	**0.325**	**0.799**	0.415
CatBoost	Pre-2022	—	—	—	—	**0.625**	0.367	0.326
CatBoost	Pre-2023	—	—	—	—	—	**0.633**	0.443
CatBoost	Pre-2024	—	—	—	—	—	—	0.586
XGBoost	Pre-2018	0.926	0.553	0.467	0.341	0.208	0.485	0.579
XGBoost	Pre-2019	—	0.397	0.399	0.368	**0.208**	0.368	**0.679**
XGBoost	Pre-2020	—	—	**0.511**	0.276	0.196	0.543	0.458
XGBoost	Pre-2021	—	—	—	0.402	0.267	0.618	0.679
XGBoost	Pre-2022	—	—	—	—	0.250	**0.443**	0.679
XGBoost	Pre-2023	—	—	—	—	—	0.525	0.350
XGBoost	Pre-2024	—	—	—	—	—	—	0.442
Logistic (L2)	Pre-2018	0.852	0.700	0.413	0.236	0.139	0.277	0.771
Logistic (L2)	Pre-2019	—	0.667	0.358	0.280	0.132	0.244	0.360
Logistic (L2)	Pre-2020	—	—	0.347	0.281	0.133	0.302	0.367
Logistic (L2)	Pre-2021	—	—	—	0.264	0.162	0.258	0.278
Logistic (L2)	Pre-2022	—	—	—	—	0.202	0.348	0.296
Logistic (L2)	Pre-2023	—	—	—	—	—	0.360	0.320
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.305
Spatial-FE Logistic	Pre-2018	0.642	0.603	0.387	0.296	0.113	0.362	**1.000**
Spatial-FE Logistic	Pre-2019	—	**0.694**	0.403	0.341	0.127	0.416	0.374
Spatial-FE Logistic	Pre-2020	—	—	0.440	0.334	0.110	0.420	0.379
Spatial-FE Logistic	Pre-2021	—	—	—	0.285	0.134	0.304	0.305
Spatial-FE Logistic	Pre-2022	—	—	—	—	0.138	0.342	0.287
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	0.471	0.298
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	0.306
TabNet	Pre-2018	0.990	0.399	**0.802**	0.226	**0.267**	0.295	0.380
TabNet	Pre-2019	—	0.364	**0.506**	0.373	0.174	0.273	0.351
TabNet	Pre-2020	—	—	0.309	0.235	0.167	0.173	0.170
TabNet	Pre-2021	—	—	—	**0.503**	0.244	0.454	**0.775**
TabNet	Pre-2022	—	—	—	—	0.216	0.349	0.461
TabNet	Pre-2023	—	—	—	—	—	0.254	0.390
TabNet	Pre-2024	—	—	—	—	—	—	**0.667**
Anchor Regression (Causal)	Pre-2018	**0.994**	0.452	0.500	**0.432**	0.133	0.343	0.525
Anchor Regression (Causal)	Pre-2019	—	0.539	0.500	0.314	0.171	0.237	0.406
Anchor Regression (Causal)	Pre-2020	—	—	0.273	0.331	**0.208**	0.230	**0.661**
Anchor Regression (Causal)	Pre-2021	—	—	—	0.331	0.267	0.303	0.575
Anchor Regression (Causal)	Pre-2022	—	—	—	—	0.155	0.427	**0.722**
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	0.356	**0.469**
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	0.557

Table 29: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	Anchor Training	2018	2019	2020	2021	2022	2023	2024
CatBoost	Pre-2018	6.307	**11.904**	8.346	6.944	27.818	**29.693**	33.407
CatBoost	Pre-2019	—	7.861	8.512	**8.671**	15.429	**38.734**	21.212
CatBoost	Pre-2020	—	—	8.910	**7.579**	**22.500**	**31.965**	23.132
CatBoost	Pre-2021	—	—	—	10.000	**35.100**	**40.130**	22.332
CatBoost	Pre-2022	—	—	—	—	**67.500**	18.460	17.538
CatBoost	Pre-2023	—	—	—	—	—	**31.825**	23.825
CatBoost	Pre-2024	—	—	—	—	—	—	31.503
XGBoost	Pre-2018	6.084	8.301	8.267	7.302	22.500	24.384	31.130
XGBoost	Pre-2019	—	5.961	7.069	7.871	**22.500**	18.497	**36.505**
XGBoost	Pre-2020	—	—	**9.053**	5.909	21.214	27.279	24.635
XGBoost	Pre-2021	—	—	—	8.613	28.800	31.047	36.505
XGBoost	Pre-2022	—	—	—	—	27.000	**22.270**	36.505
XGBoost	Pre-2023	—	—	—	—	—	26.381	18.839
XGBoost	Pre-2024	—	—	—	—	—	—	23.740
Logistic (L2)	Pre-2018	5.595	10.494	7.317	5.040	15.058	13.895	41.432
Logistic (L2)	Pre-2019	—	10.001	6.345	5.997	14.308	12.240	19.330
Logistic (L2)	Pre-2020	—	—	6.149	6.003	14.400	15.172	19.708
Logistic (L2)	Pre-2021	—	—	—	5.648	17.532	12.945	14.964
Logistic (L2)	Pre-2022	—	—	—	—	21.808	17.475	15.901
Logistic (L2)	Pre-2023	—	—	—	—	—	18.082	17.204
Logistic (L2)	Pre-2024	—	—	—	—	—	—	16.381
Spatial-FE Logistic	Pre-2018	4.218	9.048	6.859	6.338	12.214	18.184	**53.750**
Spatial-FE Logistic	Pre-2019	—	**10.412**	7.139	7.306	13.708	20.905	20.116
Spatial-FE Logistic	Pre-2020	—	—	7.794	7.138	11.868	21.117	20.372
Spatial-FE Logistic	Pre-2021	—	—	—	6.092	14.464	15.282	16.381
Spatial-FE Logistic	Pre-2022	—	—	—	—	14.914	17.199	15.444
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	23.689	16.004
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	16.452
TabNet	Pre-2018	6.498	5.979	**14.191**	4.837	**28.800**	14.814	20.409
TabNet	Pre-2019	—	5.458	**8.952**	7.985	18.750	13.706	18.850
TabNet	Pre-2020	—	—	5.471	5.037	18.000	8.687	9.113
TabNet	Pre-2021	—	—	—	**10.758**	26.308	22.792	**41.656**
TabNet	Pre-2022	—	—	—	—	23.318	17.513	24.785
TabNet	Pre-2023	—	—	—	—	—	12.753	20.988
TabNet	Pre-2024	—	—	—	—	—	—	**35.833**
Anchor Regression (Causal)	Pre-2018	**6.526**	6.774	8.850	**9.241**	14.400	17.229	28.219
Anchor Regression (Causal)	Pre-2019	—	8.087	8.850	6.718	18.514	11.934	21.799
Anchor Regression (Causal)	Pre-2020	—	—	4.832	7.088	**22.500**	11.556	**35.513**
Anchor Regression (Causal)	Pre-2021	—	—	—	7.092	28.800	15.205	30.906
Anchor Regression (Causal)	Pre-2022	—	—	—	—	16.714	21.461	**38.819**
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	17.877	**25.233**
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	29.946

Table 30: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Panel A: Maximum Absolute PR-AUC							
Pre-2018	0.994 (Causal)	0.794 (Trees)	0.802 (Deep)	0.432 (Causal)	0.267 (Deep)	0.591 (Trees)	1.000 (Linear)
Pre-2019	—	0.694 (Linear)	0.506 (Deep)	0.405 (Trees)	0.208 (Trees)	0.771 (Trees)	0.679 (Trees)
Pre-2020	—	—	0.511 (Trees)	0.354 (Trees)	0.208 (Causal)	0.636 (Trees)	0.661 (Causal)
Pre-2021	—	—	—	0.503 (Deep)	0.325 (Trees)	0.799 (Trees)	0.775 (Deep)
Pre-2022	—	—	—	—	0.625 (Trees)	0.443 (Trees)	0.722 (Causal)
Pre-2023	—	—	—	—	—	0.633 (Trees)	0.469 (Causal)
Pre-2024	—	—	—	—	—	—	0.667 (Deep)
Panel B: Maximum Relative PR-AUC Lift							
Pre-2018	6.53 (Causal)	11.90 (Trees)	14.19 (Deep)	9.24 (Causal)	28.80 (Deep)	29.69 (Trees)	53.75 (Linear)
Pre-2019	—	10.41 (Linear)	8.95 (Deep)	8.67 (Trees)	22.50 (Trees)	38.73 (Trees)	36.51 (Trees)
Pre-2020	—	—	9.05 (Trees)	7.58 (Trees)	22.50 (Causal)	31.96 (Trees)	35.51 (Causal)
Pre-2021	—	—	—	10.76 (Deep)	35.10 (Trees)	40.13 (Trees)	41.66 (Deep)
Pre-2022	—	—	—	—	67.50 (Trees)	22.27 (Trees)	38.82 (Causal)
Pre-2023	—	—	—	—	—	31.82 (Trees)	25.23 (Causal)
Pre-2024	—	—	—	—	—	—	35.83 (Deep)

Table 31: **Temporal predictive drift: PR-AUC decay by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	0.869	0.476	**0.574**	**0.371**	0.111	0.388	**0.950**
CatBoost	Pre-2019	—	0.632	**0.539**	0.283	0.083	**0.486**	0.804
CatBoost	Pre-2020	—	—	0.414	**0.266**	0.125	**0.494**	0.747
CatBoost	Pre-2021	—	—	—	0.275	**0.200**	**0.556**	0.525
CatBoost	Pre-2022	—	—	—	—	**0.500**	**0.645**	**0.590**
CatBoost	Pre-2023	—	—	—	—	—	0.388	0.538
CatBoost	Pre-2024	—	—	—	—	—	—	0.799
XGBoost	Pre-2018	0.867	0.533	0.464	0.359	0.200	**0.950**	0.887
XGBoost	Pre-2019	—	0.490	0.361	**0.389**	0.091	0.360	**0.950**
XGBoost	Pre-2020	—	—	0.385	0.217	0.111	0.442	**0.917**
XGBoost	Pre-2021	—	—	—	0.357	0.143	0.476	**0.679**
XGBoost	Pre-2022	—	—	—	—	0.250	0.402	0.521
XGBoost	Pre-2023	—	—	—	—	—	0.318	**0.617**
XGBoost	Pre-2024	—	—	—	—	—	—	0.585
Logistic (L2)	Pre-2018	0.684	**0.721**	0.292	0.198	0.200	0.332	0.608
Logistic (L2)	Pre-2019	—	**0.686**	0.318	0.200	**0.200**	0.237	0.271
Logistic (L2)	Pre-2020	—	—	0.331	0.175	**0.200**	0.295	0.273
Logistic (L2)	Pre-2021	—	—	—	0.165	**0.200**	0.253	0.243
Logistic (L2)	Pre-2022	—	—	—	—	0.250	0.225	0.238
Logistic (L2)	Pre-2023	—	—	—	—	—	0.255	0.217
Logistic (L2)	Pre-2024	—	—	—	—	—	—	0.276
Spatial-FE Logistic	Pre-2018	0.475	0.621	0.333	0.169	0.167	0.335	0.679
Spatial-FE Logistic	Pre-2019	—	0.538	0.352	0.197	0.167	0.412	0.287
Spatial-FE Logistic	Pre-2020	—	—	0.383	0.148	0.100	0.412	0.229
Spatial-FE Logistic	Pre-2021	—	—	—	0.147	0.167	0.412	0.243
Spatial-FE Logistic	Pre-2022	—	—	—	—	0.250	0.295	0.228
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	0.398	0.140
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	0.178
TabNet	Pre-2018	**0.889**	0.707	0.365	0.182	0.100	0.245	0.729
TabNet	Pre-2019	—	0.623	0.313	0.182	0.100	0.124	0.500
TabNet	Pre-2020	—	—	**0.587**	0.183	0.005	0.028	0.056
TabNet	Pre-2021	—	—	—	**0.406**	0.083	0.552	0.608
TabNet	Pre-2022	—	—	—	—	0.053	0.067	0.131
TabNet	Pre-2023	—	—	—	—	—	**0.402**	0.469
TabNet	Pre-2024	—	—	—	—	—	—	**0.830**
Anchor Regression (Causal)	Pre-2018	0.797	0.068	0.126	0.321	**0.250**	0.205	0.255
Anchor Regression (Causal)	Pre-2019	—	0.303	0.422	0.226	0.077	0.267	0.350
Anchor Regression (Causal)	Pre-2020	—	—	0.248	0.233	0.077	0.430	0.583
Anchor Regression (Causal)	Pre-2021	—	—	—	0.288	0.091	0.251	0.461
Anchor Regression (Causal)	Pre-2022	—	—	—	—	0.250	0.222	0.406
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	0.279	0.472
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	0.384

Table 32: **Temporal predictive drift: PR-AUC lift by algorithm (Optimal Entropy-Based Discretization).**

Model	**Anchor Training**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
CatBoost	Pre-2018	6.111	7.692	**11.287**	**13.243**	24.000	19.472	**51.062**
CatBoost	Pre-2019	—	10.211	**10.604**	10.096	18.000	**24.427**	43.224
CatBoost	Pre-2020	—	—	8.141	**9.475**	27.000	**24.803**	40.153
CatBoost	Pre-2021	—	—	—	9.824	**43.200**	**27.944**	28.219
CatBoost	Pre-2022	—	—	—	—	**108.000**	**32.417**	**31.727**
CatBoost	Pre-2023	—	—	—	—	—	19.472	28.928
CatBoost	Pre-2024	—	—	—	—	—	—	42.925
XGBoost	Pre-2018	6.099	8.617	9.120	12.808	43.200	**47.737**	47.703
XGBoost	Pre-2019	—	7.913	7.108	**13.876**	19.636	18.071	**51.062**
XGBoost	Pre-2020	—	—	7.570	7.741	24.000	22.194	**49.271**
XGBoost	Pre-2021	—	—	—	12.747	30.857	23.929	**36.505**
XGBoost	Pre-2022	—	—	—	—	54.000	20.180	27.995
XGBoost	Pre-2023	—	—	—	—	—	15.977	**33.172**
XGBoost	Pre-2024	—	—	—	—	—	—	31.418
Logistic (L2)	Pre-2018	4.810	**11.645**	5.739	7.050	43.200	16.707	32.698
Logistic (L2)	Pre-2019	—	**11.075**	6.250	7.147	**43.200**	11.913	14.589
Logistic (L2)	Pre-2020	—	—	6.519	6.235	**43.200**	14.814	14.671
Logistic (L2)	Pre-2021	—	—	—	5.884	**43.200**	12.721	13.078
Logistic (L2)	Pre-2022	—	—	—	—	54.000	11.287	12.810
Logistic (L2)	Pre-2023	—	—	—	—	—	12.812	11.646
Logistic (L2)	Pre-2024	—	—	—	—	—	—	14.840
Spatial-FE Logistic	Pre-2018	3.343	10.031	6.545	6.034	36.000	16.842	36.505
Spatial-FE Logistic	Pre-2019	—	8.691	6.930	7.030	36.000	20.696	15.444
Spatial-FE Logistic	Pre-2020	—	—	7.530	5.296	21.600	20.696	12.318
Spatial-FE Logistic	Pre-2021	—	—	—	5.227	36.000	20.698	13.050
Spatial-FE Logistic	Pre-2022	—	—	—	—	54.000	14.836	12.280
Spatial-FE Logistic	Pre-2023	—	—	—	—	—	20.010	7.533
Spatial-FE Logistic	Pre-2024	—	—	—	—	—	—	9.564
TabNet	Pre-2018	**6.256**	11.421	7.171	6.499	21.600	12.323	39.193
TabNet	Pre-2019	—	10.056	6.152	6.503	21.600	6.236	26.875
TabNet	Pre-2020	—	—	**11.535**	6.519	1.000	1.411	2.990
TabNet	Pre-2021	—	—	—	**14.474**	18.000	27.717	32.698
TabNet	Pre-2022	—	—	—	—	11.368	3.374	7.060
TabNet	Pre-2023	—	—	—	—	—	**20.205**	25.233
TabNet	Pre-2024	—	—	—	—	—	—	**44.632**
Anchor Regression (Causal)	Pre-2018	5.610	1.101	2.488	11.465	**54.000**	10.300	13.688
Anchor Regression (Causal)	Pre-2019	—	4.896	8.304	8.067	16.615	13.400	18.812
Anchor Regression (Causal)	Pre-2020	—	—	4.875	8.316	16.615	21.617	31.354
Anchor Regression (Causal)	Pre-2021	—	—	—	10.274	19.636	12.599	24.785
Anchor Regression (Causal)	Pre-2022	—	—	—	—	54.000	11.167	21.831
Anchor Regression (Causal)	Pre-2023	—	—	—	—	—	14.028	25.382
Anchor Regression (Causal)	Pre-2024	—	—	—	—	—	—	20.656

Table 33: **Max-of-Family dominance (Optimal Entropy-Based Discretization).**

Anchor Training	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
			Panel A: Maximum Absolute PR-AUC				
Pre-2018	0.889 (Deep)	0.721 (Linear)	0.574 (Trees)	0.371 (Trees)	0.250 (Causal)	0.950 (Trees)	0.950 (Trees)
Pre-2019	—	0.686 (Linear)	0.539 (Trees)	0.389 (Trees)	0.200 (Linear)	0.486 (Trees)	0.950 (Trees)
Pre-2020	—	—	0.587 (Deep)	0.266 (Trees)	0.200 (Linear)	0.494 (Trees)	0.917 (Trees)
Pre-2021	—	—	—	0.406 (Deep)	0.200 (Linear)	0.556 (Trees)	0.679 (Trees)
Pre-2022	—	—	—	—	0.500 (Trees)	0.645 (Trees)	0.590 (Trees)
Pre-2023	—	—	—	—	—	0.402 (Deep)	0.617 (Trees)
Pre-2024	—	—	—	—	—	—	0.830 (Deep)
			Panel B: Maximum Relative PR-AUC Lift				
Pre-2018	6.26 (Deep)	11.65 (Linear)	11.29 (Trees)	13.24 (Trees)	54.00 (Causal)	47.74 (Trees)	51.06 (Trees)
Pre-2019	—	11.07 (Linear)	10.60 (Trees)	13.88 (Trees)	43.20 (Linear)	24.43 (Trees)	51.06 (Trees)
Pre-2020	—	—	11.54 (Deep)	9.47 (Trees)	43.20 (Linear)	24.80 (Trees)	49.27 (Trees)
Pre-2021	—	—	—	14.47 (Deep)	43.20 (Linear)	27.94 (Trees)	36.51 (Trees)
Pre-2022	—	—	—	—	108.00 (Trees)	32.42 (Trees)	31.73 (Trees)
Pre-2023	—	—	—	—	—	20.20 (Deep)	33.17 (Trees)
Pre-2024	—	—	—	—	—	—	44.63 (Deep)